



Tracking country Innovation Performance: The Innovation Output Indicator 2024

The Innovation Output Indicator 2024

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Abstract

This report presents the 2024 edition of the Innovation Output Indicator (IOI), a composite indicator published by the European Commission since 2013 to offer an output-focused metric of innovation performance at the country and EU levels. The IOI measures countries' capacity to derive economic benefits from innovation by tracking the extent to which innovative ideas reach the market, create knowledge-intensive jobs, and increase country technological capability. The IOI was developed in response to a 2011 European Council request to improve innovation progress tracking. It also serves as a flagship indicator for assessing the progress of the underlying objectives of the European R&I Framework Programmes and contributes to the European Commission's Directorate-General for Research and Innovation (DG RTD) biennial assessment of innovation performance in the European Union, complementing the European Innovation Scoreboard.

The report presents the latest results for the IOI composite metric and its underlying indicators for 46 countries, including European Union (EU) Member States (MSs) and selected EFTA, OECD and emerging economies. According to the IOI, Sweden, Ireland and Germany are the best performers in the EU and are followed by Finland and Luxembourg. Sweden outperforms the other EU countries in terms of IP applications, Ireland performs very well in terms of employment in knowledge intensive industries, while Germany performs the best across the EU in terms of its domestic value added content of its knowledge-intensive manufacturing exports. On the other hand, Romania, Latvia and Bulgaria reported the lowest performance among EU countries. The largest improvements in the IOI between 2013 and 2023 were recorded by Lithuania and Estonia, with the majority of EU MSs also showing an improvement in their IOI performance. The EU27 leads Canada, Australia, and China, but it continues to lag behind its main economic partners, namely the United States, South Korea and Japan. The EU27's performance has improved over time, albeit at the same or at a slower pace than its main economic partners. This suggests that the United States, South Korea, and Japan are maintaining or increasing their lead over the EU27, whereas China is rapidly narrowing the gap.

This report also includes an extension of the IOI to the regional level for the EU27. The first, exploratory edition of the Regional Innovation Output Indicator (RIOI) utilizes both established and novel datasets to evaluate the performance of EU NUTS-2 regions in terms of innovation outputs. It provides a snapshot of regional activity, covering technological and non-technological innovation, as well as regional contributions to value added in trade, employment, and innovation activities. The results for the RIOI showcase the regional divide between the south-east and north-west - of the EU in terms of innovation output performance, with regions of Central and Western EU countries such as Germany, Sweden, Belgium, and Netherlands performing in general better than those of Eastern and Mediterranean EU countries. Exceptions in this pattern are capital and major city regions in the Eastern and Mediterranean Europe, which excel compared to the remaining regions of their country. These results are confirmed by a spatial autocorrelation analysis which showcases the clustering of high-performing regions in North-western Europe and low-performing regions in the South-east. The five best performing NUTS-2 regions in terms of the RIOI are Stockholm, Oberbayern, Berlin, Praha, and Ile-de-France, while Germany and Sweden showcase respectively eight and four regions in the top 20 performers.

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1 Introduction

The Innovation Output Indicator (IOI) is a composite indicator that aims to support policy makers by offering an output-oriented metric of innovation performance at the country and EU levels. As a monitoring tool with a recognised policy relevance and impact, it has been developed by the European Commission since 2013 in response to the European Council's request to develop a "single integrated indicator to allow a better monitoring of progress in innovation" (European Council, 2011 and 2012). The IOI measures a country's capacity to derive economic benefits from innovation by tracking the extent to which innovative ideas reach the market, create knowledge-intensive jobs, and increase technological capability. The Commission uses the IOI to assess progress against the underlying objectives of the European R&I Framework Programmes (see, e.g., European Commission, 2024a). The information provided by the IOI complements other benchmarking tools, such as the European Innovation Scoreboard (European Commission, 2024b), which is also produced by the European Commission, and it feeds into the Commission's flagship biennial assessment of innovation performance in the European Union (European Commission, 2024c).

The IOI focuses exclusively on innovation output and monitors a reduced set of dimensions to support policy-makers in establishing new or reinforced actions to remove bottlenecks that prevent innovators from translating ideas into products and services that can be successful on the markets. Its measurement framework was introduced in the 2013 Communication and Staff Working Document (European Commission, 2013) and further refined in 2014, 2016, 2017 and 2023.

This report presents the latest results of the IOI. Compared to the previous edition of the index, the present version uses the same framework for 6 indicators and four components, while it features updated data for most of the IOI component indicators. It covers 46 economies, featuring EU Member States and selected benchmark countries which include members of the EFTA, OECD and BRICS groups as well as additional emerging economies. The latest data used in this report refer to 2023 but for some indicators latest available data refer to 2022 or earlier (see **Annex 8** for further details). The empirical results of the 2024 IOI show that, despite the progress made during the last decade, the EU27 continues to underperform compared to its main economic partners such as the United States, South Korea, Japan, and the UK.

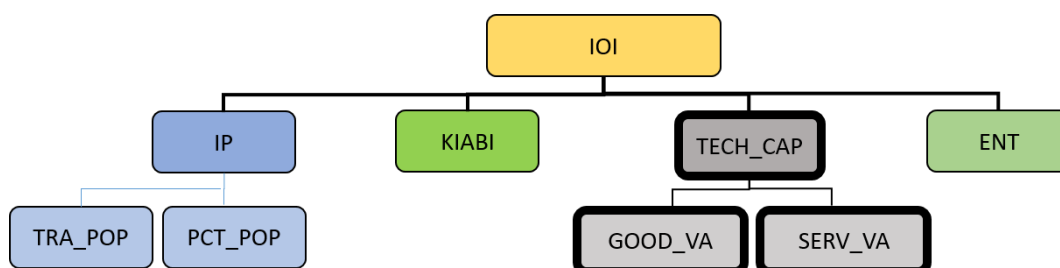
The present IOI report features, for the first time, an extension of the IOI to the regional level for the 27 EU countries. This first and exploratory edition of the Regional Innovation Output Indicator (RIOI) builds on the national IOI framework and uses established and novel datasets to assess the performance of EU NUTS-2 regions in terms of innovation outputs. It provides a snapshot of regional activity in terms of technological and non-technological innovation, regional contribution to value added in trade, employment, and innovative enterprises. This has the potential to complement the information provided in other regional benchmarking tools, such as the Regional Innovation Scoreboard (RIS), developed also by the European Commission (European Commission, 2023a). The RIOI covers 241 of the 242 EU NUTS-2 regions and the latest data used for its development refer to 2023 but for some indicators latest available data refer to 2022 or earlier (see **Annex 9**). The empirical results for the RIOI confirm the innovation divide captured at national level in the IOI. In addition, they document regional disparities in terms of innovation performance between high-performing Central and Northern European regions on the one hand, and below-average performing regions in Eastern Europe and the Mediterranean on the other hand.

The report is structured as follows. The next section presents the structure of the IOI. Section 3 discusses country performance in terms of the composite score, both when comparing EU Member States with one another and when comparing the EU as a whole with other benchmark countries, and in relation to each IOI component. Section 4 presents the structure and empirical results of the RIOI for 241 EU NUTS-2 regions. Section 5 includes a short description of contextual indicators to help understand performance differences across countries. Section 6 concludes. The description of the dataset and the methods applied to compute the aggregate measures, alongside the description of a series of robustness and sensitivity tests of the IOI and the RIOI to the methodological choices are provided in **Annex 1** and **Annex 2** for the IOI and in **Annex 5** and **Annex 6** for the RIOI.

2 Measuring country innovation performance: The IOI

The Innovation Output Index 2024 uses the revised IOI structure adopted in 2023 (Bello et al., 2023) and used in the previous two editions of the index (Bello et al., 2023 and 2024). This includes a total of six indicators aggregated into four components, including intellectual property (*IP*), knowledge-intensive activities in business industries (*KIABI*), domestic technology capacity (*TECH_CAP*), and innovative-active firms (*ENT*). These are depicted in **Figure 1** and are described in more detail in **Annex 8**.

Figure 1. The Innovation Output Indicator Framework



Source: Authors. Note: Indicator definitions are as follows, *TRA_POP* is trademark classes per million population, *PCT_POP* is patents filed under the patent cooperation treaty per million population, *GOOD_VA* is domestic value added content of medium-high and high-tech manufacturing exports as a share of total manufacturing exports. *SERV_VA* is the domestic value added content of knowledge-intensive service exports as a share of total service exports. *ENT* is the share of innovation-active firms on the total number of firms. *KIABI* measures the number of persons employed in knowledge-intensive business industries within total employment.

The IP component aims to capture a country's technological and non-technological innovation outputs. It consists of the arithmetic average of two indicators: the number of patent applications filed under the Patent Cooperation Treaty (*PCT_POP*) and the number of trademark classes in total resident trademark applications¹ (*TRA_POP*), each denominated by population (in millions).

The first indicator (*PCT_POP*) relies on patent data. It is well-known that patent data have a number of drawbacks and may not fully capture the whole set of innovation activities. Not all inventions, for instance, are patentable since they may not meet the patentability criteria (Dernis et al., 2001; Choi et al., 2007) or some organisations may prefer other mechanisms to protect their technologies (e.g., secrecy) (OECD, 2009). However, the use of patent information as a proxy of innovation output is quite established in the innovation literature (Maurseth & Verspagen, 2002; Barzotto et al., 2019; Corradini et al., 2021) and they remain one of the most reliable indicators of innovating activity. The requirements of novelty, industrial applicability, and non-obviousness to deem an invention patentable have indeed been argued to be strongly related to innovative activity.

The limitations of patent data as a measure of innovation outputs are particularly evident in sectors with a low propensity to patent, such as services (Scherer, 1983; Arundel and Kabla, 1998; Brouwer and Kleinknecht, 1999). As of the 2023 framework revision, the *IP* component includes a second indicator, which is based on trademark data (*TRA_POP*). Information on trademarks offers another viewpoint on the multifaceted phenomenon of innovation, in particular one that is less focused on the technological side. Trademarks have been shown to convey information on non-technological innovations (Mendonça et al., 2004; Millot 2009), such as innovation in services or marketing innovation, as well as to better capture innovation in small and medium-sized enterprises (SMEs) than traditional innovation indicators (Mendonça et al., 2004).

¹ A trademark application is filed by an applicant who can be resident or non-resident in the country of application. For example, a Belgian resident may want to protect their trademark in Germany (non-resident application). Each trademark application can cover multiple goods and services which are categorized into classes. Trademark classes are designed for international comparison and therefore suitable for the IOI.

The second component (*KIABI*) measures the number of persons employed in knowledge-intensive business industries as a share of total employment. It aims to capture the structural orientation of the business economy towards knowledge-intensive activities. This component has remained unchanged in the latest revision of the IOI and it features in the index's previous editions (Vertesy and Damioli, 2020; Bello et al., 2023; Bello et al., 2024).

The third component, *TECH_CAP*, aims to capture the contribution of domestic technology capacity to the country's exports in knowledge-intensive manufacturing and service sectors. It is composed as the arithmetic average of two indicators: domestic value added content of medium-high and high-tech manufacturing exports as a share of total manufacturing exports (*GOOD_VA*), and domestic value added content of knowledge-intensive service exports as a share of total service exports (*SERV_VA*). The *TECH_CAP* indicator makes use of trade in value added data², which measures flows related to the value that is added by a country in the production of any good or service that is exported (OECD, 2013). Compared to similar metrics used in previous editions of the IOI, which make use of data on gross exports,³ the use of value added data is able to differentiate between exports that reflect domestic technology capabilities and those that rely more on foreign inputs, such as those that are the result of mere imported components assembling. In addition, in a world of highly interconnected global value chains traditional measures of trade based on gross exports face certain limitations as they record flows of good and services every time they cross borders and can lead to double counting and misguided policy decisions (OECD, 2013). The use of value-added data avoids this. A higher value in *TECH_CAP* indicates a lower foreign content of the country's exports in these industries.

The fourth and last component of the IOI is *ENT*, which is the share of innovation-active firms on the total number of firms. This indicator captures both product and business process innovations (i.e., process, marketing and organisation innovation), as well as other innovation activities by firms (e.g., R&D or training) which have not yet resulted in a product or a business process innovation. It also reflects innovation activity across less knowledge-intensive economic sectors of a country, which are not captured by the *KIABI* or the *TECH_CAP* component.

² <https://www.oecd.org/sti/ind/measuring-trade-in-value-added.htm>

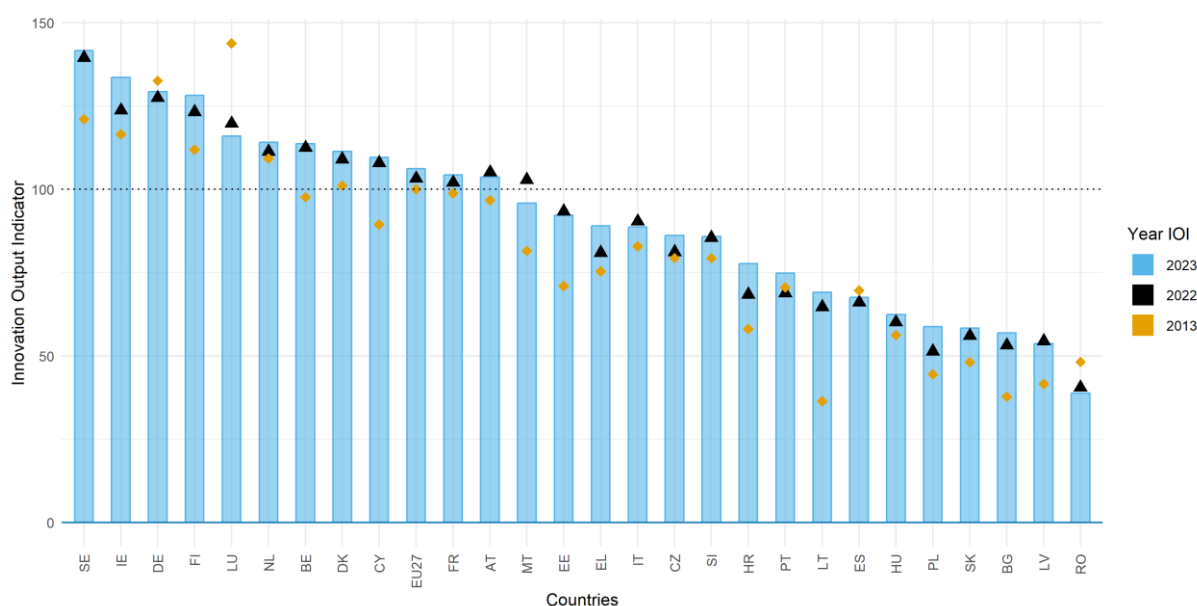
³ This was the case for the precedent of the *TECH_CAP* component in previous editions of the IOI, namely competitiveness of knowledge-intensive goods and services (*COMP*), which was defined as the arithmetic average of two indicators, namely the share of medium and high-tech goods in total exports (*GOOD*) and the share of knowledge-intensive services in the total service exports (*SERV*). The *COMP* indicator was adopted in the IOI to measure the technological competitiveness of countries, i.e. their ability to commercialize the results of research and development (R&D) and innovation in international markets. It also reflects specialization in medium- and high-technology products, which have been found to be positively associated with economic growth, productivity and welfare, as well as with high value-added and well-paid employment (e.g. Hausmann et al 2007, Yoo 2008, Falk 2009).

3 Results of the IOI 2024

3.1 Country performance in composite scores

This section presents the IOI 2024 scores obtained as the arithmetic average of the four components described in the previous section. The methodology for obtaining the IOI is described in detail in **Annex 1**. **Figure 2** shows IOI scores in 2013, 2022 and 2023 for the 27 EU MS. IOI scores are normalized as to benchmark the IOI value of the EU27 in 2013 to 100. Sweden is the country performing the best among the EU27, followed by Ireland, Germany, and Finland. On the opposite side of the ranking are Romania, Latvia, Bulgaria, and Slovakia. In total, 11 of the 27 Member States perform in 2023 better than the EU27. These include Western and Northern European countries, alongside Cyprus (see the bars in **Figure 2**)

Figure 2. IOI scores normalized to the EU27 score in 2013.



Source: Authors' calculation

The trends across the 2013-2023 decade⁴ reveal that for the majority of the EU Member States (23 out of 27) innovation output performance as measured by the IOI has improved (see also **Table 10** in **Annex 4**). The largest improvements in the normalised IOI scores are observed in Lithuania (32.6 percentage points) and, though to a somewhat lesser extent, in Estonia, Sweden, Cyprus and Croatia (from 21.2 to 19.6 percentage points). On the other hand, IOI scores declined between 2013 and 2023 only in four EU countries. This decline was particularly marked for Luxembourg, more nuanced in Romania, while it is very small for Germany and Spain.

A more detailed look over the 2013-2023 trends indicates that, over the last decade, low and high-performing EU countries progressed in more or less the same pace and that convergence between them has been relatively slow. Particularly, the average change in the IOI scores between 2013 and 2023 is 8.2 in absolute terms and 8.8% in relative terms for Member States above the median 2013 IOI score (which is represented by Czech Republic, for which the absolute increase in IOI is 6.8 and the relative one is 8.6%). On the other hand, Member States below the median exhibited on average a change in their IOI score that is equal to 11.4 in absolute terms and 23.6% in relative terms, which however is largely driven by the

⁴ Given the number of methodological changes performed across the different editions of the IOI, the comparison of changes across years should be based on the latest version, which back casts IOI scores using the current methodology up to 2013, rather than using the previous IOI editions.

improvements in Lithuania (without considering Lithuania, the absolute increase in IOI for these countries is 9.6, slightly above that of the countries below the median).

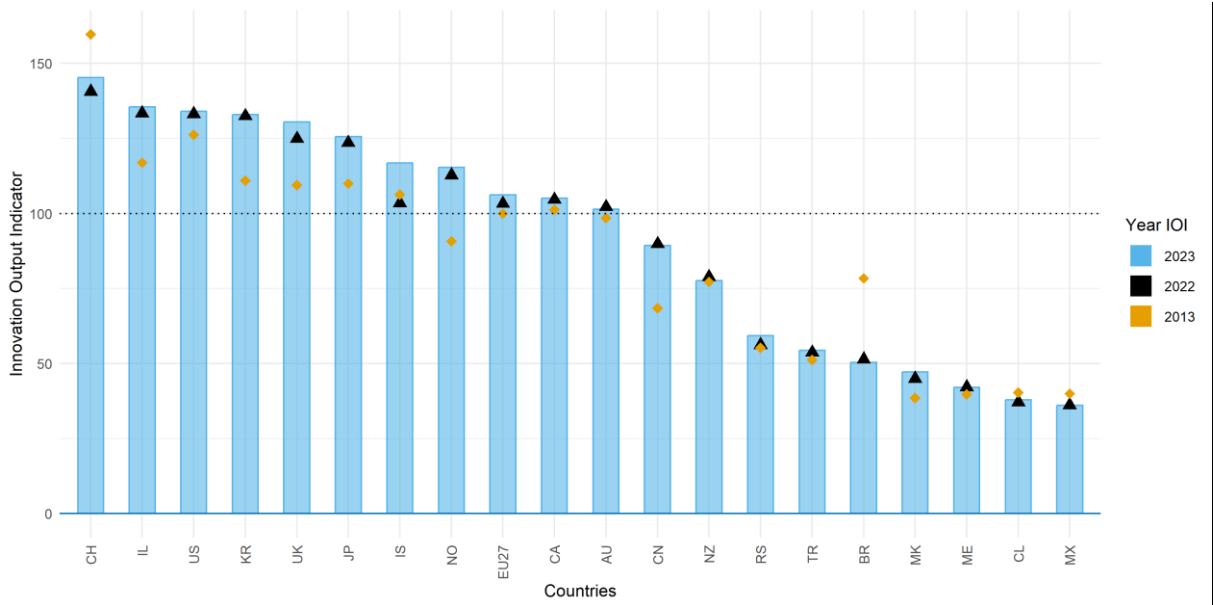
Box 1. EU aggregate figures considered in the IOI report

All country and regional performance scores presented in this and the next section and in the report’s Annexes benchmark country performance with respect to the performance of the EU27 as a single entity. For the IP, ENT and KIABI indicators, EU27 performance is constructed as the weighted average of the 27 country performances. This is equivalent to constructing, for example, the EU27 KIABI by adding the numerators and denominators of the 27 country KIABI ratios and then dividing them to obtain the EU27 aggregate score. The same is done for the TECH_VA indicators but with the difference that these exclude intra-EU trade to consider the EU27 as a single entity trading with the rest of the world. This measure (denoted as EU27 and used as such throughout the report) will be used to compare the IOI scores of the EU as a single entity with non-EU economies. Its use is in line with the benchmarking done in other monitoring frameworks such as the European and the Regional Innovation Scoreboard.

In the figures of individual indicators shown in Section 3.2, the *simple, unweighted*, average of the 27 EU countries (denoted as “EU Average”) is also presented, to facilitate a comparison of individual EU Member States performance with that of the “average EU country”. The TECH_VA estimates of EU Average include both extra and intra-EU trade as they are constructed from country-level TECH_VA values.

Figure 3 compares the performance of the EU27 in terms of innovation outputs with a selection of extra-EU countries including EFTA, OECD, BRIC members, and EU candidate countries. The EU27 benchmark excludes intra-EU trade, making it suitable to compare scores of the EU as a single entity with non-EU countries (see **Box 1**). The Figure provides IOI scores for 2013, 2022 and 2023 normalised by the scores of the EU27 for 2013. Switzerland is the best performer for 2023 and, as is evident from **Figure 3**, also for 2013 and 2022. Switzerland leads a group of countries with similar IOI scores, namely Israel, the United States, South Korea, United Kingdom, and Japan. The EU27 is ahead of a group of countries, which include Canada, Australia, China, and New Zealand, although differences in IOI scores are small in the case of Canada and Australia. These are followed by a group of additional emerging economies which however show markedly lower performance. These are Serbia, Türkiye, Brazil, North Macedonia, Montenegro, Chile, and Mexico.

Figure 3. Comparison with world / benchmark countries: IOI score normalized to the EU27 score in 2013.



Source: Authors’ calculation

Looking at trends in the 2013-2023 period, Norway exhibited the largest absolute improvement in the IOI (from 91% of the EU27 score in 2013 to 115% in 2023), followed by South Korea, United Kingdom, and China (for which the IOI improved from 68% of the EU27 score in 2013 to 89% in 2023). The EU27 exhibited

a nuanced improvement in innovation output performance as measured by the IOI of about 6.2% (its IOI is 106.2% of its 2013 value). This is similar to that of the US (6.15% or from 126.3% to 134% of the EU27 IOI in 2013). On the other hand, Brazil exhibited the most marked performance decline (-28.1 in absolute terms), which arises from a decline in its performance across various IOI indicators (share of innovation-active enterprises, patent applications, and, to a lesser extent, domestic value added in medium-high and high-tech manufacturing exports). Switzerland exhibited a sized performance decline of 14.4 in absolute terms (9% in relative terms), owed mostly to a 28% decline in the share of innovation-active enterprises), while the remaining countries exhibited smaller variations in their performance in the 2013-2023 period.

The performance scores for non-EU countries should be read with some caution. Despite the use of matching guidelines provided by, for example, Eurostat for the US and Japan⁵, differences in industrial classification and coverage may imply that KIABI scores are not fully comparable between non-EU countries and the EU27. Moreover, the presence of missing indicators is somewhat more pronounced for non-European countries, in particular for Montenegro (with missing GOOD_VA, SERV_VA and ENT), North Macedonia and Serbia (GOOD_VA and SERV_VA), and China (KIABI), Canada, Israel and Mexico (ENT).⁶ For comparisons across time, differences in data source (OECD, Eurostat and National Offices) within the same country are present in a few countries for ENT (Switzerland, United Kingdom) and KIABI (Japan, North Macedonia, South Korea, Türkiye, United Kingdom).

Box 2: Tracking performance change in the IOI over time

Benchmarking the IOI performance with respect to a fixed benchmark (the EU27 for 2013) allows for both cross-sectional as well as intertemporal comparisons of IOI scores. It can tell us whether a country (*c*) has improved or not in the last decade, and where it stands with respect to other countries, both in 2013, and in 2023. The extent of country improvement or deterioration over time can be gauged by simply dividing the two scores, i.e. as $IOI_c^{2023}/IOI_c^{2013}$. A ratio above one indicates country improvement over time, while one below one indicates deterioration. This estimate of change however does not directly inform the reader on the initial position of the country with respect to the benchmark or the change in the innovation gap or surplus between the country and the benchmark. In other words, it does not show whether the country has approached the benchmark, widened an existing gap with it, or fallen further behind it.

To obtain this information on the gap between the country and the benchmark itself, we plot in **Figure 4** the changes in countries' IOI over time (the ratio $IOI_c^{2023}/IOI_c^{2013}$) against their initial position with respect to the EU27 benchmark. Countries coloured in green (yellow) exhibited an improvement of their performance between 2013 and 2023 that was larger (smaller) than the 6.2% improvement recorded for the EU27 benchmark (dashed horizontal line). On the other hand, for countries coloured in red the ratio $IOI_c^{2023}/IOI_c^{2013} < 1$ which means that their IOI declined compared to their 2013 score. The vertical solid line at point $IOI_c^{2013} = 100$ splits countries into those performing better than the EU27 in 2013 and those lagging behind it.

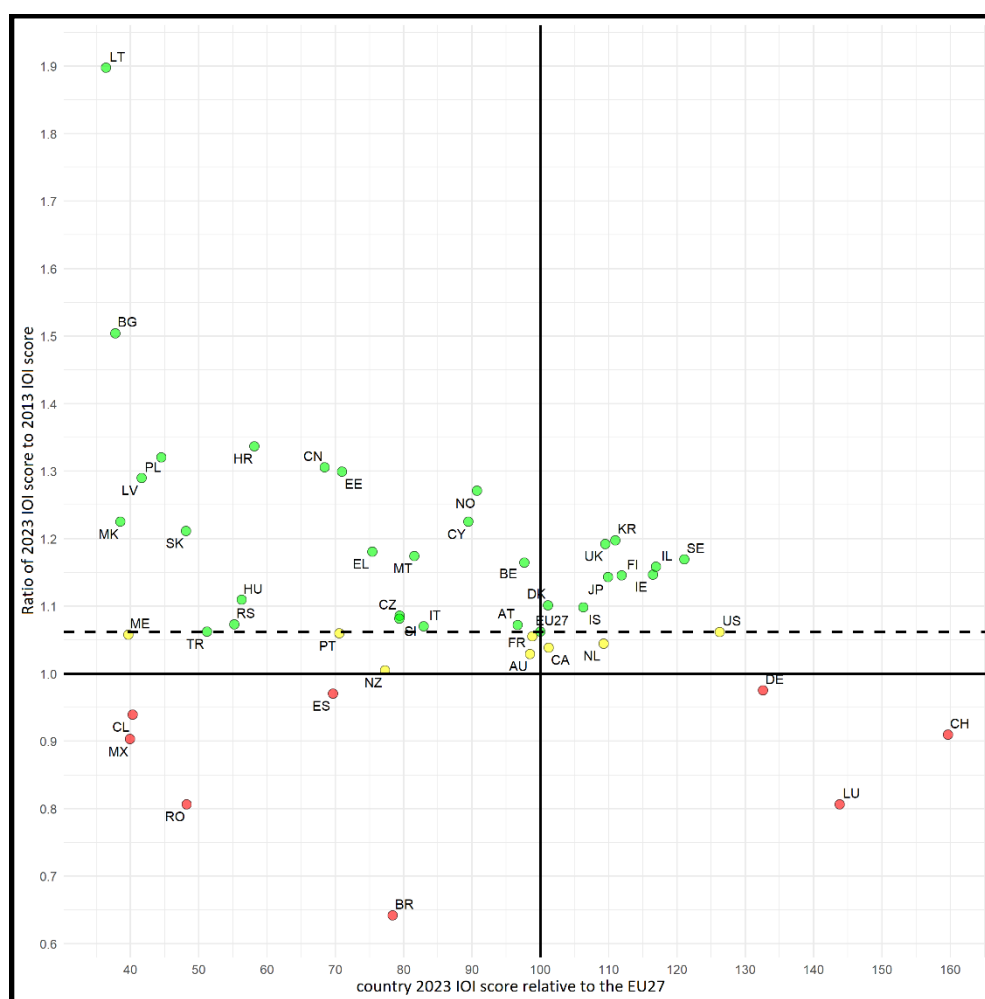
All the EU Member States that were lagging behind the EU27 performance in 2013, with the exception of France, Spain, Portugal and Romania, have improved their IOI performance at a faster pace than the EU27 benchmark, thereby narrowing the gap by 2023. France and Portugal have improved their performance but less than the EU27 (by 5.5% and 5.9% respectively), while Spain's and Romania's performance has fallen further behind to that of the EU27 benchmark. This fall was more pronounced for Romania and quite negligible for Spain.

For what concerns the EU27 competitors, the US has experienced growth over the last decade comparable to that of the EU27, resulting in a relatively stable performance gap in innovation output. In contrast, other international competitors, such as South Korea, Japan, the UK, and China, have demonstrated larger performance increases than the EU27. Notably, China's rapid progress has led to a significant narrowing of the performance gap with the EU27 over the past decade (see **Figure 3**). Conversely, South Korea, Japan, and the UK have further widened their existing performance gap with the EU27.

⁵ See https://ec.europa.eu/eurostat/cache/metadata/Annexes/htec_esms_an_8.pdf

⁶ See **Annex 1** for details on how data are imputed to build the final IOI in these cases.

Figure 4: Performance change in IOI scores between 2013 and 2023



Source: Authors' calculation.

Note: Dashed line represents the ratio of IOI score in 2023 to IOI score in 2013 for the EU27. Countries coloured in green (yellow) exhibited an improvement of their performance between 2013 and 2023 that was larger (smaller) than the one in EU27. Countries coloured in red exhibited a decline in their IOI in 2023 compared to their 2013 score.

3.2 Analysis by indicator

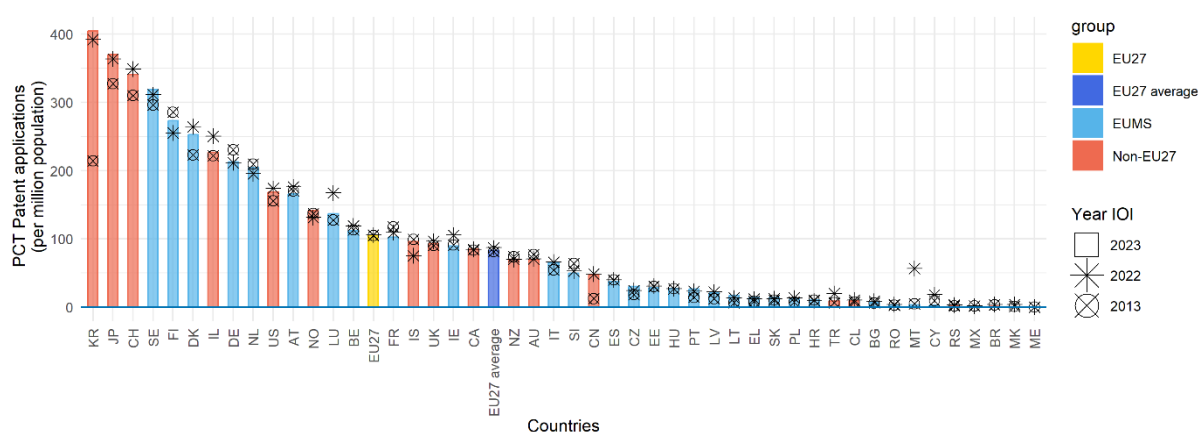
A composite indicator, such as the IOI, provides a concise summary of performance and serves as a starting point for examining trends and performance at the indicator level. This, in turn, can yield a more nuanced understanding of the underlying factors driving strong or weak innovation output. This subsection examines the performance of countries over time for each of the six indicators that comprise the IOI. The six figures below depict country performance for the years 2013, 2022, and 2023⁷. They also enable comparison of the 27 EU countries (light blue bars) with the “average EU country” (dark blue bar) (See Box 1). Alternatively, the performance of the EU27 as a single entity (gold bar) can be compared to that of non-EU countries (coral-coloured bars).

Figure 5 shows per-capita PCT patent applications (PCT_POP) for the 46 countries included in the IOI for this edition. South Korea leads the 2023 ranking, followed by Japan, Switzerland, and Sweden. These four countries have registered more than 300 patent applications per million citizens in 2023. Mexico, Brazil, North

⁷ In the Figures in this section, years refer to the IOI reference period (Year IOI) and correspond to a zero, one, two, or a 3-year shift relative to the years of the observed data due to carrying forward of missing observations for the most recent years (see **Annex 8**).

Macedonia, and Montenegro lag at the tail of the distribution having registered less than 2 PCT patent applications (per million in population). With regards to changes between 2013 and 2023, the majority of the considered countries (28 out of 46) exhibited an increase in patent activity per million population, although EU average has remained stable over this period. The largest improvements (more than 80%) are observed in China, Lithuania, South Korea and Portugal. A group of EU countries follows with milder but still large increases above 30% (Greece, Latvia, Czech Republic, Poland, and Slovakia), although their performance in 2023 is still below the average EU country. The patenting activity of the EU27 has remained stable during this decade at 105 patents per million population, lagging behind its main competitions, such as the US and Japan.

Figure 5. Number of patent applications filed under the Patent Cooperation Treaty per million in population (PCT_POP)



Source: See Annex 8.

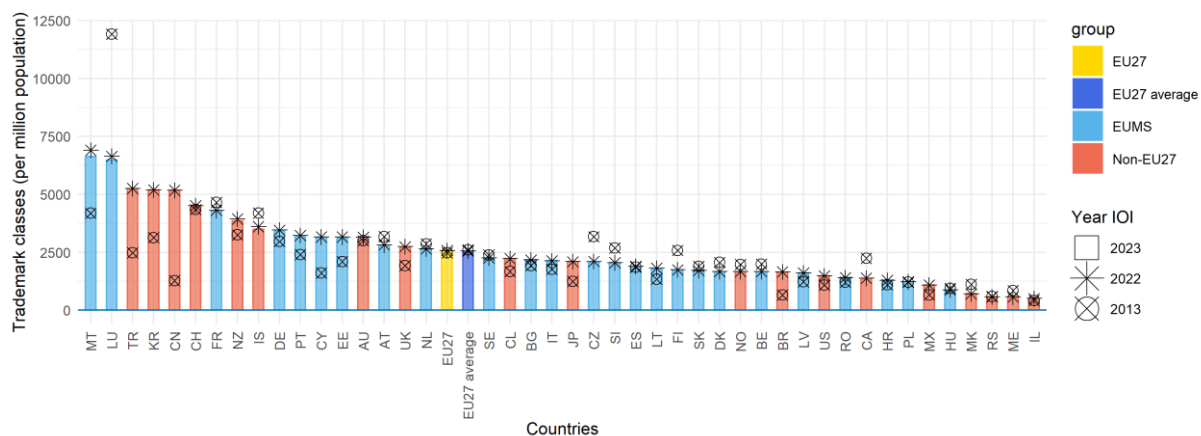
Figure 6 depicts the number of trademark classes⁸ of resident applications per capita (TRA_POP) across countries, with the exception of Greece, Ireland and North Macedonia for which this indicator is not available⁹. In 2023, Malta leads the ranking, overtaking Luxembourg which ranks second but topped the ranking in 2013.¹⁰ These are followed by a group of non-EU countries (Türkiye, South Korea, China, and Switzerland), among which China exhibited the largest improvement between 2013 and 2023, more than tripling its trademark registrations per million population. Compared to their ranking in terms of patents, these countries perform relatively better in terms of registered trademarks, but good performance in this indicator for some of them could also be linked to the presence of more favourable IPR-related regulations and procedures, and the relative importance of certain economic sectors. For example, high propensity sectors for trademarks like business services and advertisement have a substantial share in Luxembourg, while the gaming and software sectors are relevant in Malta (European Commission, 2022). At the opposite side of the 2022 ranking, Montenegro, Serbia, and Israel show the lowest performance, with less than 600 trademark classes per capita. Brazil, Türkiye, and Cyprus experienced marked improvements over time, managing to double their trademark registrations per million population. Trademark registration for the EU27 as a whole increased by only 3.4%, reaching about 2565 trademarks per capita in 2023, well above some of its competitions, such as the US.

⁸ Trademarks are registered with specific class codes – one for each type of product or service the trademark applies to. A single trademark application can cover many trademark classes.

⁹ Annex 1 describes how data are imputed to build the final IOI for these countries.

¹⁰ Luxembourg reached their highest point in 2016 with about 13 thousand trademarks per million population. Ever since there is a continuous decrease in this indicator, which is driven primarily from the decline in registered trademarks (-43% from 2016) rather than the population increase in the country (12% from 2016).

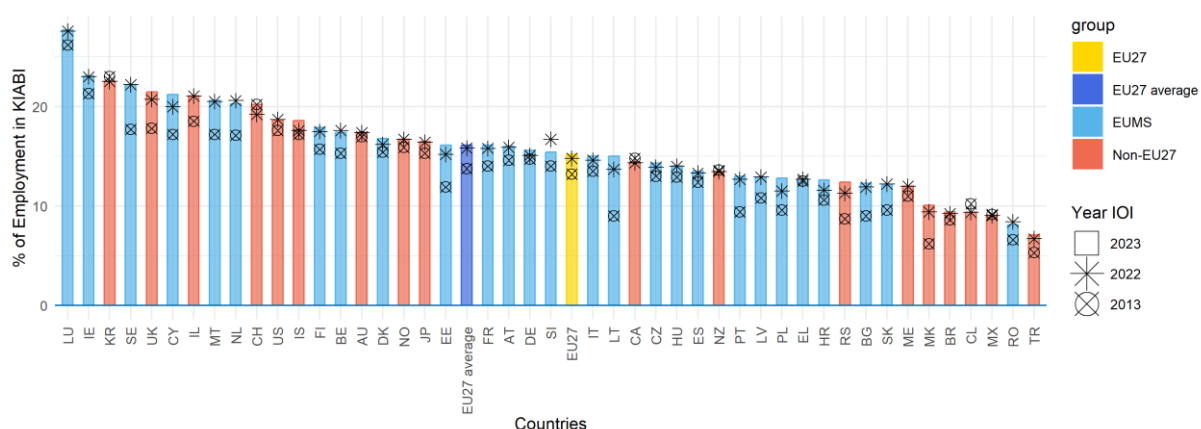
Figure 6. Number of trademark classes of resident applications per million in population (TRA_POP)



Source: See Annex 8.

Figure 7 shows the share of employment in knowledge-intensive activities in business industries (KIABI) in total employment except for China for which this indicator is not available. Luxembourg tops the ranking across the whole decade, with a marked lead over Ireland, South Korea, Sweden, and the UK, which rank in the 2nd to 5th positions. Performance on this indicator relates to the structure of a country's economy. Countries with strong financial and/or ICT service sectors tend to perform better than the rest. Brazil, Chile, Mexico, Romania and Türkiye are found at the bottom of the ranking for 2023, having less than 10% of employment in knowledge-intensive activities. The share of employment in KIABI increased in the vast majority of the considered countries (41 out of 45) between 2013 and 2023. Lithuania, Sweden, Estonia and Cyprus exhibited the largest improvements, increasing their share of employees in knowledge-intensive activities in business industries by more than four percentage points in a decade. On the other hand, this share declined, however slightly (less than 1 percentage point), between 2013 and 2023 for Mexico, New Zealand, Canada, South Korea, and Chile. For the EU27, the gradual shift of employment towards knowledge-intensive activities is a recognised megatrend, particularly for services (see, e.g., Cedefop, 2021). Between 2013 and 2023, KIABI increased for all EU Member States (ranging from an increase of 4.5 percentage points in Sweden to 0.2 in Greece), while the EU27 exhibited an increase of 2 percentage points (from 13.2% to 15.2%).

Figure 7. Employment in knowledge intensive activities in business industries (KIABI) as percentage of total employment

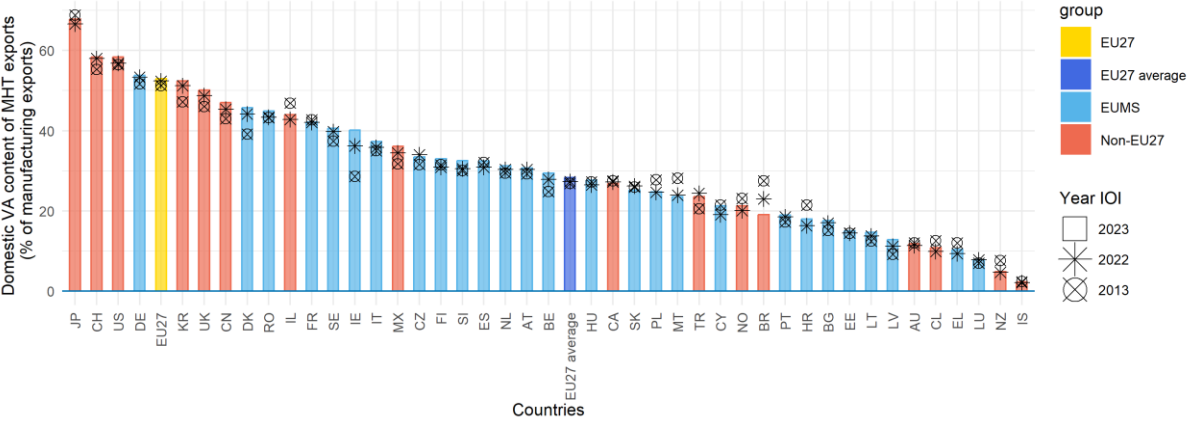


Source: See Annex 8.

The graphs in **Figure 8** and **Figure 9** depict the domestic value added content of exports, respectively in high and medium-high technology manufacturing (as a percentage of total manufacturing exports) and knowledge-intensive service industries (as a percentage of total service exports). Data on these indicators are not available for Montenegro, North Macedonia and Serbia (see **Annex 3** for more details). Japan is the country that, with a marked lead, records the highest domestic contribution to the value of its high and medium-high technology manufacturing exports –more than 60%– for all years between 2013 and 2023. It is

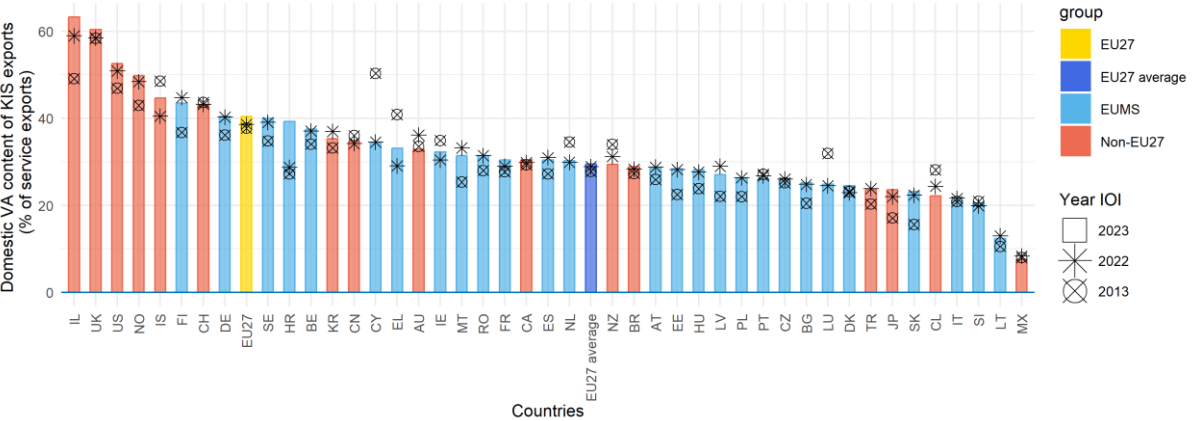
followed by Switzerland, the United States, Germany, South Korea and the UK. This reveals the lower dependency on foreign inputs in knowledge-intensive exports for these countries, for all of which more than 50% of the value added in knowledge-intensive manufacturing exports is created by the domestic industry. In terms of services, a group of international EU partners lead the ranking: Israel and the United Kingdom are the top performers with more than 60% of the value added to their knowledge-intensive service exports stemming from domestic industries, and they are followed by the United States and Norway, for which *SERV_VA* is larger than 50%. On the opposite side of the 2023 rankings, Luxembourg, New Zealand, and Iceland show the lowest domestic contribution to exports in high and medium-high tech manufacturing industries, and Slovenia, Lithuania, and Mexico do so in knowledge-intensive service exports.

Figure 8. Domestic value added content of exports in high and medium-high technology manufacturing (MHT) industries as percentage of total manufacturing exports



Source: See Annex 8.

Figure 9. Domestic value added content of exports in knowledge-intensive service industries as percentage of total service exports

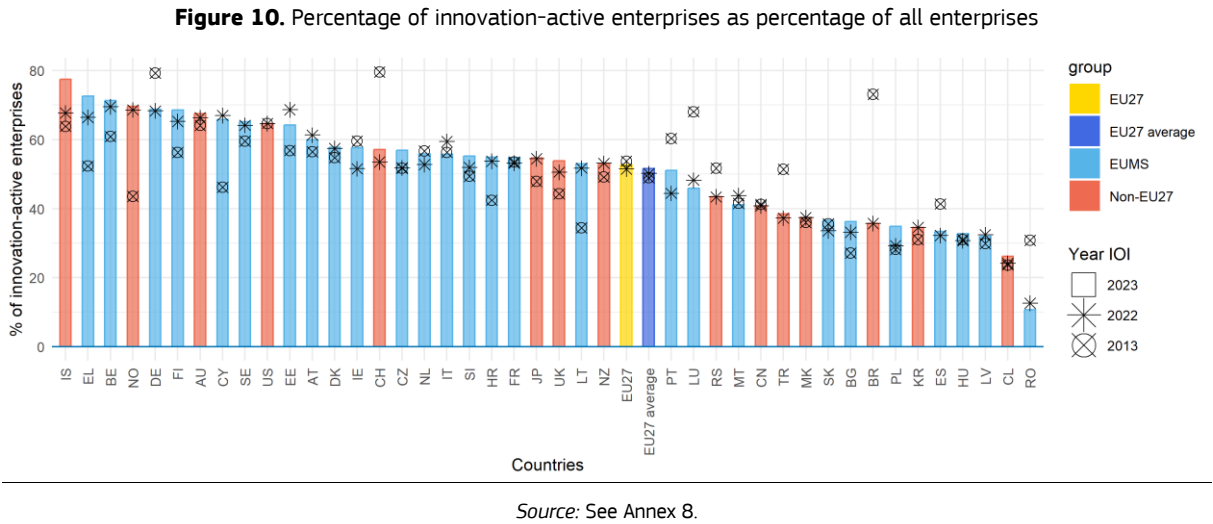


Source: See Annex 8.

Between 2013 and 2023, most of the considered countries (30 out of 44 for *GOOD_VA* and 31 out of 44 for *SERV_VA*) increased their domestic contribution to knowledge-intensive exports. Volatility is more pronounced in services than in manufacturing industries, but the average change in country shares has been similar, with a 1.1 percentage point change for *GOOD_VA* and a 1.7 percentage point change for *SERV_VA*. Ireland experienced the largest improvement in manufacturing industries, while Israel and Croatia showed the most significant improvement in services, with all three countries increasing their domestic value-added content of exports by more than 10 percentage points. In contrast, Brazil experienced a marked decline of more than 5 percentage points. Similarly, Cyprus, Greece, Luxembourg, and Chile exhibited a decline in the share of domestic value added in knowledge-intensive service exports. Overall, the EU27 relies less on foreign inputs in its knowledge-intensive manufacturing exports (53% in 2023) compared to its knowledge-intensive service exports (40% in 2023), and has increased its domestic contribution over the last decade, with a 1.7

percentage point increase in *GOOD_VA* and a 2.6 percentage point increase in *SERV_VA*. The EU's performance is similar to that of South Korea and the USA in the manufacturing sector. In contrast, it exceeds that of China, Korea, Japan, and Australia in the service sector, where it lags behind Israel, the USA, and the UK. It is also noteworthy that the scores for the EU27 as a single entity are significantly higher than those for the EU average, highlighting a substantial reliance of EU Member States on intra-EU trade.

Figure 10 presents the percentage of innovation-active enterprises, with the exception of Canada, Israel, Mexico, and Montenegro, for which this data is not available. In 2023, Iceland had the largest share of enterprises engaging in innovation activities, followed by Greece, Belgium, Norway, Germany, and Finland. In contrast, Latvia, Chile, and Romania had the lowest shares. As shown in the figure, this indicator exhibits the highest volatility over the 2013-2023 period among the six indicators considered in the IOI. Notably, there are nine countries that experienced a significant decline (more than 7%) in the share of innovation-active enterprises during this decade, while 11 countries saw an increase of more than 7%. This volatility reflects the dynamic nature of business activity, as well as potential methodological changes in national surveys. Substantial improvements of over 15 percentage points were observed in Norway, Greece, Cyprus, and Lithuania, whereas similar-sized declines occurred in Brazil, Switzerland, Luxembourg, and Romania. For the EU27 group, the share of enterprises engaging in innovation activities has remained relatively stable over time, with a small decline of 1 percentage point (from approximately 54% to 53%) between 2013 and 2023.



The analysis above documents a high heterogeneity in country performance across the various dimensions of innovation taken into account in the IOI. Such heterogeneity across the indicators comprising a composite index is frequently an attestation of the added value of combining these dimensions to obtain a composite measure of the socio-economic phenomenon under study (in this case innovation output activity), which helps to highlight other aspects of the phenomenon not immediately apparent from the analysis of individual indicators (Saisana *et al.*, 2022).

Within the EU, Sweden performs relatively better in terms of PCT patent applications and employment in knowledge-intensive activities in business industries compared to the other indicators, and performs relatively worse only in terms of trademarks. Ireland performs better in terms of employment in knowledge intensive activities as compared to other IOI indicators, while Germany does so in terms of the domestic value added content of exports in high and medium-high tech manufacturing and the share of innovative enterprises. On the other hand, Finland’s performance is best in terms of the share of PCT patent applications and domestic value added content of exports in knowledge-intensive services.

A similar degree of variety is observed among the EU’s international partner countries. Switzerland, which tops the IOI 2023 ranking of all 46 countries, excels in all IOI components, except for innovation-active enterprises. Similarly, the United States achieves its best performance in the domestic value added content of knowledge-intensive exports. In contrast, South Korea and Japan exhibit a mix of strong performances, with both countries standing out in terms of PCT patent applications. Additionally, South Korea has a high share of employment in knowledge-intensive activities, while Japan excels in the domestic value added content of manufacturing exports. These strengths highlight the excellence of South Korea and Japan’s high-tech manufacturing sectors in distinct ways.

The analysis of individual indicators can also provide insight into the performance gap between the EU27 and its main international economic partners, as noted in the previous section. By comparing the EU27's performance (gold bar) with that of non-EU countries (coral-colored bars), it becomes clear that the EU27's weaker performance is primarily due to its lower share of employment in knowledge-intensive business industries, where it is outranked by the US, South Korea, Japan, and the UK. Additionally, the EU27 lags behind the US, South Korea, and Japan in terms of PCT patent applications, although its performance is similar to that of the UK. Furthermore, the EU27 also trails behind the US and the UK in terms of the share of domestic value added in knowledge-intensive services, with a share of 40% compared to 53% and 60%, respectively. On the other hand, the EU27 performs relatively well compared to its international partners in terms of domestic value added in high and medium-high technology manufacturing, with a share of 53% that is close to those of South Korea (52%), the UK (50%), although it still lags behind the US (58%) and Japan (67%). The EU27 also performs similarly to the UK and Japan in terms of the share of enterprises engaging in innovation activities, with a share of 53% compared to 54% and 55%, respectively, although it still falls behind the US (65%).

Over the last decade, China has made significantly faster progress than the EU in most IOI indicators, underscoring its rapid emergence as a major player in the global innovation landscape. For instance, China's *KIABI* and *GOOD_VA* shares increased by approximately 4 percentage points each between 2013 and 2023, which is twice the corresponding increase in EU27 shares (2 and 1.7 percentage points, respectively). In contrast, China's dependence on foreign inputs knowledge-intensive exports has increased, as reflected by a 1.4 percentage point decline in its *SERV_VA* share, while the EU27 has reduced its reliance on foreign inputs in the production of knowledge-intensive services over the past decade, as evidenced by a 2.6 percentage point increase in its *SERV_VA* share.

3.3 Contextual indicators

A set of contextual indicators is provided in **Annex 9** to better understand differences across EU countries in relation to the IOI, as well as to its single components. The selected indicators refer to the following macro-economic dimensions: economic performance, structure of the economy, entrepreneurship and innovation.

The best performing countries in the IOI, namely Sweden, Ireland and Germany were also among the top 10 performers in terms of GDP per capita (see **Table 19** in Annex 9) in 2023 confirming the positive link between innovation and economic performance. They, however, differ in terms of structural orientation of their economy. Manufacturing, for instance, is more important for the German economy than it is for the Swedish economy, which is more service-oriented. Romania, which is at the bottom of the IOI ranking, is the EU country with the lowest share of employment generated in the service sector.

Enterprises' birth rates are also quite heterogeneous across EU countries. In 2020, they ranged from 4.6% in Greece, 5.4% in Austria and 6.5% in Italy to 14.1% in Malta and 18.1% in Lithuania. Among the four top innovators, as measured by the IOI, only Finland performed close to the EU average (nearly 9%), whereas the other three countries reported lower birth rates. Over the past years, there has been a steady decline in business dynamics in the EU, raising concerns about the implications for innovation. As shown in the Annex 9 (**Table 21**), birth rates have declined in 20 out of the 27 EU countries over the period 2011–2020. This decline is documented if enterprises' births are measured as a share of the total number of active firms or as a share of total employment. Several factors, including the demographic factors, the reduction in knowledge diffusion, the combination of increasing mark-ups and changes in market structure, and the Covid-19 pandemic in recent years, can explain this decline (European Commission, 2022).

Finally, fast-growing enterprises¹¹ can also be a key driver of economic growth and knowledge creation. In this regard, the EU has been found to lag behind its main international competitors (European Commission, 2021). Fast-growing enterprises only represent 12% of EU total employment in 2022, although large differences exist across countries. The share of fast-growing enterprises ranged from slightly over 3% in Cyprus to around 22% in Ireland in 2021.

¹¹ High-growth is defined by an annual average employment growth of 10% over a three-year period.

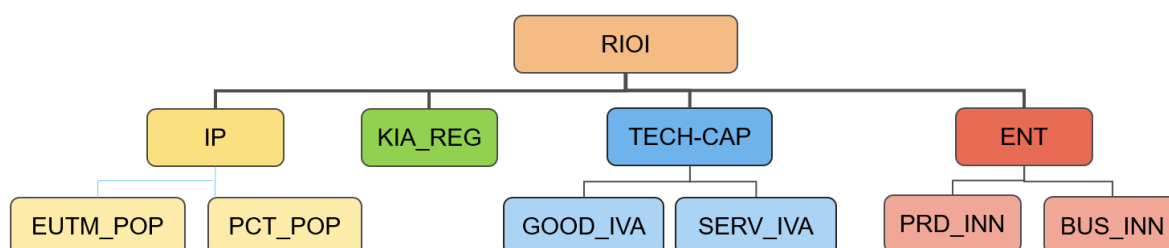
4 The Regional Innovation Output Indicator

4.1 The RIOI framework

Activities, relations, and collaborations taking place at a sub-national level are increasingly being recognised as crucial facilitators of innovation within a country. Innovation has a significant regional dimension as it tends to be highly concentrated in specific places (European Commission, 2024d). The European Commission recognises and supports the potential of regions to act as innovation incubators through several programmes, including the European Structural and Investment Funds and Horizon Europe programmes. These aim to foster innovation, promote a fairer Europe and facilitate a transition to a green and digital economy by funding infrastructure, R&D, education, training and entrepreneurship. As such, measuring the innovation output of regions is a crucial tool to monitor progress towards these goals. This becomes even more demanding in the face of amplifying regional disparities in R&I performance across the EU (European Commission, 2024d).

Despite the political interest, monitoring tools on EU regional innovation are scarce. An example includes the Regional Innovation Scoreboard produced by the European Commission (European Commission, 2023a), which is a regional extension of the European innovation scoreboard (EIS), assessing the innovation performance of European regions on a limited number of indicators. This section extends the IOI at the regional level for EU27 NUTS-2 regions and aims to present a first measure of innovation output at sub-national level. This experimental edition of the Regional Innovation Output Indicator (RIOI) builds on the IOI conceptual framework presented in **Figure 1**, making necessary adjustments to it due to data availability at the regional level. The conceptual framework of the RIOI includes a total of seven indicators aggregated into four components, which are depicted in **Figure 11** and are described in more detail in **Annex 9**.

Figure 11. The Regional Innovation Output Indicator Framework



Source: Authors. Note: Indicator definitions are as follows, *EUTM_POP* is EU trademark classes per million population, *PCT_POP* is patents filed under the patent cooperation treaty per million population, *KIA_REG* measures the number of persons employed in medium-high and high-technology manufacturing and knowledge-intensive services within total employment. *GOOD_IVA* is domestic value added content of international medium-high and high-tech manufacturing exports as a share of total international manufacturing exports. *SERV_IVA* is the domestic value added content of knowledge-intensive international service exports as a share of total international service exports, *PRD_INN* is the share of SMEs introducing product innovations on the total number of SMEs, and *BUS_INN* is the share of SMEs introducing business process innovations on the total number of SMEs.

The IP component, namely Intellectual Property (*IP*), aims to capture a region's technological and non-technological innovation outputs. It consists of the arithmetic average of two indicators: the number of patent applications filed under the Patent Cooperation Treaty (*PCT_POP*) and the number of trademark applications filed for at EUIPO (*EUTM_POP*), each denominated by population (in millions). The first indicator (*PCT_POP*) relies on patent data aggregated at the level of the NUTS-2 region from OECD REGPAT database, while the second indicator (*EUTM_POP*) is based on data from trademark applications to the EU Patent Office (EUIPO).¹²

¹² Data on EUTM trademark applications are available by Eurostat for up to 2016. More recent data are collected by DG RTD for the purposes of the RIS and were made available to JRC for the purposes of developing the RIOI.

The second component (*KIA_REG*) measures the number of persons employed in medium-high and high-technology manufacturing and knowledge-intensive service industries as a share of total regional employment. It captures the structural orientation of the regional business economy towards knowledge-intensive activities. This definition of knowledge-intensive regional employment is also used in the RIS monitoring framework.¹³

The third component, *TECH_CAP*, captures the contribution of domestic (in this case, *regional*) technology capacity to the region's *international* exports in knowledge-intensive manufacturing and service sectors. It is composed as the arithmetic average of two indicators: domestic value added content of medium-high and high-tech manufacturing *international* exports as a share of total manufacturing *international* exports (*GOOD_IVA*), and domestic value added content of knowledge-intensive service *international* exports as a share of total service *international* exports (*SERV_IVA*). These indicators are developed using data on regional trade in value added available from the European Commission's FIGARO REG Input-Output Tables (see **Box 3**). A higher value in the *TECH_CAP* component suggests that the particular NUTS-2 region is relying less on inputs from abroad in its exported knowledge-intensive goods or services, or alternatively, that the domestic (regional) technology contributes more to the value of its international knowledge-intensive exports. Lack of data availability prevented the use of intra-country trade data between different regions, so the extent that a region depends on inputs of other regions from the same country to develop its knowledge intensive international exports is not captured. However, this definition – which focuses on international trade – is the exact counterpart of the corresponding indicators of the national IOI, which also focus on *international* trade. This focus on international trade resonates also with the various definitions of regional competitiveness, which these indicators aim to capture. Such definitions place a larger importance on a region's ability to produce goods and services which meet the needs of *international* markets (see, e.g., European Commission, 1996, Kitson *et al.*, 2004).

The fourth component of the RIOI is *ENT*, which is calculated as the arithmetic average of two indicators, *PRD_INN* which is the SMEs that have introduced product innovations as a share of the total number of SMEs in the region, and *BUS_INN* which captures the SMEs that have introduced business process innovations as a share of the total number of SMEs in the region. As such, the *ENT* component can capture both product and business process innovations (i.e., process, marketing and organisation innovation), while reflecting product and business process innovation activity across all economic sectors of a country, including those of less knowledge-intensive nature which are not captured by the *KIA_REG* or the *TECH_CAP* component.

The RIOI framework resembles closely the national IOI framework and has also benefited from a detailed literature review of existing frameworks monitoring aspects related to innovation at the regional level.¹⁴ The two frameworks differ in (i) the source of trademark data (WIPO for the IOI and EUIPO for the RIOI), (ii) the sectors included in the *KIABI* and *KIA_REG* components (see footnote 13), and the *ENT* component, which in the national IOI is a single indicator capturing the share of firms that are *innovation-active*, i.e., they have either introduced product and/or business process innovations or engage in other innovation which have not yet resulted in a product or a business process innovation. This share is measured across all firms regardless of their size, whereas the two indicators of the RIOI *ENT* component focus on SMEs.¹⁵

¹³ The *KIA_REG* indicator differs from the one used in national IOI in that it includes NACE Rev. 2 sectors C20, C27-C30, H50, N80, O, P, Q, R91-R93 and excludes B09, C19, and N79.

¹⁴ This review regarded EU regional monitoring frameworks such as the RIS, the Regional Competitiveness Index, the Regional Entrepreneurship Development Index, as well as regional innovation frameworks from third countries such as the US. The results of the review confirmed that patent and trademark counts are the basic source for variables reflecting technological and non-technological innovation. Also, employment shares in groups of knowledge-intensive business industries are an indispensable part in Indicators capturing the structural orientation of the business economy, and the definitions used are quite similar to that used in the RIS. In addition, when data on enterprise innovation activities are reflected in European frameworks, this is done using EU CIS data. Lastly, our review did not reveal any regional index making use of trade in value added data, or regional trade data in general. This gap was addressed in the RIOI by relying on the FIGARO-REG Input-Output Tables (see **Box 2**).

¹⁵ This is done for confidentiality reasons. CIS data at the regional level are collected from National Statistical Offices for DG RTD for the purposes of the RIS and were made available to JRC for the purposes of developing the RIOI. For more details on this data collection, please refer to the RIS technical reports (European Commission, 2023b).

Box 3. The European Commission's FIGARO-REG Input-Output Tables

Input-Output tables and their multiregional extensions have been a cornerstone of economic analysis and modelling, traditionally applied at the national level. These analyses are particularly valuable for evaluating the effects of economic shocks, inter-industry relationships, and, more recently, for quantifying the value added content of international trade and of final demand of countries. By leveraging suitable tools, these analyses can be modelled in a way that not only enhances the theoretical understanding of the economies' performance but also provides practical, applied insights, ultimately enriching our knowledge to improve economic decision-making.

However, the escalating complexity of global trade, together with the increasing concentration of economic activities within countries, poses significant challenges in evaluating economic, social, and environmental performance accurately. This challenge is particularly pronounced when analysing the economic flows within regions of a country, where the intricacies of global trade and the uneven distribution of economic activities can distort our understanding of the metrics of economic performance. As global trade volumes and complexities continue to rise, and economic activities become more dispersed within countries, the assessment of economic, social, and environmental performance at the subnational level becomes increasingly nuanced, highlighting the need for more refined and regional approaches to evaluation.

To address this challenge, the Joint Research Centre (JRC) has developed a comprehensive multiregional input-output table, known as FIGARO-REG (García-Rodríguez *et al.*, 2023), building on the FIGARO intercountry Input-Output tables (Rueda-Cantuche & Rémond-Tiedrez, 2019) published by Eurostat as official statistics since 2021. FIGARO-REG is a detailed database that is consistent with FIGARO, using 2017 as a benchmark year, and provides a more granular understanding of the economic landscape of each specific economic area at the NUTS2 level. This database enables the observation of the consequences of economic linkages, weaknesses, dependencies, and potential improvements in economic performance, while also offering a complete view of bilateral trade flows. FIGARO-REG comprises 240 NUTS2 regions from the European Union (EU), 48 regions from Norway and the United Kingdom together, and 16 countries plus a rest of the world region. It also provides an unprecedented level of detail with 56 industries, allowing for a nuanced view of economies beyond national borders in the EU. This database enables researchers and policymakers to conduct economic analysis at the subnational level, gaining a deeper understanding of the unique characteristics of each region and developing targeted policies to support economic growth and development. Furthermore, FIGARO-REG facilitates the analysis of both intraregional and interregional effects, providing a powerful tool for informing evidence-based decision-making.

CARMEN (Computable Analysis of the Regional Multipliers of the European Economy) is the modular economic model approach used to exploit the potentialities of FIGARO-REG. It encompasses several modules to perform specific and customised economic analysis. The main value added of CARMEN is that it allows for a deep dive analysis of economic linkages, taking advantage of the granularity of FIGARO-REG at territorial and industrial level analysing economic impacts, evaluating short-term economic effects of external shocks, delivering detailed outcomes of the composition of the regional value added or obtaining global value chains indicators.

Several studies have been conducted utilising the CARMEN framework, each with different objectives and focuses, all sharing a common emphasis on subnational analysis within the EU. These analyses have explored a range of topics, including the value-added composition of the European automotive industry cluster (European Commission, 2024e), the direct and indirect employment impacts of the coal transition (European Commission, 2025a), and the vulnerability of the EU regions to non-EU protectionist trade policies (European Commission, 2025b). These studies demonstrate the versatility and applicability of the CARMEN framework in examining complex economic issues at the subnational level, providing valuable insights for policymakers and stakeholders.

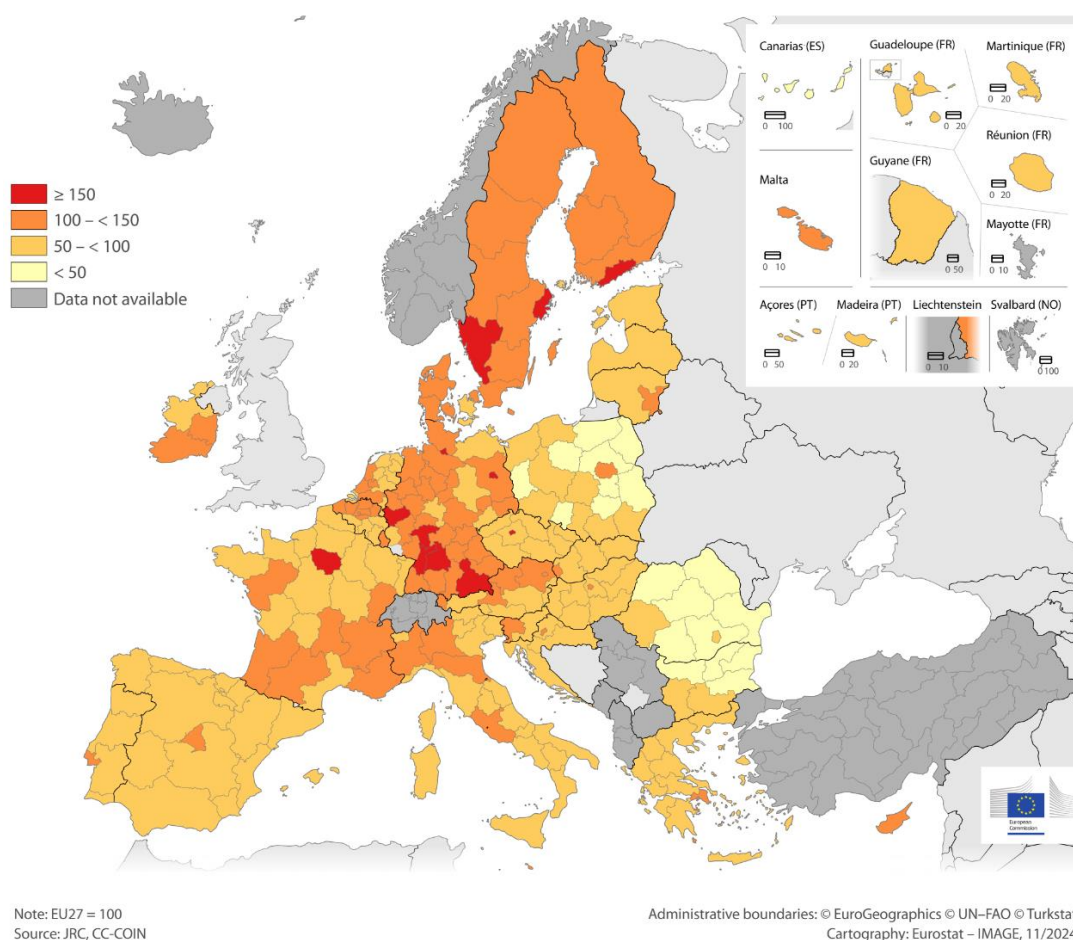
The CARMEN framework also shows its capacity analysing global value chains (GVCs) at the regional level, where the production of goods and services often involves complex networks of inputs from multiple countries worldwide. The elaboration of regional indicators capturing the intricate flows of goods and services within the GVCs overcome the limitation of the national ones, providing a more nuanced and accurate understanding of international trade by accounting for the value added generated in each territorial unit due to the consumption of goods and services worldwide. By doing so, these GVC indicators offer a more detailed and precise picture of commercial relationships between regions, enabling the analysis of global production distribution, networks, and supply chains. This, in turn, can provide valuable insights into the organisation and dynamics of GVCs, ultimately informing policy decisions and business strategies that seek to optimise regional participation in global trade.

4.2 Results of the RIOI 2024

Figure 12 presents the results of the first, experimental edition of the RIOI. The RIOI is presented as a snapshot in time, making use of the latest available data for each indicator (see Annex 8), and is also benchmarked against the EU27 score for 2023. The strong orange and red colours correspond to regions that score above the EU27 and the yellow and pale orange to regions that score below.

In total, there are 92 NUTS-2 regions (38%) for which the RIOI is above the EU27 benchmark, while the top 5% (12 regions, coloured red in **Figure 12**) have an RIOI which is greater than the EU27 benchmark by 50%. The best performing regions in terms of innovation output are found in Central and Northern European countries including Germany, France, Belgium, the Netherlands, Denmark, Sweden, and Finland. Stockholm (SE11), Oberbayern (DE21), Berlin (DE30), and Praha (CZ01) regions lead the RIOI ranking. These are followed by Ile-de-France (FR10), Helsinki-Uusimaa (FI1B), Stuttgart (DE11), Köln (DEA2), Karlsruhe (DE12), Hamburg (DE60), Västsverige (SE23), and Darmstadt (DE71). Two regions from Germany, two from Sweden, and one region from Austria, Belgium, Ireland, and Netherlands complete the top-20 best performing regions in terms of the RIOI (see **Table 1**). The detailed ranking of all 241 regions is given in **Table 17** in **Annex 7**.

Figure 12. RIOI scores normalized to the EU27 score in 2023



Source: Authors' calculation through EC [IMAGE](#) tool

These regions thrive due to a combination of factors, including high-skilled workforce, attractiveness to foreign investments, and the presence of critical infrastructure and institutions that foster innovation, such as innovation hubs, academic institutions, and research centres (European Commission, 2024f). For instance, the Paris-Saclay cluster located in Île-de-France region (FR10) brings together academic institutions, research centres, and leading corporations to facilitate knowledge transfer and collaboration in areas like Artificial Intelligence, biotechnology, and energy to facilitate knowledge transfer. In contrast, Praha (CZ01) region benefits from a unique concentration of state institutions, numerous universities, research institutions, and a

diverse range of companies, from start-ups to established international and local firms, as well as well-developed infrastructure, and a large pool of skilled workers.

Mediterranean and Eastern European regions are below the EU27 score with innovation output being the weakest in certain regions of Bulgaria, Romania, and Poland. Columns 4 to 6 of **Table 1** show the 20 weakest performing regions in terms of the RIOI. Three Romanian regions are ranked at the bottom of the RIOI distribution. The 20 least performing regions include also eight regions from Poland, four from Bulgaria, and two from Spain. However, there are certain South and Eastern European regions that stand out and perform relatively better than other regions within their respective countries. These include those comprising country capitals or major cities, such as Comunidad de Madrid (ES30), Centro (PT17) including Lisbon, Lazio (IT14), Attiki (EL30), Cyprus (CY00), Warszawski stołeczny (PL91) including Warsaw, and Sostinės regionas (LT01), which includes the capital city of Vilnius. These regions outperform the EU27 in terms of innovation output (see **Figure 12**).

Table 1. The 20 best and the 20 least performing regions in terms of the RIOI.

rank	NUTS-2 code	Name	rank	NUTS-2 code	Name
1	SE11	Stockholm	222	ES63	Ciudad de Ceuta
2	DE21	Oberbayern	223	PL62	Warmińsko-mazurskie
3	DE30	Berlin	224	ES70	Canarias
4	CZ01	Praha	225	PL61	Kujawsko-pomorskie
5	FR10	Ile-de-France	226	PL81	Lubelskie
6	FI1B	Helsinki-Uusimaa	227	BG31	Северозападен (Severozapaden)
7	DE11	Stuttgart	228	RO12	Centru
8	DEA2	Köln	229	PL43	Lubuskie
9	DE12	Karlsruhe	230	BG34	Югоизточен (Yugoiztochen)
10	DE60	Hamburg	231	BG33	Североизточен (Severoiztochen)
11	SE23	Västsverige	232	BG32	Северен централен (Severen tsentralen)
12	DE71	Darmstadt	233	PL84	Podlaskie
13	DK01	Hovedstaden	234	PL52	Opolskie
14	IE06	Eastern and Midland	235	RO11	Nord-Vest
15	NL41	Noord-Brabant	236	RO31	Sud-Muntenia
16	BE31	Prov. Brabant Wallon	237	PL72	Świętokrzyskie
17	DE25	Mittelfranken	238	PL92	Mazowiecki regionalny
18	AT13	Wien	239	RO22	Sud-Est
19	SE22	Sydsverige	240	RO41	Sud-Vest Oltenia
20	SE12	Östra Mellansverige	241	RO21	Nord-Est

Source: Authors' calculation.

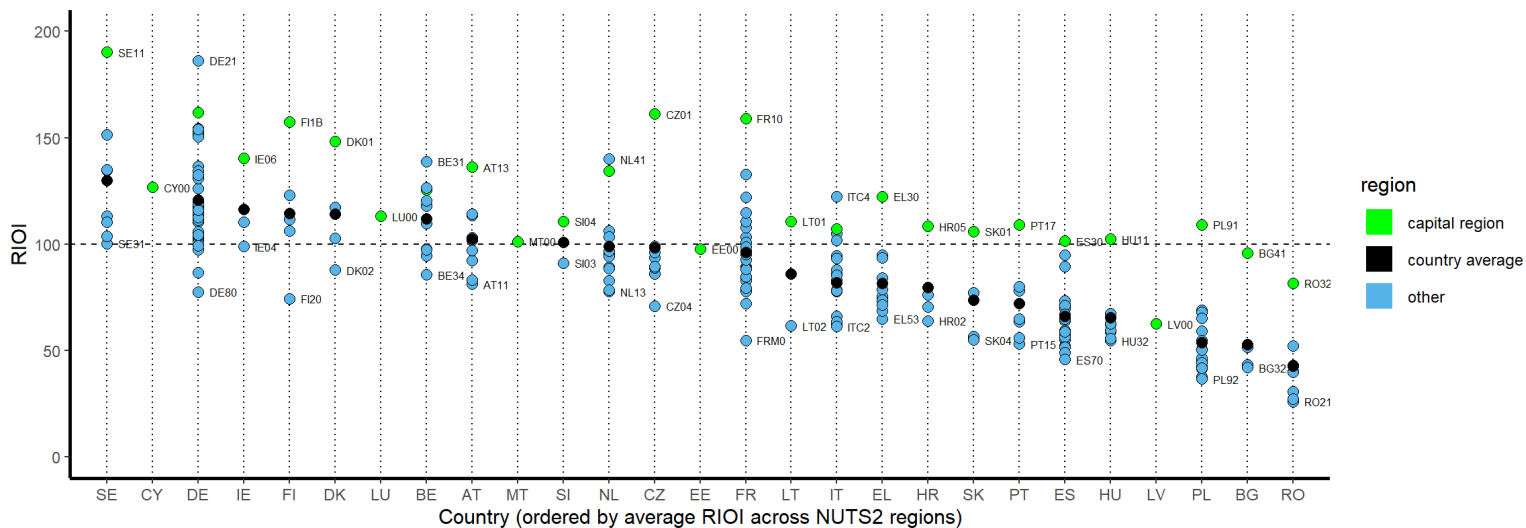
A more detailed look and the comparison of individual indicators is quite useful as it helps identifying regional strengths and performance gaps and thus provides input for more informed policy design at the regional level. With regards to the top-20 regions in **Table 1**, Stockholm (SE11) performs relatively better in terms of KIA_REG (ranked 2nd across the 241 regions with 68% of its employment being in knowledge-intensive business industries) and registered trademarks, with 722.5 per million inhabitants (336% of the EU27 value). On the other hand, its performance is weaker in terms of BUS_INN. Oberbayern (DE21) is the most relatively balanced top-20 region, excelling in terms of its domestic value added to knowledge-intensive international manufacturing and service exports (67.7% and 62.6% respectively) and intense patenting activity (15.9 patents per million inhabitants, 336% of the EU27 value). Berlin (DE30) and Praha (CZ01) lead the regional ranking in terms of domestic value added in knowledge-intensive international manufacturing and service exports (76.1% and 77.1% respectively).

Regional performance across individual indicators is not balanced and even top-performing RIOI regions can underperform in a certain area. For example, Brabant Wallon (BE10) ranks 1st in terms of employment in knowledge-intensive business industries (68.3%). However, it ranks slightly below EU27 in terms of domestic value-added in knowledge-intensive international service exports (93% of the EU27 value, 149th). Similarly, the Eastern and Midland Irish region (IE06) is a top-performer when it comes to adding domestic value to its knowledge-intensive exports (ranked 3rd in both SERV_IVA and GOOD_IVA), but its patenting activity is rather weak (it did not have any registered patents and thus ranks at the bottom along with other 68 regions that have zero patents).

With regards to the 20 weakest regions in **Table 1**, most of them perform below the EU27 score for all the considered indicators, and in many cases they are ranked among the weakest performers. For example, the ranks of the Świętokrzyskie (PL72) region in terms of the seven RIOI indicators range from 191st to 238th highlighting its relatively poor performance across all dimensions of innovation output considered in the RIOI. On the other hand, less strong regions may perform well in a few or in one indicator (European Commission, 2024d), suggesting a comparative advantage or effective policies in these domains. This is the case for the Romanian Sud-Muntenia (RO31) and Sud-Vest Oltenia (RO41) regions, which are among the weakest performers in terms of all the RIOI indicators except for their domestic value added to knowledge-intensive international manufacturing exports, where they are ranked 30th and 62nd respectively¹⁶.

Figure 13 summarizes regional performance by country, highlighting each country's capital city region (green coloured dots), country average performance (black coloured dots), and each country's best and worst performing regions by their NUTS code. With the exception of Germany, Belgium, Netherlands, and Italy, capital city regions are the best in terms of innovation output within a country. This stylised fact points towards the known over-concentration of innovation activities in capital and other major cities. In addition, in some countries, *strong* innovation output performance is dispersed across multiple regions. Notably, all regions in Sweden and the majority of regions in Belgium, Finland, Germany, Denmark, and Ireland exceed the EU27 score (indicated by the dashed line). In contrast, countries such as Bulgaria and Romania have all regions performing below the EU27 score. Additionally, in the Czech Republic, Greece, Portugal, and Spain, only one NUTS-2 region - the one containing the capital city - outperforms the EU27 score. In the remaining countries, such as Austria, the Netherlands, and France performance is balanced around the EU27 benchmark, with at least a third of their regions surpassing the EU27 score.¹⁷

Figure 13. RIOI scores by country, normalized to the EU27 score in 2023



Source: Authors' calculation

Regarding within-country differences in regional performance, two distinct patterns emerge. In the first pattern, observed in most countries, one region –most notably the capital region– significantly outperforms its domestic counterparts in terms of innovation output. Examples of this pattern include Austria (AT13), Bulgaria (BG41), the Czech Republic (CZ01), Finland (FI1B), France (FR10), Greece (EL30), Romania (RO32), and Slovakia (SK01), in which the capital region's lead over the other regions is quite extensive. Even in Sweden, where all regions perform relatively well, SE11 stands out as a top performer. In contrast, the second pattern is characterized by a more dispersed performance across regions, without a single standout performer. This is

¹⁷ 86% (33/38) of the NUTS-2 regions in Germany, 80% (4/5) in Denmark and Finland, 67% (2/3) in Ireland, 63% (7/11) in Belgium, and 55% (5/9) in Austria outperform the EU27. The same holds for 33% (4/12) of the regions in the Netherlands and 31% (8/18) in France.

the case in countries such as Belgium, Spain, and to a lesser extent Italy. Slovenia, which consists of only two regions, exhibits a unique situation, with both regions performing similarly.

“Centralisation” of regional innovation performance in the capital city region is more severe in the South and Eastern European countries. In Bulgaria, Yugozapaden (BG41) tops the within-country RIOI ranking, ranked 109th in terms of the RIOI, while the second-best region (BG42-Yuzhen tsentralen) is ranked 221st. In the Czech Republic, Praha (CZ01) is ranked 4th while CZ02 (*Střední Čechy*) is 95th. In Lithuania there is a difference of 137 rank positions between *Sostinės regionas* (LT01, 61st) and *Vidurio ir vakarų Lietuvos regionas* (LT02, 198th). In Poland, despite Małopolska (PL21) –the country’s 2nd best region and 178th in total in terms of the RIOI– includes the economic centre and business hub of Kraków and features a strong industrial economy (European Commission, 2024f), it still lags quite much behind the *Warszawski stołeczny* (PL91) region (which includes Warsaw and is ranked 66th) in terms of innovation output. Similarly large differences can be observed in Greece, Hungary, Portugal, and Romania. On the other hand, in Germany, Finland and France, the differences in performance between the top-performing region and its immediate follower are much less intense.

Figure 13 also illustrates that the strong performance of countries like Sweden, Ireland, and Germany, and the weak performance of countries like Bulgaria and Romania, as reflected in the national IOI (**Figure 2**), is mirrored at the regional level. This suggests that in top-performing countries, strong national-level performance is often underpinned by robust performance in most of their regions. Conversely, countries at the lower end of the national IOI distribution tend to exhibit weaker performance across most of their regions. This implies that top-performing countries have favourable framework conditions and policies that promote innovation in a relatively balanced manner across their territories. Moreover, these findings reveal a strong correlation between national and regional IOI performance, particularly among the top and bottom-performing EU countries. This correlation suggests that national-level strengths and weaknesses in innovation are often reflected at the regional level, highlighting the importance of coordinated policies and framework conditions that support innovation across all regions.

4.3 Additional analysis

Regional clusters of innovation outputs

Figure 12 revealed a concentration of highly performing regions in terms of innovation output located in Northern and Central Europe, including Finland, Sweden, Denmark, Ireland, Germany, Austria, France, and the northern part of Italy. This may underscore the existence of spill-over effects, which could be interpreted as strong regional innovation output in a particular region influencing neighbouring regions positively. Similarly, a spatial cluster of low values emerges in the Iberian Peninsula, southern Italy, Greece, Bulgaria, Romania, Hungary, Czech Republic, Poland, and the Baltic States. According to recent qualitative research results, a region’s geographical location is linked to the existence of barriers to innovation activity (European Commission, 2024f).

This subsection presents the results of a spatial autocorrelation analysis capturing the spatial dependence among regions in the RIOI and its component indicators. It aims to examine in a robust manner whether and for which regions proximity plays a role in their innovation output performance (i.e., whether there is a tendency for regions with similar scores to be located close to each other and vice versa), how “far” does spatial dependence stretch, and whether exceptions exist, namely “outlying” regions performing poorly in a broader territory of innovation output prosperity or excelling –in a relative manner– in a neighbourhood of poor performers. Spatial interaction among regions and the potential presence of regional spill-overs in competitiveness can be evaluated using Exploratory Spatial Data Analysis (ESDA) (Annoni & Kozovska, 2011). The main objective of ESDA is to identify the structure of the spatial correlation that best describes the data, focusing specifically on spatial autocorrelation and spatial heterogeneity.¹⁸ To examine whether the RIOI scores exhibit spatial autocorrelation, this sub-section adopts the global Moran’s I index (Anselin, 1995) employing a distance-based spatial weight matrix with the number of nearest neighbours set to 10.¹⁹ This

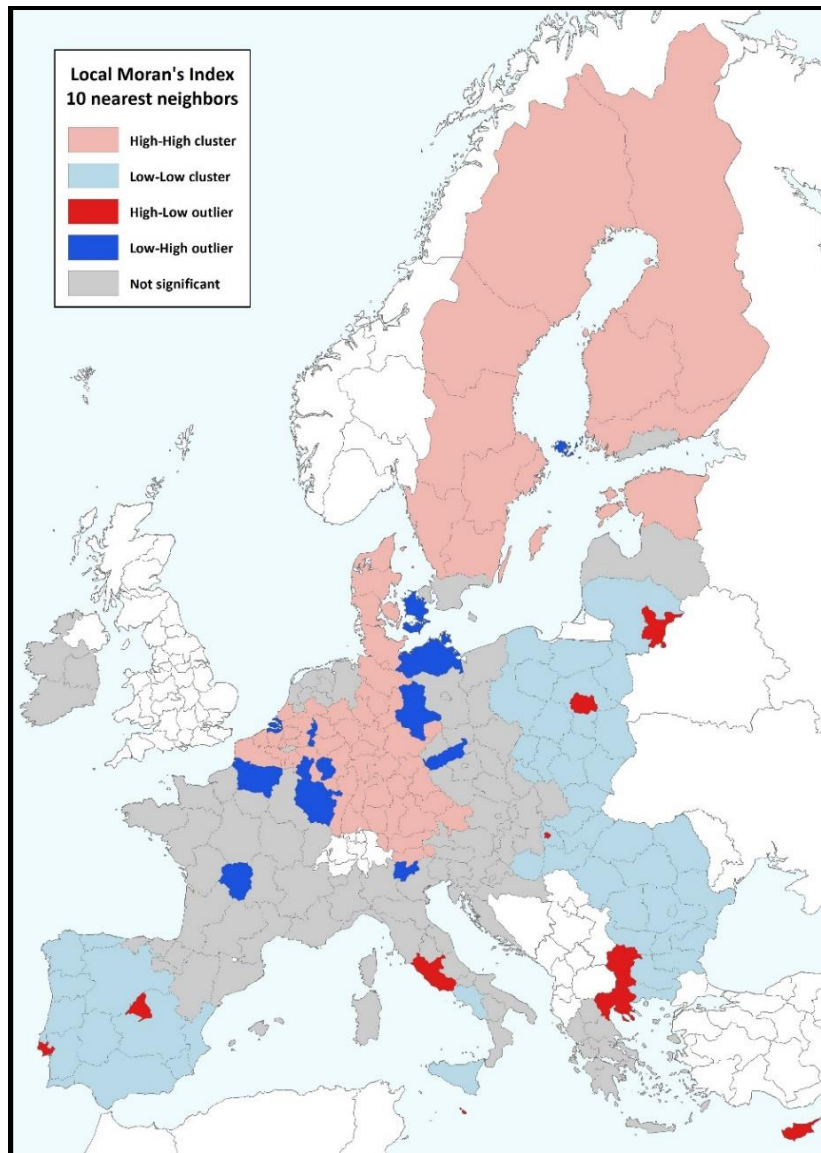
¹⁸ The interested reader is referred to Cressie (1984), Haining (2003) and Anselin et al. (2007) for more methodological details.

¹⁹ This type of matrix is preferred due to the geographical distribution of EU regions and their adjacency particularities, especially due to the presence of certain peninsulas and several islands. Note that the nearest neighbours are defined based on the geographical distance between regions irrespective of the country in which they belong. This means that the nearest neighbours for e.g., a region in the north of France may include more regions of Belgium and Netherlands than other French regions. In general, the number of neighbours used in

index takes values from -1 (maximum negative autocorrelation) to 1 (maximum positive autocorrelation) and indicates the extent to which observations are spatially grouped (positive spatial autocorrelation), dispersed (negative spatial autocorrelation), or distributed randomly across space.

The values of the global Moran's I index for the RIOI is 0.50, a figure suggesting a clear indication that innovation output is positively spatially autocorrelated across regions. This result is statistically significant at the 1% level (z-score is equal to 18.93). Positive correlation exists also at the level of the individual indicators of the RIOI, albeit its level varies (the corresponding global Moran's I values range between 0.17 and 0.66). Spatial clustering is stronger for KIA_REG, BUS_INN, and PRD_INN, and weaker -although still statistically significant- for EUTM_POP and SERV_IVA. These results confirm the claim that there is indeed spatial dependency across regions in terms of innovation output.

Figure 14. Clustering of high and low regional innovation output in the EU.



Source: Authors' calculation

empirical analysis of EU NUTS2 regions varies from 5 (Annoni & Kozovska, 2011) to 15 (Artelaris *et al.*, 2024). Our results below are robust for this range of neighbours.

Figure 14 uses the Anselin Local Moran's I (Anselin, 1995) index to cluster the EU regions into groups representing hot-spot areas of higher RIOI scores (pink-coloured regions), cold-spot areas of lower RIOI scores (those coloured in pale blue), and spatial outliers. These are either regions with low RIOI values surrounded by one or more regions with high innovation output performance (blue-coloured regions) or regions excelling in a neighbour of mediocre performance (coloured in red) and vice versa (Andrews et al., 2020). Regions in grey do not belong to a particular cluster. **Figure 14** depicts significant clusters of positive spatial autocorrelation, meaning that there are neighbouring regions in which innovation output performance is similarly high, suggesting potential gains and spillover effects from their territorial proximity. As expected, the main cluster of “high-high” regions expands in the central and northern EU regions. In total, this cluster includes regions from eleven different countries: the three Nordic countries (Denmark, Sweden, and Finland), one Baltic State (Estonia), Belgium, Netherlands, Luxembourg, and adjacent countries like France, Germany, Austria, and Italy. Most “high-high” regions originate from Germany and Belgium, while almost all Nordic regions are included in this cluster. On the other hand, there are three low-rate clusters. The most prominent one covers Eastern Europe and includes Greek, Bulgarian, Romanian, Hungarian, Czech, Slovakian, Polish, and Lithuanian regions. A second cluster emerges in the Iberian Peninsula, while the least significant one is composed of southern Italian regions.

The results regarding the outliers, namely regions that are considered overachievers in low-rated neighbourhoods or under-achievers located close to high-rated clusters, are also quite interesting. In particular, the spatial autocorrelation evaluation corroborates that certain key regions in the South and Eastern Europe appear to systematically differentiate themselves from adjacent regions that show poor innovation output performance. In most cases, these regions include country capitals or major cities, such as Lazio (IT14), Comunidad de Madrid (ES20), Área Metropolitana de Lisboa (PT17), Budapest (HU11), Sostinės regionas (LT01), and Warszawski stolenczy (PL91), all of which host their country's capital. These regions naturally accumulate human capital and entrepreneurial activity, which is however often at the expense of the remaining regions of their country. This can be the result of the relative centralisation of activities in these countries, with key institutions and development efforts concentrated in the capital or a few major cities, in combination with insufficient public sector support. The latter may be manifested in terms of lack of funding but also legislative constraints and bureaucratic processes that act as a “red tape” for potential innovative companies, especially. For example, recent results from qualitative research point to insufficient support to businesses to innovate from the public sector in Polish regions, but in this case the financial incentives for innovation would mainly be provided at national level SMEs (European Commission, 2024f). On the other hand, centralisation of activities can also concern the academic sector and the labour market, where a normal consequence is depopulation and a local brain-drain in peripheral regions. In Hungary, many young people choose to continue their education or start their careers in the capital city Budapest or abroad. This trend diminishes the local workforce's potential, especially in sectors that demand high levels of knowledge. The same is also the case for regions in Northern Italy, with people in their “productive age” leaving towards centre or Northern Italy or abroad to study and work. (European Commission, 2024g). In another example, Sostinės regionas (LT01) captures approximately three-quarters of the total FDIs in Lithuania. This is ought to several reasons, among which a very effective public body (“Invest in Lithuania”) responsible for attracting FDI. However, this is a national and not a regional agency, highlighting the relative absence of similar agencies at the regional level that would try to emphasize investments in the country's other NUTS2 region.

On the other hand, the low-performing regions that are adjacent to the high-rated cluster are mainly located close to and around its external border. This fact practically justifies the formation of the “high-high” cluster of regions with strong innovation outputs performance. Most of these “low-high” regions are part of Germany France, Belgium and the Netherlands.²⁰ Some of these experience transitions challenging their innovation performance, such as Limburg (NL42), in which declines in the mining industry and shifts in the region's major chemical industry hub affect the region's innovation landscape (European Commission, 2024g).

Finally, in grey coloured regions, innovation output performance does not appear to relate significantly to the performance of their nearest neighbours. In statistical terms, this is an intermediate situation between “high-

²⁰ The full list of “high-low” positive outlier regions includes CY00-Kýpros, LT01-Sostinės regionas, PL91-Warszawski stołeczny, PT17-Área Metropolitana de Lisboa, IT14-Lazio, HU11-Budapest, ES30-Comunidad de Madrid, MT00-Malta, BG41-Yugozapaden, and EL52-Kentriki Makedonia. The “low-high” negative outlier regions are NL42-Limburg (NL), FR22-Picardie, FR12-Limousin, DK02-Sjælland, DEB2-Trier, DEE0-Sachsen-Anhalt, ITH2-Provincia Autonoma di Trento, BE34-Prov. Luxembourg (BE), FRF3-Lorraine, NL34-Zeeland, DE80-Mecklenburg-Vorpommern, FI20-Åland, and CZ04-Severozápad.

high” clusters of similar good performance and “low –low” clusters of similarly bad performance. For countries like Ireland, Greece, and large parts of France, in which most of the nearest neighbours belong to the same country and most regions are grey coloured, this suggests the existence of regional disparities in their innovation outputs, in the sense that a cluster of similarly high performances is not found among their regions. In other words, potential gains and spill over effects from innovation activity of neighbouring regions may not be realised or not be that intense. These disparities are however not so amplified as in countries such as Poland, the Czech Republic, and Spain. These include a “low-low” cluster with a “high-low” outlier, suggesting that a single region considerably outperforms all other regions of the country (see **Figure 13**).

Index validation

This subsection conducts a validation analysis of the first, experimental version of the RIOI. As with any composite indicator, the RIOI summarizes a concept that is unobservable, in the sense that it is not directly measurable by a single metric. Given that, the precision with which the RIOI scores approximate regional performance in this target concept –innovation output performance– can be evaluated by examining the concordance of RIOI scores with two groups of variables: (i) alternative measures of regional innovation performance or a quite similar concept for the same units (the EU NUTS2 regions), and (ii) metrics of observable phenomena that are part of or closely related to regional innovation output performance. These can be variables representing the inputs to innovation activities, as well as measures of the framework conditions in EU regions that facilitate the process of innovation. A strong relationship between the RIOI and such metrics could provide further evidence that the RIOI is indeed a good approximation of regional innovation output performance.

Table 2 presents the correlations of the RIOI and its underlying components with two other summary measures of concepts related to innovation at the regional level, namely the 2023 Summary Index of the RIS (European Commission, 2023a) and the latest edition of the Regional Competitiveness Index for 2022 (RCI, Dijkstra *et al.*, 2023) (upper panel), and a collection of variables representing inputs to the innovation process and variables representing framework conditions fostering innovation (lower panel).²¹

Table 2. Table of correlations between the RIOI and its components, other regional indices, and contextual indicators.

	RIOI	RIOI pillars			
		IP	ENT	KIABI	TECH_CAP
Summary Innovation Index 2023 (RIS)	0.899	0.767	0.674	0.787	0.644
Regional Competitiveness Index 2022 (RCI)	0.802	0.681	0.452	0.807	0.567
RCI Innovation sub index	0.815	0.696	0.537	0.794	0.504
RCI Technological readiness pillar	0.496	0.442	0.150	0.693	0.279
RCI Business sophistication pillar	0.735	0.549	0.724	0.562	0.417
RCI Innovation pillar	0.844	0.787	0.485	0.730	0.599
Regional GDP per capita	0.693	0.664	0.322	0.601	0.569
GERD (% OF GDP)	0.703	0.584	0.444	0.619	0.543
BERD (% OF GDP)	0.689	0.637	0.376	0.584	0.572
GOVERD (% OF GDP)	0.434	0.251	0.289	0.342	0.458
Population (25-64) with tertiary education (%)	0.558	0.550	0.226	0.626	0.336

Source: Eurostat (rd_e_gerdreg, edat_lfse_04, nama_10r_2gdp), & authors' calculation. Note: all correlations are statistically significant.

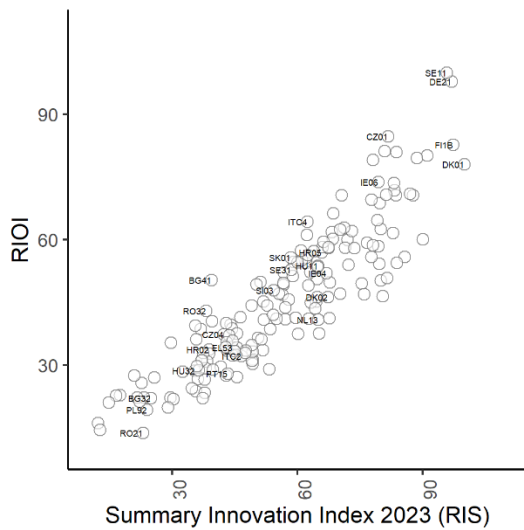
Regarding the correlations in the upper panel, the table shows that the RIOI is well-correlated with both the RIS and the RCI (correlation coefficients 0.89 and 0.80 respectively) and with the underlying components of RCI, particularly so with its Innovation sub index and component (correlation coefficients 0.81 and 0.84 respectively). The four components of the RIOI are also well-correlated with these indices. Most correlations are above 0.5, suggesting a good level of concordance between these alternative measures of innovation

²¹ The selection of the variables in the lower panel was informed from the “Input” and “Framework Conditions” Pillars of the RIS and similar Pillars in other composite indices identified through the literature review conducted for the RIOI framework.

performance. The smallest correlation is the one between the ENT component of the RIOI and the technological readiness pillar of the RCI (0.15). It is worth noting that this correlation is partly driven by indicators that are common to all three indices. For instance, KIABI is also included in RCI, whereas KIABI, PRD_INN and BUS_INN are also used in RIS. Additionally, information on trademarks and patents is incorporated into the RIS and RCI, although it is normalized using a different denominator.

Despite these similarities, these metrics provide insights into different facets of regional innovation performance. Specifically, the RIOI includes indicators related to innovation outputs, whereas both RIS and RCI examine distinct and additional dimensions of the regional innovation system. This can be seen by their differences for individual regions. **Figure 15** plots the RIOI against the RIS Summary Innovation Index (both indices are normalised to range from 0 to 100). The best and least-performing region in terms of the RIOI is indicated by name, as in **Figure 13**.

Figure 15. Score of the RIOI and RIS Summary Innovation Index



Source: Authors' calculation and RIS (European Commission, 2023a).

There are several cases of regions performing differently in terms of the two indices. Such differences can be partly explained by the fact that the RIOI focuses on innovation output while the RIS Summary Index also considers a region’s investments and variables constituting framework conditions, such as the share of population with tertiary education. For example, when considering both inputs and outputs of the innovation process, through the RIS, the top-performing regions SE11, DE21, FI1B, and DK01 perform quite similarly. However, when focusing only on innovation output performance, by means of the RIOI, Stockholm (SE11) and Oberbayern (DE21) clearly outperform the other two. In a similar fashion, BG41 and IE04 perform similarly in innovation output (the RIOI), while IE04 outperforms BG41 in the RIS Summary index. This points towards more investments and/or better framework conditions in the Irish region compared to the Bulgarian one. Similar discrepancies between the two indices also emerges in the case of other regions (e.g., comparing BG41 with HU32 or ITC4 with NL13). These show the added value of RIOI and its potential to act in a complementary fashion with other existing measures of innovation performance.

The lower panel of **Table 2** shows the correlation of the RIOI and its components with five variables, namely regional GDP per capita, gross domestic expenditure in R&D (GERD), the R&D expenditure conducted by the business sector (BERD) and the government sectors (GOVERD) as a share of regional GDP, and the share of population aged 25 to 64 having completed tertiary education. R&D expenditure as a share of GDP is widely regarded as a vital input to the innovation process (it is also part of the RIS “Investments” Pillar), while higher GDP per capita and a larger share of higher education graduates are viewed as facilitators of innovation.

Most of the correlations between the RIOI and these variables are very strong (above 0.5) and the same holds for the RIOI components. Particularly, the RIOI, and its IP, KIABI, and TECH_CAP components are highly correlated with GDP per capita, and GERD and BERD as a share of GDP. ENT, which captures product and process innovations, is the component which results to be less correlated with the five contextual variables. This can be partly explained, at least for the R&D expenditure, by the fact that the underlying indicators of the

ENT component capture the innovations of SMEs, which in general conduct less business R&D compared to large firms.

The results in this sub-section and in particular the correlations in **Table 2** show that this first, experimental version of the RIOI is in accordance with regional performance in other summary measures measuring similar concepts to the one captured by the RIOI, while it is also well-correlated with variables known to facilitate stronger innovation outputs. This suggests that the RIOI framework is a good approximation of regional performance in terms of innovation outputs.

5 Conclusions

This report presents the 2024 edition of the Innovation Output Indicator (IOI), an output-oriented measure of innovation performance at the country and EU levels which assesses countries' capacity to derive economic benefits from innovation. It tracks the extent to which innovative ideas reach the market, create knowledge-intensive jobs, and increase country technological capability.

Chapter 3 presents the latest figures for the composite index and its underlying indicators for 46 countries, including the 27 EU Member States, EU candidate states, EFTA and selected BRICS, OECD and emerging economies. According to the IOI, Sweden, Ireland, and Germany are the best-performing EU Member States in 2023 and are followed by Finland and Luxembourg. Sweden outperforms the other EU countries in terms of IP applications, Ireland does so in terms of employment in knowledge intensive industries, while Germany outperforms the other EU Member States in terms of its domestic value added content of its knowledge-intensive manufacturing exports. Conversely, Romania, Latvia and Bulgaria recorded the lowest performance among EU member states. The EU27 leads Canada, Australia, and China, but it continues to lag behind its main economic partners, namely the United States, South Korea and Japan. These results are mainly due to its relatively weaker performance compared to international partners in terms of the share of knowledge-intensive employment in business industries, patent applications, and the share of domestic value added in knowledge-intensive services firms. The EU27's performance has improved over time, albeit at a slower pace than its main economic partners. This suggests that the United States, South Korea, and Japan are increasing their lead over the EU27, whereas China is rapidly narrowing the gap.

Chapter 4 of this report extends, for the first time, the IOI to the regional level of the 27 EU Member States. The newly developed Regional Innovation Output Indicator (RIOI) builds on the national IOI framework and uses established and novel datasets to assess the performance of 241 EU NUTS-2 regions in terms of innovation output. Its goal is to provide a snapshot of regional activity in terms of technological and non-technological innovation, regional contribution to value added in trade, employment, and innovative enterprises. According to the RIOI scores, the best performing regions form a cluster stretching from Eastern France, the Netherlands, Belgium, and German regions, to Denmark, Sweden and Finland. By contrast, clusters of lower-performing regions form in the Mediterranean countries and Eastern Europe, with noteworthy exceptions of certain important regions excelling in neighbourhoods of weak performance. The RIOI results are complemented by a spatial autocorrelation analysis that confirms the existence of regional disparities in terms of innovation performance in the EU and the formation of clusters of high-performing Central and Northern European regions on the one hand, and below-average performing regions in Eastern Europe and the Mediterranean on the other hand.

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List of abbreviations and definitions

Code	Country
AT	Austria
AU	Australia
BE	Belgium
BG	Bulgaria
BR	Brazil
CA	Canada
CH	Switzerland
CN	China
CL	Chile
CY	Cyprus
CZ	Czech Republic
DE	Germany
DK	Denmark
EE	Estonia
EL	Greece
ES	Spain
EU27	European Union
FI	Finland
FR	France
HR	Croatia
HU	Hungary
IE	Ireland
IL	Israel
IS	Iceland
IT	Italy
JP	Japan
KR	South Korea
LT	Lithuania
LU	Luxembourg
LV	Latvia
ME	Montenegro
MK	North Macedonia
MT	Malta
MX	Mexico
NL	Netherlands
NO	Norway
NZ	New Zealand
PL	Poland
PT	Portugal
RO	Romania
RS	Serbia
SE	Sweden
SI	Slovenia
SK	Slovakia
TR	Türkiye
UK	United Kingdom
US	United States

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Annexes

Annex 1. The IOI methodology

The IOI 2024 dataset and index development

The IOI 2024 dataset includes 506²² country-year observations covering the 27 European Union Member States, EFTA countries (Switzerland, Iceland, Norway), EU candidates (Montenegro, North Macedonia, Serbia, Türkiye), selected OECD countries (Australia, Canada, Chile, Israel, Japan, South Korea, Mexico, New Zealand, United Kingdom, United States of America) and two BRICS members (Brazil and China) over the time period 2013 until 2023. It also includes an observation for the 27 EU MS as a single economic zone (EU27). The dataset includes six indicators: PCT_POP, TRA_POP, KIABI, GOOD_VA, SERV_VA and ENT.

Comparison with earlier IOI reports

It must be stressed that comparisons with results from previous IOI reports are not possible. Although the methodology in the current report is the same as in the previous two IOI reports, results for the same year, e.g. 2021 in the IOI 2023 report (Bello et al., 2024) and 2021 in this year's report, are different due to: (i) updated data being published; (ii) by adding new data at the end of the time series for each indicator and removing data at the start of the time series, the highest and lowest scores used for calculating normalised scores across all countries and all years for an indicator can change, directly impacting these normalised scores, and (iii) the rolling 11-year period affects the imputation of missing data, with some countries that in earlier reports were reporting data for an indicator not doing so in later reports, and hence, imputations are used instead.

Data availability: The majority of missing data is imputed using the last known value. That is, in case data was missing on one country-year, the last known value was imputed from a previous year. If data was missing between two non-missing data points, the average of those adjacent years was used. In case no last known values are available, the imputation was done using the MICE package in R (Van Buuren and Groothuis-Oudshoorn, 2011; Van Buuren, 2018), which implements multivariate imputation by chained equations. For the IOI 2024, the JRC imputed the KIABI indicator for China, the ENT indicator for Canada, Israel, Mexico, and Montenegro, the trademark indicator for Greece, and Ireland, and the GOOD_VA and SERV_VA indicators for Montenegro, North Macedonia, and Serbia using MICE. All indicator data shown in **Section 3.2** and descriptive statistics and correlations in this Annex exclude the MICE imputed data. Please refer to **Table 9** in annex 3 for statistics of the data including the MICE imputations. **Figure 24** up to **Figure 29** in Annex 3 show the available data for each indicator by country and year.

Outlier treatment: No outliers were detected using the criteria set out in JRC/OECD (2008), according to which (absolute) skewness and (absolute) kurtosis are considered excessive when they simultaneously cross the threshold-values of 2 and 3.5, respectively (Groeneveld and Meeden, 1984). Descriptive statistics for the non-normalized IOI 2024 dataset are shown in **Table 3**.

Table 3. Descriptive Statistics of IOI indicators for the 2023 data [before treatment and normalisation]

Indicator	Min	Max	Mean	Median	Std	Skew	Kurt	N. non-missing	% non-missing
GOOD_VA	2.33	67.8	30.2	29.4	15.8	0.35	-0.51	43	93%
SERV_VA	7.57	63.3	32.1	30.4	11	0.82	1.52	43	93%
KIABI	7.1	27.7	15.7	15.4	4.45	0.36	-0.01	45	98%
PCT_POP	0	405	95.1	48.7	111	1.34	0.88	46	100%
TRA_POP	493	6630	2510	2080	1530	1.11	0.77	44	96%
ENT	10.7	77.4	51.1	54.7	14.8	-0.44	-0.21	42	91%

Standardization: After imputation, the data was standardized over the last eleven years of all countries (country-year observations). The min-max standardization is applied across all years and ensures the maximum score of 10 corresponds to the maximum observed value over the 11 years while the minimum

²² That is, 46 countries for 11 years (2013 to 2023).

score, zero, corresponds to the lowest observed value over the 11 years. Descriptive statistics for the standardised indicators are provided in **Table 4**.

Weighting: All indicators and pillars of the IOI are equally weighted within their aggregate. Unequal weights were used in previous editions of the IOI so that the IOI was statistically balanced in its underlying components. After the methodology update implemented in the 2023 edition of the index, this was not necessary.

Aggregation: All indicators were added up using the arithmetic mean to form pillars. All pillars of the IOI were added up using the arithmetic mean using equal weights.

Table 4. Normalized descriptive statistics of the IOI 2024 indicators.

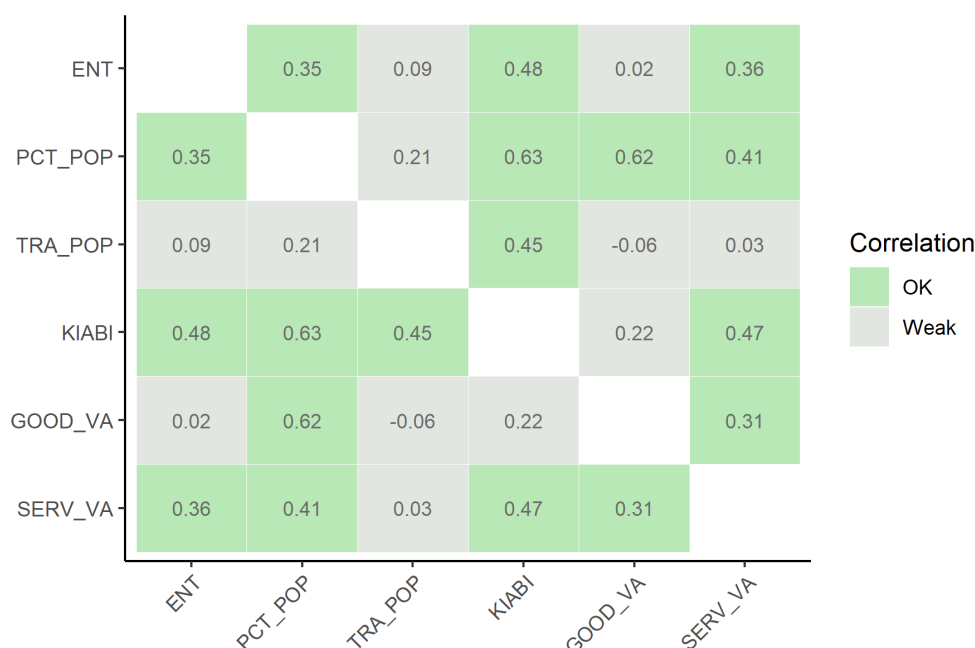
Indicator	Min	Max	Mean	Median	Std	Skew	Kurt	N. non-missing
GOOD_VA	0.06	9.83	4.23	4.11	2.36	0.35	-0.51	43
SERV_VA	0.03	10	4.41	4.12	1.97	0.82	1.52	43
KIABI	0.8	10	4.67	4.55	1.99	0.32	-0.02	44
PCT_POP	0	10	2.4	1.22	2.75	1.31	0.8	45
TRA_POP	0.09	4.93	1.71	1.36	1.19	1.12	0.78	43
ENT	0.07	9.69	5.9	6.42	2.14	-0.44	-0.21	42

Source: Authors' calculation

Correlations between normalised indicators and the structural coherence of the IOI

The JRC/OECD guidelines on composite indicators (JRC/OECD, 2008) recommends correlations between 0.30 and 0.92 for any two indicators that share the same pillar. An adequate correlation ensures a minimum level of information (in this case at least 9%) transmitted from indicators to aggregates. **Figure 16** shows that the indicators of the intellectual property component (TRA_POP and PCT_POP) do not meet these criteria and have a slightly weaker correlation. Nevertheless, both indicators are included in the IOI to cover innovations in both goods and services. The correlation between the indicators of the Technology Capacity component (GOOD_VA and SERV_VA) are above the 0.3 threshold.

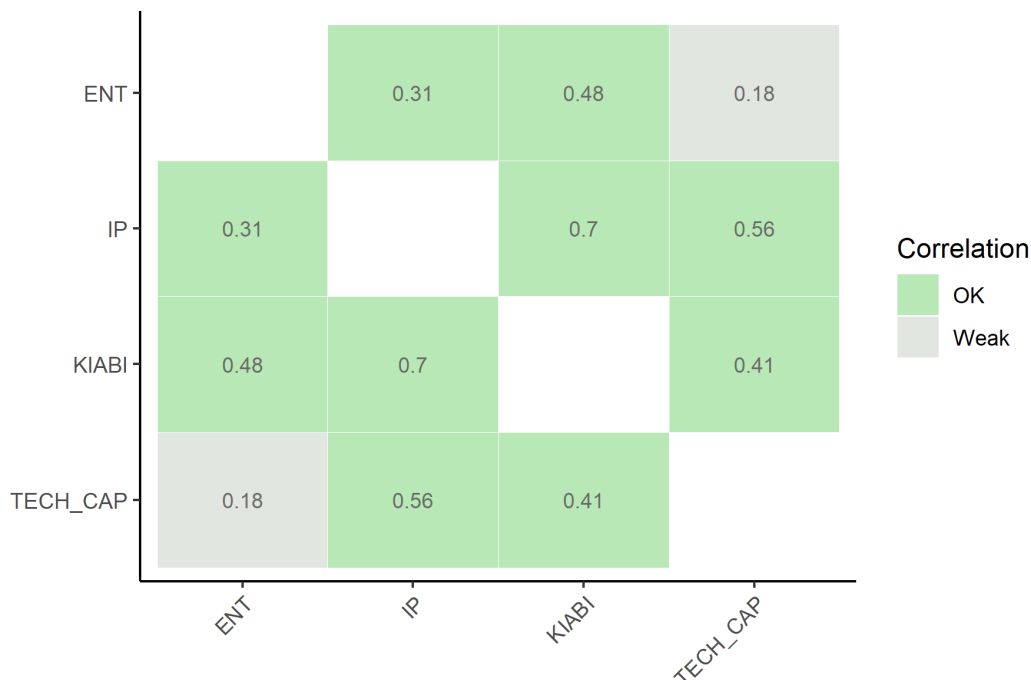
Figure 16. Pairwise Pearson correlations between indicators



Source: Authors' calculation

Figure 17 shows correlations between the pillars of the IOI. Overall, the correlation structure is adequate. Only the correlation between TECH_CAP and ENT is below the threshold of 0.3, while that between IP and ENT is slightly above it, suggesting a lower rate of information transmission from ENT to the IOI compared to other indicators.

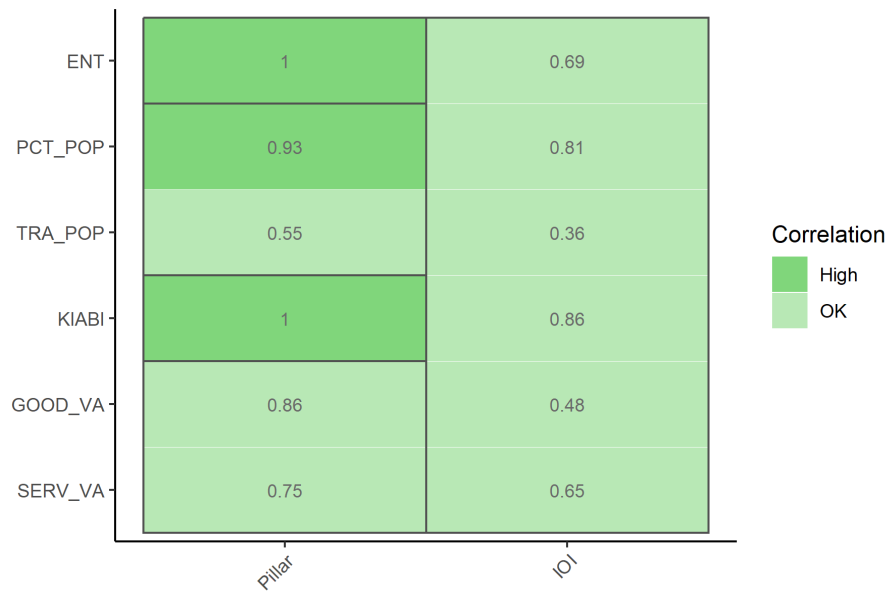
Figure 17. Pairwise correlations between Pillars



Source: Authors' calculation

Figure 18 depicts the pairwise correlations between indicators with their respective pillars and the IOI. From the left column, we see that the PCT_POP and TRA_POP indicators are both well correlated with the IP pillar, and the same holds for the GOOD_VA and SERV_VA indicators and the TECH_CAP pillar. The right column shows indicator correlations with the IOI. The strength of the correlation between indicator and aggregates reflects more directly the amount of information an indicator transmits to the index. All six correlations of individual indicators with the IOI fall within the recommended guideline, providing evidence that the IOI effectively summarises the information from its indicators and pillars. However, the individual indicators' impact to the overall IOI scores appears to vary. The KIABI and PCT_POP indicators have the strongest correlation with the IOI while TRA_POP has the lowest, followed by GOOD_VA, meaning that PCT_POP has a greater impact than TRA_POP on the IOI scores. On the other hand, the correlations of GOOD_VA and SERV_VA with the IOI differ less from one another, suggesting that the two components of the TECH_CAP pillar have a more balanced contribution to the IOI scores.

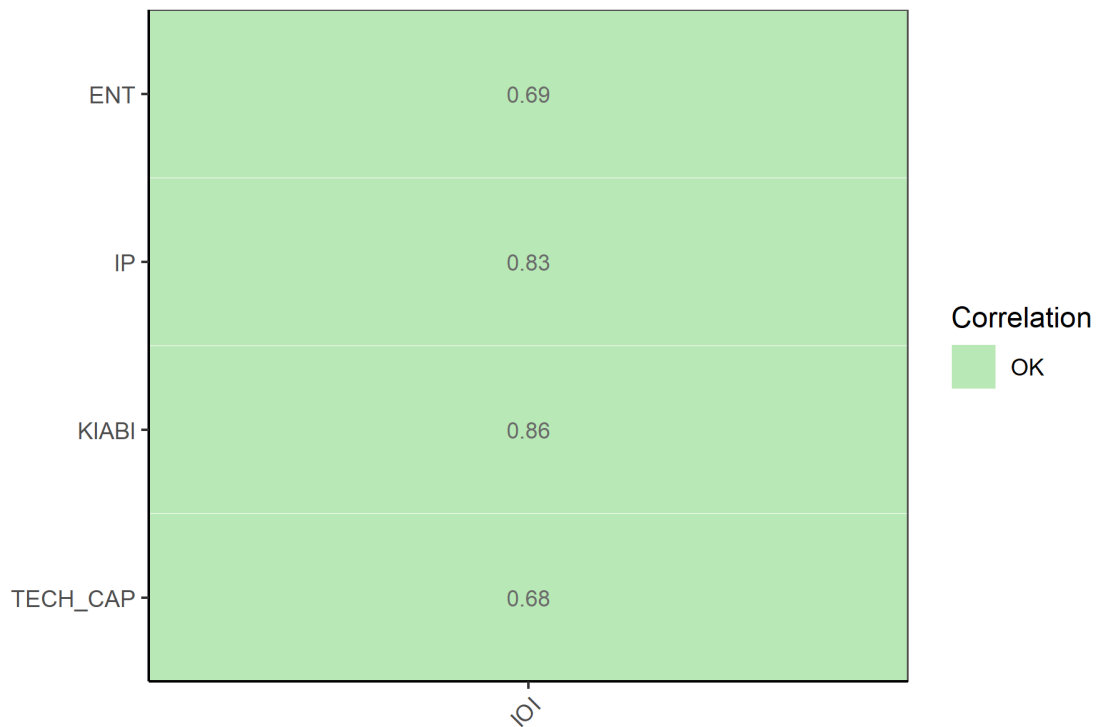
Figure 18. Pairwise correlations between indicators and their respective pillars and with the IOI



Source: Authors' calculation

Figure 19 depicts the correlations between pillars and the IOI. At this level, which is the most important level of aggregation, we see that all pillars have an adequate correlation with the IOI. The closeness between the correlations (ranging from 0.68 to 0.86) indicates that each pillar contributes information to the IOI to a rather similar degree and there is no domination of the IOI by a specific Pillar. The TECH_CAP and ENT pillars have a slightly lower correlation, as a result of their low correlation with the other pillars. Nevertheless, all correlations are adequate and similar, providing evidence that the IOI is a good summary measure of the information contained in the six considered indicators.

Figure 19. Pairwise correlations between pillars and the IOI.



Source: Authors' calculation

Annex 2. Robustness of IOI ranks and validation of results

Uncertainty analysis

Methodological choices are part of constructing a composite indicator. Developers make these choices using expert opinion, statistical theory or other practical considerations. However, there often exist equally valid alternatives to the choices being made for a specific indicator. In this sense, methodological choices introduce some uncertainty, as users may ask whether rankings are robust to these alternative specifications. The ranking of any composite indicator should not be heavily dependent (i.e., there should not be much uncertainty) to a particular methodology chosen during its development. This section investigates to what extent modelling choices influences the ranking of countries in the IOI.

The uncertainty analysis systematically changes four different modelling assumptions: the normalisation, imputation, the weights and aggregation. **Table 5** lists the model specifications used for the nominal IOI and the alternatives considered in the uncertainty analysis. Mean imputation is the alternative method to test for uncertainty in the treatment of missing data. Mean imputation is a frequently-used imputation method in composite indicators. It substitutes missing data for an indicator with the mean of the country's remaining indicators in the corresponding aggregation level. The standardization of indicators with a mean of five and a standard deviation of 1.5 is the alternative normalisation method. Previous versions of the IOI used this standardization formula. The geometric mean is the alternative aggregation method considered. The Compared to the arithmetic mean that implies full substitutability across the aggregated dimensions, geometric mean rewards more the countries with a balanced performance across indicators compared to those with unbalanced profiles. Lastly, the uncertainty analysis varies the aggregation weights randomly around +/-20% from their nominal values.

Table 5 Uncertainty parameters: missing values, normalisation, aggregation and weights

Source of Uncertainty	Original	Alternative
I. Treatment of missing values	EM algorithm	Mean imputation
II. Normalisation of indicators	Min-max across last 10 years	Z-score (mean = 5, SD = 1.5) across last 10 years
III. Aggregation formula at pillar level	Arithmetic mean	Geometric mean
IV. Pillar weights	Equal weights	+/- 20%

Source: Authors

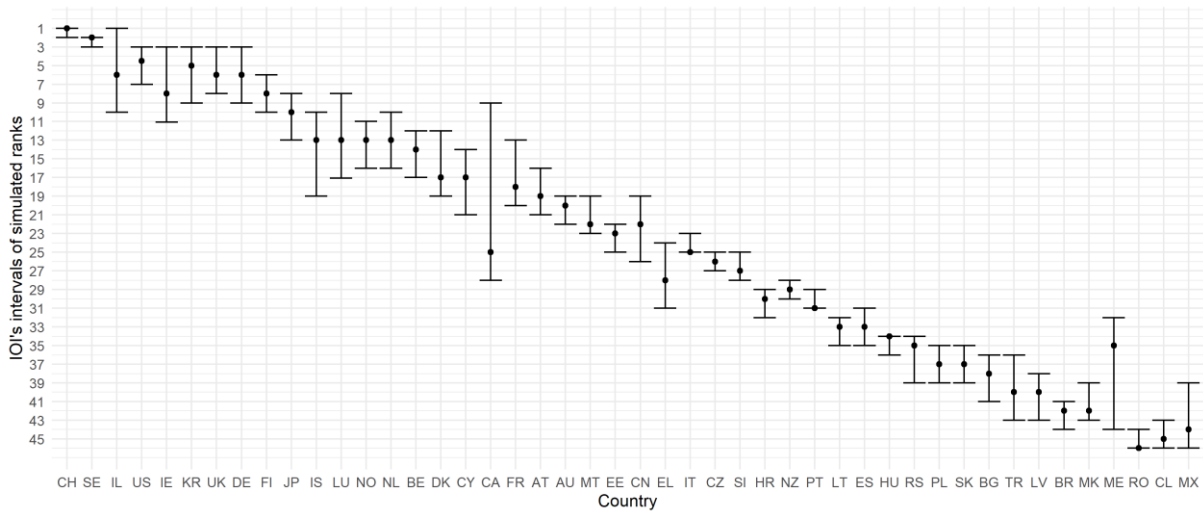
All the combinations (original and alternative) of the normalisation, imputation and aggregation steps result in eight alternative versions of the IOI. The analysis consists of computing each of these versions 50 times using a randomly selected set of weights from a distribution of weights with random perturbation of up to 20% of their original value. This results in 4000 versions of the IOI. The statistics of interest - plotted in **Figure 20** and shown in **Table 6**- are the median rank, the 5th and 95th percentile rank for each country across the 4000 versions.

As **Figure 20** shows, the ranks of the IOI are robust against changes in methodology. The dots in **Figure 20** are the median rank across the simulations. These are ordered closely to the original ranking of the IOI as indicated by the country order on the x-axis. In particular, 91.3% (42/46) of countries' median rank is less than three ranks away from the nominal ranking. These are cases like South Korea, United Kingdom, Italy, Croatia, and Poland, which do switch ranks, but only by a few places. There are three countries for which the median rank is three positions away from the nominal rank (Israel, Ireland, and Greece). Lastly, Montenegro and Canada are the only country where the median IOI rank deviates more than three positions from the nominal rank (7 and 8 rank positions respectively). The likely cause for the differences in these five countries is the existence of missing data for these countries, and consequently, to the imputation process, in which the default modelling choice (the EM imputation) takes into account the country's performance in a set of additional indicators highly correlated with the indicator to be imputed. For Montenegro, this results in lower imputed values than the alternative imputation function (the mean imputation), which assigns to each the missing value the arithmetic mean of the non-missing values for that indicator, while for other countries such as Canada (which does not have data for ENT) the reverse occurs.

The confidence interval (CI) provides another indication of the uncertainty in the ranking due to modelling choices. The confidence interval reflects whether modelling alternatives result in a wide variety of rankings. In the uncertainty analysis, confidence intervals are computed as the 5th and 95th percentile of the observed distribution of ranks across simulations.

The CI indicates highly stable rankings for 15 out of 46 countries, which have a CI range of zero to three. An additional 25 countries have a CI range between four and seven, indicating some uncertainty, but still allowing accurate inference. Lastly, there are only six countries (Canada, Iceland, Ireland, Israel, Luxembourg, and Montenegro) that have a CI of greater than seven positions, which indicate a larger uncertainty in their ranking and their ranks should be interpreted with some caution. **Table 6** shows the median ranks, confidence intervals, and deviations from the nominal rank.

Figure 20. Median rank in the Innovation Output Indicator across simulations with alternative methods.



Source: Authors' calculation.

Table 6. Table of the Median rank, Q5 and Q95 across simulations and nominal rank.

Nominal IOI Rank	Country	Median Rank	Q5	Q95	CI Range	Deviation from Nominal
1	CH	1	1	2	1	0
2	SE	2	2	3	1	0
3	IL	6	1	10	9	-3
4	US	4.5	3	7	4	-0.5
5	IE	8	3	11.05	8.05	-3
6	KR	5	3	9	6	1
7	UK	6	3	8	5	1
8	DE	6	3	9	6	2
9	FI	8	6	10	4	1
10	JP	10	8	13	5	0
11	IS	13	10	19	9	-2
12	LU	13	8	17.05	9.05	-1
13	NO	13	11	16	5	0
14	NL	13	10	16	6	1
15	BE	14	12	17	5	1
16	DK	17	12	19	7	-1
17	CY	17	14	21	7	0
18	CA	25	9	28	19	-7
19	FR	18	13	20	7	1
20	AT	19	16	21	5	1
21	AU	20	19	22	3	1
22	MT	22	19	23	4	0
23	EE	23	22	25	3	0
24	CN	22	19	26	7	2
25	EL	28	24	31	7	-3
26	IT	25	23	25	2	1
27	CZ	26	25	27	2	1
28	SI	27	25	28	3	1
29	HR	30	29	32	3	-1
30	NZ	29	28	30	2	1
31	PT	31	29	31	2	0
32	LT	33	32	35	3	-1
33	ES	33	31	35	4	0
34	HU	34	34	36	2	0
35	RS	35	34	39	5	0
36	PL	37	35	39	4	-1
37	SK	37	35	39	4	0
38	BG	38	36	41	5	0
39	TR	40	36	43	7	-1
40	LV	40	38	43	5	0
41	BR	42	41	44	3	-1
42	MK	42	39	43	4	0
43	ME	35	32	44	12	8
44	RO	46	44	46	2	-2
45	CL	45	43	46	3	0
46	MX	44	39	46	7	2

Source: Authors' calculation.

Sensitivity Analysis

Sensitivity analysis builds on the results from an uncertainty analysis to investigate which modelling assumptions drive variation in the rankings of countries. **Table 7** shows descriptive statistics of the sensitivity analysis. It shows the (absolute) deviations between the median rank and the nominal IOI rank for each of the eight model specifications across 50 runs with random perturbations of the weights, averaged across the 46 countries.

As seen from **Table 7**, the random perturbation of the weight alone has little impact on rankings (the median rank in 89% of the countries -41 out of 46- is the same as the nominal rank). The median ranking of the remaining 5 countries is within one position of their nominal ranks. Changing one of the aggregation, imputation or normalisation functions in combination with the random weights has a small impact on the ranking. In all the three cases (see columns 2 to 4 of **Table 7**) the average change in ranks is less than two positions, and for more than 70% of the countries the median rank is up to 2 positions above or below their nominal rank. When changing more than one functions (see columns 5 to 7 of **Table 7**) the average difference between nominal and median ranks increases only slightly, while only twelve countries change three or more ranks in comparison to the nominal ranks when using all three alternative functions (normalisation, imputation, aggregation) and random perturbations of the weights. In this case, the average of the absolute differences between nominal and median ranks is only 2 positions.

Table 7. Average deviation between IOI nominal rank and median rank of eight model specifications.

Only Weights	Alternative Specification						
	Weights +						
	Agg	Imp	Norm	Agg+Imp	Agg+norm	Norm+Imp	All
0.130	1.891	1.696	0.978	2.391	1.283	1.630	2.043

Source: Authors' calculation. Note: Agg refers to alternative aggregation formula, namely the geometric mean. Imp refers to the alternative imputation formula, i.e. mean imputation. Norm refers to the alternative normalisation procedure, namely standardization (Mean = 5, S.D. = 1.5).

Index validation analysis

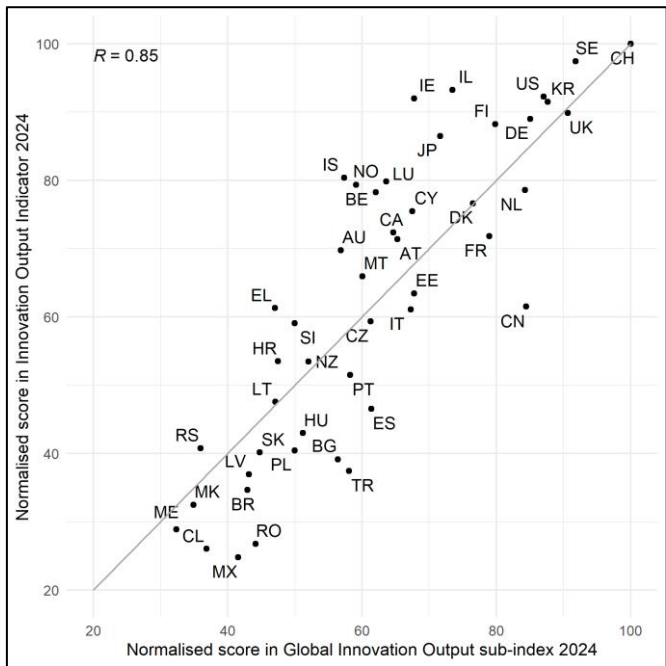
A composite indicator summarizes a multifaceted phenomenon that is not directly measurable by a single indicator or variable. As such, constructing composite indicators requires the use of metrics of observable phenomena that are part of or closely related to the composite indicator's target concept. Given that the target concept is unobservable, it is useful to validate the resulting measurement by comparing the composite indicator to other measures of the same or similar concepts. A strong relationship between independently created measures of highly similar concepts provides evidence that they do indeed measure the same underlying phenomenon in a precise way.

In this section, the latest edition of the IOI 2024 is plotted against the Global Innovation Index (GII), developed by WIPO and based on 80 indicators (WIPO, 2024), as well as the Summary Innovation Index (SII) in the European innovation scoreboard developed by the European Commission (European Commission, 2024b). The GII includes an Input and an Output sub-pillar. This section plots the IOI against both the GII and its Output sub-pillar. Each plot shows the subsample of countries that appear in both indexes and normalised versions of the indices by dividing the scores of each index with the maximum score obtained in each index. In the top left of each graph, readers can find the Pearson correlation coefficient between the two indices.

Figure 21 is a scatterplot of the IOI 2024 score and the Global Innovation Output sub-index 2024 score. Overall, there is a strong correlation between the two measures of innovation output, indicating that both measure a similar phenomenon. There are however some differences, indicated by an absolute difference in country ranks by the two indices equal to 5.22 positions. This can be attributed to the wider scope of the GII output sub-index, which monitors not only services, knowledge and technology outputs like the IOI but also creative outputs such as creative goods and services and online creativity. For instance, Israel scores highly and is ranked 7th in the "knowledge and technology outputs" dimension of the GII output sub-index, which is close to its IOI rank (3rd). However it is ranked 28th in the "Creative outputs" dimension of the GII Output sub-index. Therefore, the overall rank of Israel is much lower in the GII Output sub-index (12th) compared to its IOI rank. On the other hand, Türkiye, is ranked 39th in the IOI, which is very close to its rank in the "knowledge and technology outputs" dimension of the GII output sub-index (37th). However it is ranked 15th in the "Creative

outputs” dimension of the GII Output sub-Index, resulting in an overall rank in the GII Output sub-index (27th) larger than its IOI rank.

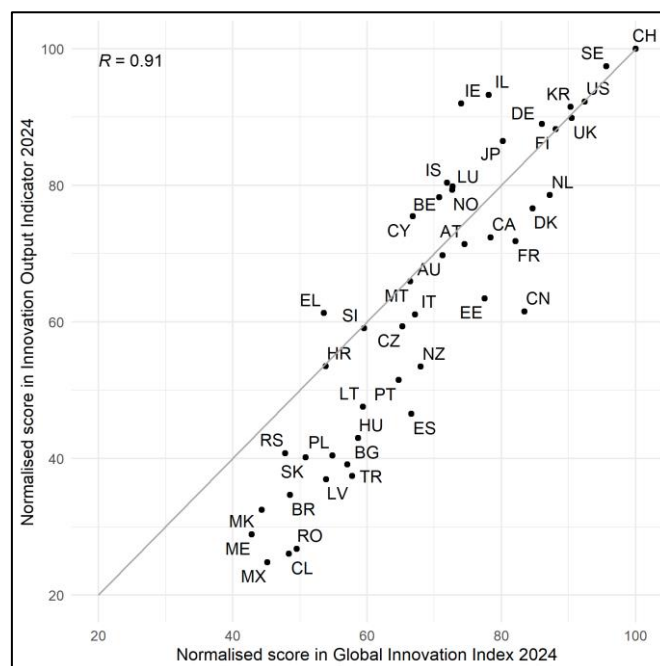
Figure 21. Score of the IOI and Global Innovation Output sub-index.



Source: Authors’ calculation and WIPO (2024).

Figure 22 shows the scatterplot of the IOI 2024 and the Global Innovation Index 2024 scores. The correlation coefficient shows a strong relationship between the two indexes, which is slightly stronger than that of the IOI with the GII output sub-index (Pearson correlation coefficient 0.91 vs. 0.85). A likely cause is the relatively strong relation between indicators considered in the GII inputs and output indicators of the IOI. For example, GII input indicators focusing on high-tech goods and services and gross expenditure on R&D by firms are strongly related to the value added and the share of innovative enterprise indicators used in the IOI. This weighs down the importance of the creative outputs on the Output sub-index and aligns the overall GII further with the IOI. Nevertheless, rank differences between the two indices are non-negligible and are on average equal to 4.6 positions. This is reflected, for example, to the ranks of Ireland (5th in the IOI but 17th in terms of the GII) and Spain (33rd in the IOI but 26th in terms of the GII).

Figure 22. Score of the IOI and Global Innovation Index.

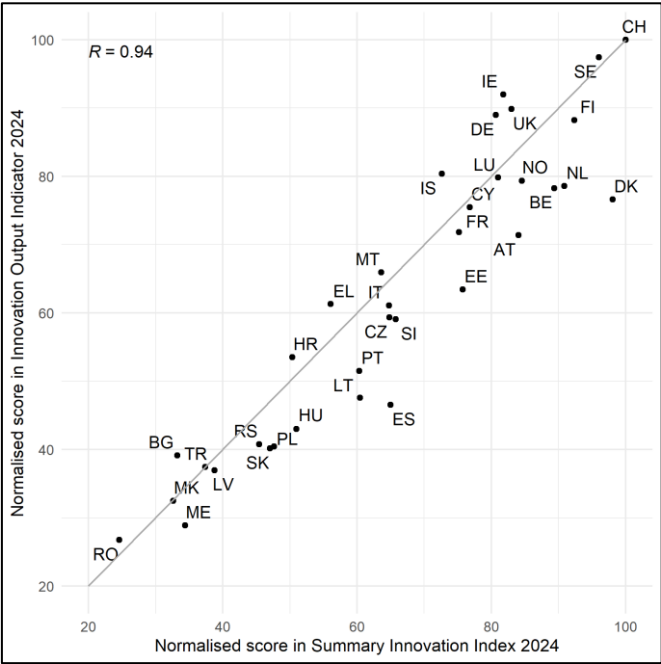


Source: Authors' calculation and WIPO (2024).

Figure 23 plots the 2024 IOI and SII scores. Similarly to the previous graphs, there is a high correlation indicating that the underlying phenomenon measured overlaps much in the two indices. Still, rank deviations among the two indices are non-negligible and are on average 3.14 positions. There are 13 countries (about 37% of the 35 considered) for which the IOI rank deviates more than 3 positions from the SII rank. This includes, for example, Austria that performs much better in terms of the SII compared to the IOI (8th vs 15th), most likely driven by its good performance on Investments where it ranks 7th. On the other hand, Germany ranks lower in SII than in IOI (12th vs 5th) most likely due to its relatively medium performance in the framework conditions and pillar of the SII (20th). Finally, it is also informative to consider for comparison the SII types of activities, i.e. the investment, framework conditions, innovation activities and impacts pillars of the SII. The latter includes three dimensions, namely employment impacts, sales impacts and environmental sustainability. As expected, the IOI correlates stronger with the impacts pillar, followed by innovation activities (**Table 8**). This is mainly driven by the high correlation with the IOI and employment impacts sub-pillar, which includes the KIABI component in addition to a second component, which is close to the ENT indicator and corresponds to the share of innovative enterprises, measured in terms of employment. As for the innovation activities pillar of the SII, this includes a sub-pillar named “intellectual assets”, which includes PCT and trademark applications, also included in the IP component of the IOI. Notice also that despite the high correlations, the deviation in ranks between the IOI and each of the components is above 3 positions in all cases but one (that of the employment impacts sub-pillar).

The results in this Section could give rise to two remarks: *First*, the IOI is sufficiently well correlated with other frameworks monitoring the concept of innovation performance, including the innovation outputs of countries. This validates the conceptual coherence of the IOI in that it verifies that it indeed provides performance estimates of the concept it is designed to, using different indicators and measurement framework. *Second*, the rank differences between the IOI and the GII, SII and their components are sufficient to highlight the added value of the IOI in the landscape of innovation monitoring tools. That is, the IOI highlights aspects of country innovation output performance which are not able to view by looking at other innovation frameworks and indices.

Figure 23. Score of the IOI and Summary Innovation Index.



Source: Authors’ calculation and European Commission (2024).

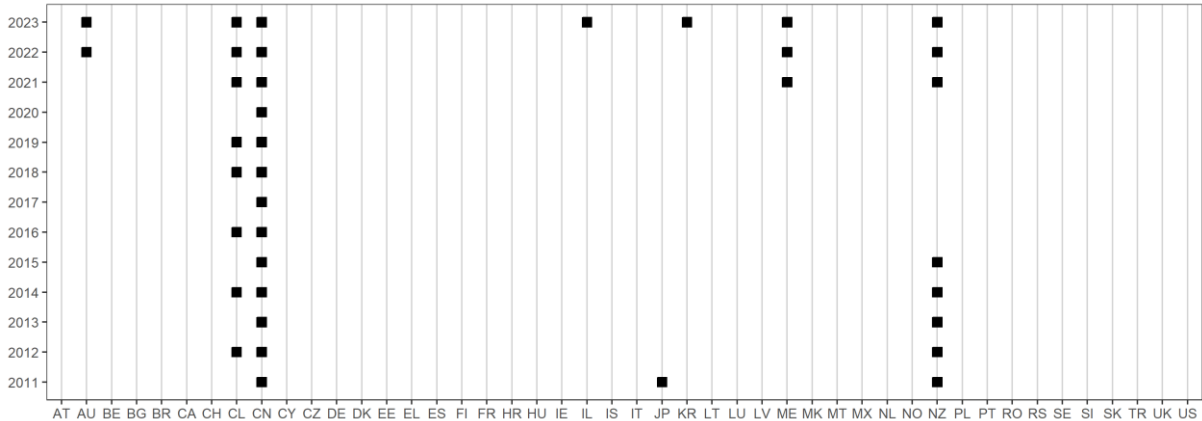
Table 8. Table of correlations and average rank deviations between EIS components and the IOI

	Investments	Framework Conditions	Innovation Activities	Impacts	Impacts sub-pillars		
					Employment Impacts	Sales Impacts	Environmental Sustainability
Correlation	0.85	0.84	0.87	0.92	0.91	0.67	0.66
Average rank deviation	4.91	4.34	4.29	3.43	2.86	6.57	7.11

Source: Authors’ calculation. Note: all correlations are significant at $p<0.001$

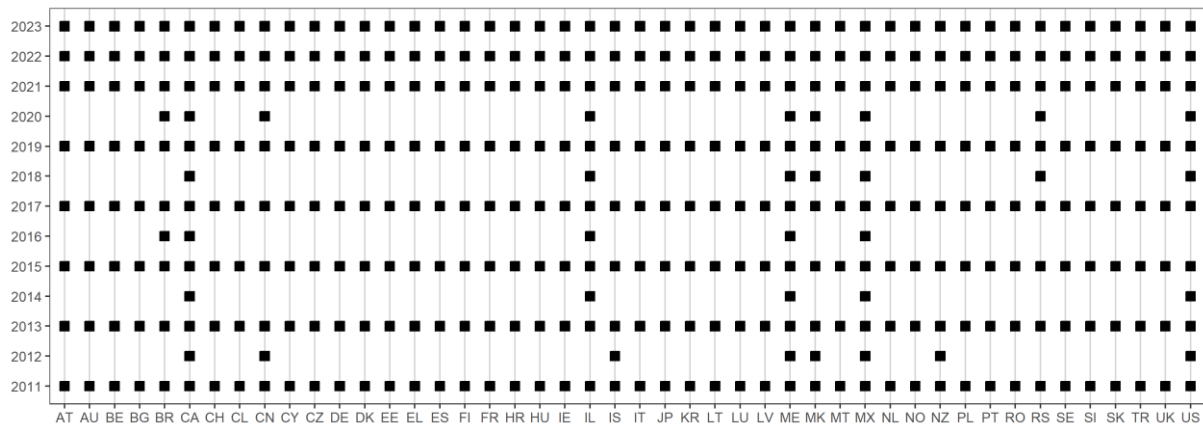
Annex 3. Missing data patterns per IOI indicator

Figure 24. Missing data years per country on KIABI



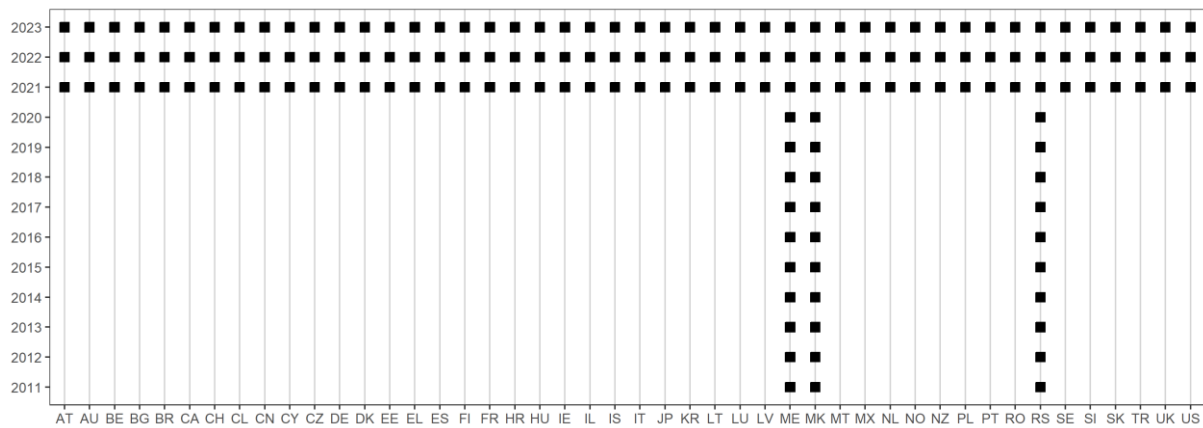
Source: Authors’ calculation. Note: Squares indicate missing data for the country-year

Figure 25. Missing data years per country on ENT



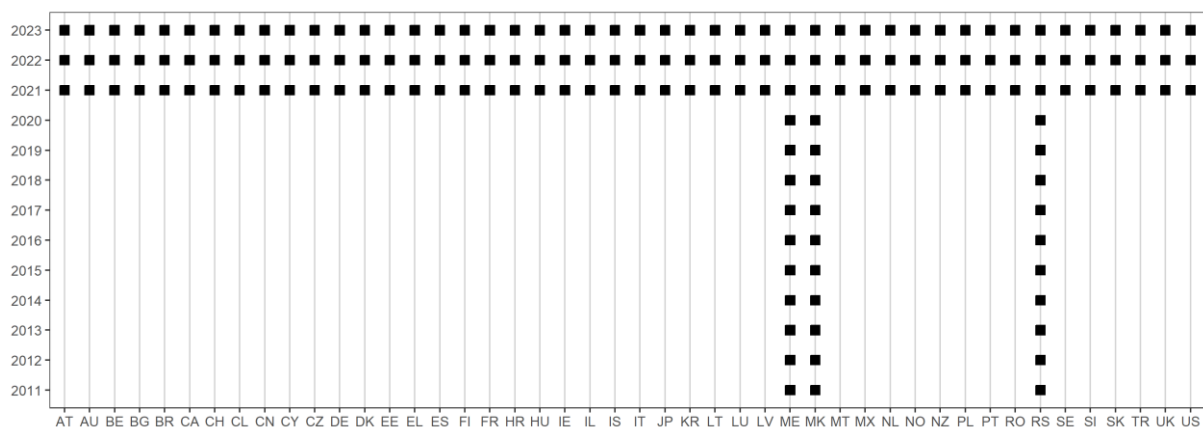
Source: Authors' calculation. Note: Squares indicate missing data for the country-year

Figure 26. Missing data years per country on SERV_VA



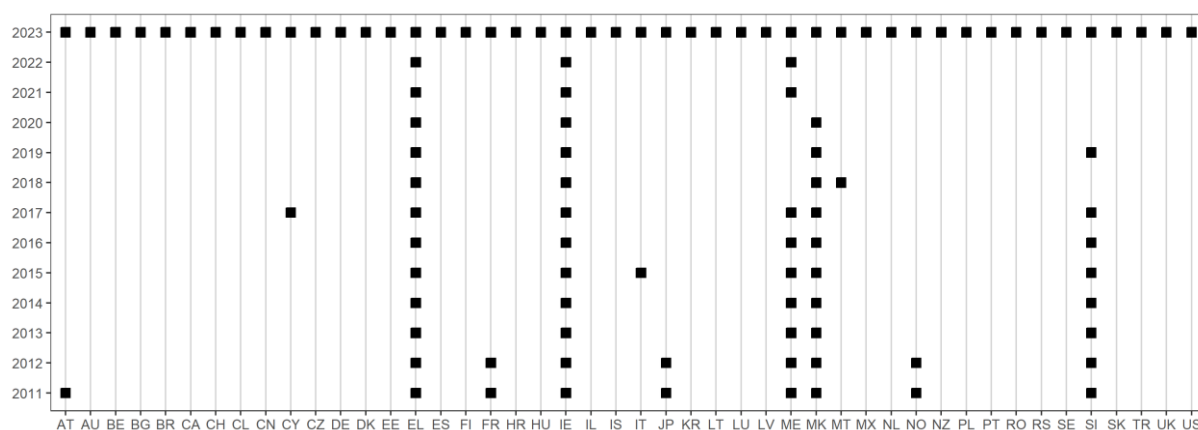
Source: Authors' calculation. Note: Squares indicate missing data for the country-year

Figure 27. Missing data years per country on GOOD_VA



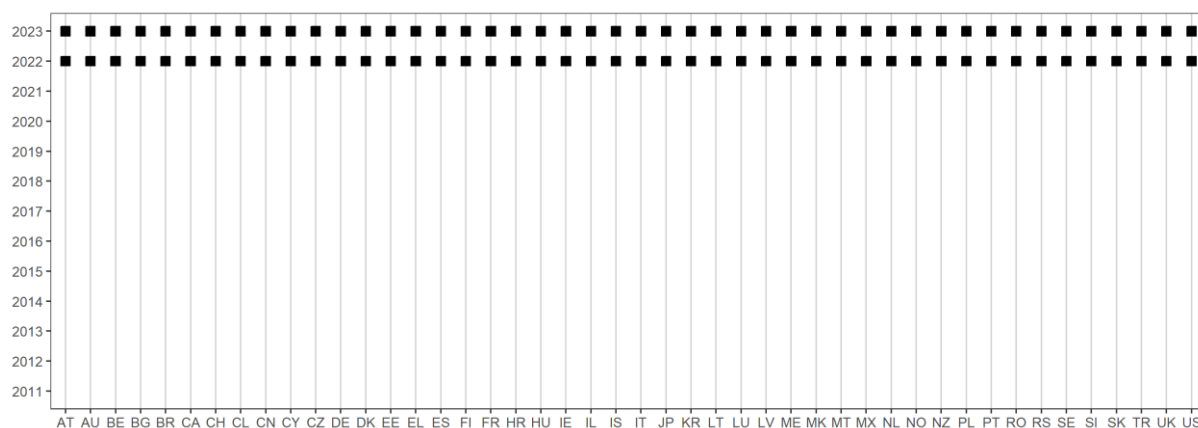
Source: Authors' calculation. Note: Squares indicate missing data for the country-year

Figure 28. Missing data years per county on TRA_POP



Source: Authors' calculation. Note: Squares indicate missing data for the country-year

Figure 29. Missing data years per county on PCT_POP



Source: Authors' calculation. Note: Squares indicate missing data for the country-year

Table 9. Normalized and Imputed descriptive statistics of the IOI 2024 and data for the year 2023

Indicator	Min	Max	Mean	Median	Std	Skew	Kurt	N. non-missing
GOOD_VA	0.06	9.83	4.08	3.78	2.37	0.41	-0.46	46
SERV_VA	0.03	10	4.27	4.01	1.99	0.85	1.41	46
KIABI	0.8	10	4.61	4.42	1.97	0.39	0.04	46
PCT_POP	0	10	2.35	1.2	2.74	1.34	0.88	46
TRA_POP	0.09	7.05	1.8	1.38	1.42	1.64	3.33	46
ENT	0.07	9.86	5.87	6.42	2.23	-0.32	-0.43	46

Source: Authors' calculation.

Annex 4. Country performance per IOI indicator

Table 10. Country performance per IOI indicator and change between 2013 and 2023.

Country	Indicator	IOI	GOOD_va	SERV_va	Patents	Trademarks	TECH_CAP	ENT	IP	KIABI
AT	Rank 2023	20.5	22	28	11	16	21	13	11	22
	Value 2023	104	4.29	3.8	4.1	1.9	4.05	7.18	3	4.78
	% Change	7.22	5.41	14.73	-2.66	-13.91	9.59	7.65	-6.53	15.05
AU	Rank 2023	22	42	17	23	15	35	8	24	15
	Value 2023	101	1.49	4.55	1.71	2.12	3.02	8.25	1.92	5.39
	% Change	3.06	-1.22	-2.62	-10.41	2.78	-2.28	6.22	-3.56	3.5
BE	Rank 2023	14.5	23	12	14	35	17	4	25	14
	Value 2023	114	4.11	5.39	2.78	1	4.75	8.81	1.89	5.4
	% Change	16.33	20.24	12.83	-0.88	-21.43	15.92	20.56	-7.31	21
BG	Rank 2023	40	38	35	40	24	40	38	36	40
	Value 2023	57	2.34	3.19	0.16	1.42	2.77	3.75	0.79	3.13
	% Change	50	18.94	36.17	0.73	16.72	28.3	53.76	14.92	89.19
BR	Rank 2023	43	35	27	46	36	32	40	41	44
	Value 2023	50	2.56	3.85	0.03	1	3.2	3.68	0.51	1.83
	% Change	-35.9	-32.99	7.88	-69.58	351.03	-13.25	-59.42	237.25	25.27
CA	Rank 2023	19	26	22	20	40	24	1	29	28
	Value 2023	105	3.72	4.12	2.09	0.76	3.92	9.86	1.43	4.07
	% Change	3.96	-2.73	4.98	0.92	-48.09	1.18	13.71	-19.44	-4.1
CH	Rank 2023	1	2	7	3	7	4	17	2	10
	Value 2023	145	8.43	6.36	8.42	3.23	7.39	6.76	5.82	6.65
	% Change	-9.38	5.81	-2	9.91	3.11	2.31	-32.4	7.94	0
CL	Rank 2023	47	43	42	39	23	46	46	35	45
	Value 2023	38	1.33	2.64	0.21	1.46	1.99	2.3	0.84	1.82
	% Change	-5	-15.92	-28.96	16.67	43.14	-25.06	18.41	39.09	-17.21
CN	Rank 2023	27	8	14	26	6	9	34	16	31
	Value 2023	89	6.73	4.88	1.18	3.79	5.8	4.41	2.48	3.69
	% Change	30.88	9.82	-4.75	276.98	435.39	3.18	-1.61	386.79	90.77
CY	Rank 2023	17	31	15	43	13	26	9	31	6
	Value 2023	110	2.89	4.83	0.07	2.18	3.86	8.01	1.13	7.1
	% Change	23.6	-1.03	-37.16	-69.48	124.37	-27.22	54.49	86.53	33.61
CZ	Rank 2023	29.5	17	34	28	27	22	18	34	29
	Value 2023	86	4.7	3.33	0.74	1.33	4.01	6.73	1.04	4.02
	% Change	8.86	6.27	4.38	61.75	-39.51	5.48	12.56	-22.07	16.88
DE	Rank 2023	8	4	8	8	11	5	6	8	23
	Value 2023	129	7.74	5.91	5.25	2.41	6.83	8.45	3.83	4.6
	% Change	-3.01	4.27	15.03	-7.87	17.94	8.67	-15.17	-1.07	9.57
DK	Rank 2023	16	9	37	6	33	16	14	9	16
	Value 2023	111	6.54	3.07	6.24	1.01	4.8	6.85	3.63	5.13
	% Change	9.9	17.64	8.68	13.29	-23.71	14.62	6.65	6.11	13.86
EE	Rank 2023	24	39	29	29	14	38	12	27	20
	Value 2023	92	1.96	3.76	0.71	2.16	2.86	7.78	1.44	4.82
	% Change	29.58	4.63	38.83	-4.75	59.61	24.85	15.78	36.69	63.64
EL	Rank 2023	27	44	16	34	19	37	3	32	37
	Value 2023	89	1.26	4.6	0.35	1.84	2.93	8.99	1.1	3.3
	% Change	18.67	-16.57	-23.23	70.03	33.31	-21.89	48.22	38.1	2.78
ES	Rank 2023	35	20	23	27	29	18	43	33	32
	Value 2023	68	4.55	4.12	0.93	1.17	4.34	3.34	1.05	3.66
	% Change	-2.86	1.01	15.9	-5.52	0.15	7.57	-25.59	-2.44	15.49
EU27	Rank 2023	18	5	9	15	21	6	28	20	25
	Value 2023	106	7.62	5.91	2.59	1.73	6.76	6.13	2.16	4.42
	% Change	6	3.51	8.72	-0.11	4.05	5.72	-2.46	1.51	25.32
FI	Rank 2023	9	18	6	5	31	11	7	7	13
	Value 2023	128	4.64	6.46	6.74	1.07	5.55	8.42	3.91	5.62
	% Change	14.29	5.05	22.91	-4.51	-38.35	14.76	26.87	-11.19	21.15

Country	Indicator	IOI	GOOD_va	SERV_va	Patents	Trademarks	TECH_CAP	ENT	IP	KIABI
	% Change	31.11	-11.54	28.26	55.48	2.94	4.56	37.68	14.87	74.42
PT	Rank 2023	33	36	33	31	12	36	30	28	34
	Value 2023	75	2.55	3.34	0.65	2.22	2.95	5.9	1.43	3.44
	% Change	5.63	11.67	-5.43	83.88	38.62	1.29	-18.43	46.79	87.8
RO	Rank 2023	46	10	20	41	39	12	48	44	47
	Value 2023	39	6.42	4.3	0.09	0.81	5.36	0.07	0.45	1.25
	% Change	-18.75	4.07	16.6	24.55	24.07	8.76	-97.58	24.11	115.38
RS	Rank 2023	37.5	34	40	44	46	39	32	47	39
	Value 2023	59	2.7	2.89	0.06	0.16	2.8	4.79	0.11	3.17
	% Change	7.27	24.27	0.89	27.46	-2.98	10.97	-19.96	3.43	108.82
SE	Rank 2023	2	13	10	4	22	10	10	4	4
	Value 2023	142	5.77	5.83	7.88	1.47	5.8	7.93	4.68	7.54
	% Change	17.36	9.05	19.13	7.69	-6.94	13.9	11.38	5.09	36.29
SI	Rank 2023	29.5	19	44	25	28	29	21	30	24
	Value 2023	86	4.57	2.36	1.23	1.31	3.46	6.49	1.27	4.51
	% Change	8.86	8.62	-2.55	-22.5	-27.62	4.54	14.8	-25.24	16.09
SK	Rank 2023	39	27	41	35	32	33	37	39	41
	Value 2023	58	3.47	2.83	0.35	1.05	3.15	3.81	0.7	3.04
	% Change	20.83	-2.94	92.7	38.02	-12.14	24.85	3.98	-3.43	58.14
TR	Rank 2023	41.5	30	38	38	4	34	35	22	48
	Value 2023	54	3.23	2.9	0.22	3.83	3.06	4.08	2.03	0.8
	% Change	5.88	15.85	25.73	36.5	131.3	20.33	-31.24	122.88	-
UK	Rank 2023	7	7	2	18	18	1	25	21	5
	Value 2023	130	7.2	9.48	2.3	1.85	8.34	6.29	2.08	7.21
	% Change	18.18	9.31	3.92	3.27	51.91	6.18	28.03	20.43	29.28
US	Rank 2023	4.5	3	3	10	38	2	11	14	11
	Value 2023	134	8.42	8.08	4.17	0.87	8.25	7.85	2.52	5.95
	% Change	6.35	3.5	14.3	8.15	57.39	8.52	0	14.31	8.37

Source: Authors' calculation. Note: EU27 denotes the EU27 as an aggregate block (excluding intra-EU trade).

Annex 5. The RIOI methodology

The RIOI 2024 dataset and index development

The dataset for the first, exploratory edition of the RIOI for 2024 includes 241 observations covering the 99.6% (241/242) of the NUTS level 2 regions of the 27 European Union Member States and the 27 EU MS as a single economic zone (EU27) for benchmarking reasons. Note that this includes five Member States (Cyprus, Estonia, Luxembourg, Latvia, and Malta) which are also classified as NUTS level 2 regions. The RIOI estimates for these five countries are not comparable to their IOI estimates shown in Section 2. This is because there are differences in the IOI and the RIOI in terms of some of the indicators used, the observation sample (which affects indicator minima and maxima that are used in the normalisation process), and the method used for imputing missing data (which was not used to impute missing data for these five countries but affected the data of other NUTS-2 regions). The dataset includes seven indicators: PCT_POP, EUTM_POP, KIABI, GOOD_IVA, SERV_IVA, BUS_INN, and PRD_INN.

Data availability: The first edition of the RIOI is developed for one year, using the last available data for each of the indicators. That is, if data for 2023 were not available for a particular indicator, data from a previous year were used. Please refer to **Annex 9** or **Table 11** which include information on the year in which data for each the seven indicators refer. The data availability threshold for inclusion of regions in the RIOI was set at 65%, in accordance with JRC-OECD (2008) guidelines. This meant that regions with data missing for three or more indicators would be excluded from the sample. This was the case for 1 NUTS-2 region (FRY5-Mayotte, see **Table 11**), which is thus not included in the RIOI. For regions for which no last known values are available for one or two indicators, the imputation was done by means of the algorithm used to develop the European Commission's RIS (see European Commission, 2023b), which imputes missing values for NUTS-2 regions with values of previous years (if available, using their ratio with the value of the parent NUTS-1 region) or with values of the parent NUTS-1 region, if regional NUTS-2 data are not available for the previous years. The JRC imputed the PRD_INN and BUS_INN indicators for 47 NUTS-2 regions (all regions mentioned in **Table 11** except the 12 regions of the Netherlands and FRY5) with the corresponding values of their parent NUTS-1 region for the same year (2020), using the RIS algorithm. All descriptive statistics and correlations

shown in this Annex exclude these imputed data. Statistics of the data including these imputations are given in **Table 18** in **Annex 6**. From these we can see that the available data after imputation for PRD_INN and BUS_INN are 229 instead of 241, because of the 12 missing values for the Dutch NUTS-2 regions. This is because regional data at the NUTS-1 level for these regions are not provided to Eurostat by means of the CIS. Note that not imputing these values results into the ENT Pillar, which is the average of PRD_INN and BUS_INN indicators, being missing for the 12 Dutch regions. This is effectively equivalent to a “shadow” imputation where the missing Pillar is imputed with the average of the regions’ scores for the other three Pillars.

Table 11: Data availability for RIOI indicators prior to imputation

Indicator	Latest available year	Data availability	Data not available for any year for:
GOOD_iVA	2017	100.0%	
SERV_iVA	2017	100.0%	
KIABI	2023	99.6%	
PCT_POP	2022	99.6%	
EUTM_POP	2022	99.6%	FRYS
PRD_INN	2020	75.2%	60 regions [AT (all 9 regions), BE (10 regions, all regions except BE10), FR (all 27 regions), NL (all 12 regions), PT (1 region), DE (1 region)]
BUS_INN	2020	75.2%	60 regions [AT (all 9 regions), BE (10 regions, all regions except BE10), FR (all 27 regions), NL (all 12 regions), PT (1 region), DE (1 region)]

Source: Authors’ calculation. Note: Data for KIABI and PCT_POP are not available for 1 NUTS-2 region in 2023 and 2022 respectively, but data for this region and previous years are available and were imputed.

Outlier treatment: The criteria set out in JRC/OECD (2008), according to which (absolute) skewness and (absolute) kurtosis are considered excessive when they simultaneously cross the threshold-values of 2 and 3.5, respectively were used to detect outlying observations. This was done using the COINr package for building Composite Indicators, developed at the JRC (Becker et al., 2022). Descriptive statistics for the non-normalized IOI 2023 dataset are shown in **Table 12**. This table excludes data imputed with NUTS-1 values. From **Table 12** we see that the skewness and kurtosis values for PCT_POP and EUTM_POP cross the JRC/OECD thresholds. In particular, the COINr package detected five and three outlying observations in PCT_POP and EUTM_POP respectively. These were treated using winsorisation, as suggested by JRC/OECD (2008) for up to five outlying observations.

Table 12: Descriptive Statistics of RIOI indicators for the 2023 data and 241 NUTS-2 regions [before treatment and normalisation]

Indicator	Min	Max	Mean	Median	Std	Skew	Kurt	N. non-missing	% non-missing
GOOD_iVA	0	76.1	26.6	24.4	16.1	0.67	0.09	241	100%
SERV_iVA	9.51	77.1	33.7	31.3	12.3	0.76	0.55	241	100%
KIABI	19.5	68.3	46.1	46.5	8.67	-0.21	-0.23	241	100%
PCT_POP	0	37.1	2.95	1.46	4.62	3.41	16.9	241	100%
EUTM_POP	0	1530	198	138	194	3.14	15	241	100%
PRD_INN	1.37	58.4	26.4	27.2	11.4	0.12	-0.28	182	76%
BUS_INN	1.28	72.8	40.4	46	16.3	-0.37	-0.61	182	76%

Source: Authors’ calculation.

Standardization: After imputation, the data was standardized using the min-max standardization, normalising the data so that the maximum score of 10 corresponds to the maximum observed value while the minimum score, zero, corresponds to the lowest observed value. Descriptive statistics for the treated and standardised indicators are provided in **Table 13**.

Weighting: Following the logic of the national IOI, all indicators and pillars of the RIOI are equally weighted within their aggregate. This results in each of the four Pillars receiving a weight of 25%, while individual indicators –except KIABI which consists of a Pillar by itself– receive a weight of 50% within their Pillar.

Aggregation: All indicators were added up using the arithmetic mean to form pillars. All pillars of the IOI were added up using the arithmetic mean using equal weights.

Table 13. Treated and normalized descriptive statistics of the IOI 2023 indicators.

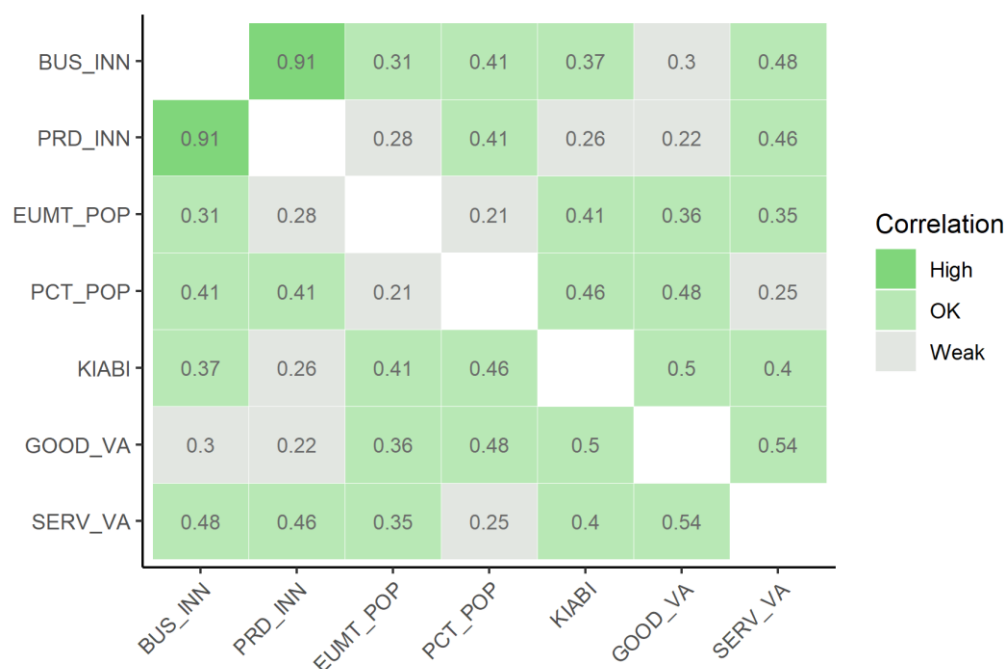
Indicator	Min	Max	Mean	Median	Std	Skew	Kurt	N. non-missing
GOOD_iVA	0	10	3.5	3.21	2.12	0.67	0.09	241
SERV_iVA	0	10	3.57	3.22	1.81	0.76	0.55	241
KIABI	0	10	5.45	5.54	1.78	-0.21	-0.23	241
PCT_POP	0	10	1.74	0.91	2.37	1.95	3.67	241
EUTM_POP	0	10	2.64	1.91	2.19	1.58	2.33	241
PRD_INN	0	10	4.39	4.53	2	0.12	-0.28	182
BUS_INN	0	10	5.47	6.26	2.28	-0.37	-0.61	182

Source: Authors' calculation.

Correlations between normalised indicators and the structural coherence of the RIOI

The conceptual coherence of the RIOI was tested using correlation analysis as done for the national IOI (see **Annex 1**). **Figure 30** shows the correlations among the seven indicators considered in the RIOI. The indicators of the intellectual property component (TRA_POP and PCT_POP) have a slightly weaker correlation than that suggested by the JRC/OECD (2008) guidelines (0.3), while the correlation between the two indicators of the Enterprise Innovativeness component (PRD_INN and BUS_INN) is very close to the 0.92 threshold. Nevertheless, all four indicators are included in the IOI to cover both technological and non-technological innovation outputs (in the case of the IP component) and innovativeness of enterprises in both the development of new products and the uptake of innovative business processes (in the case of ENT).

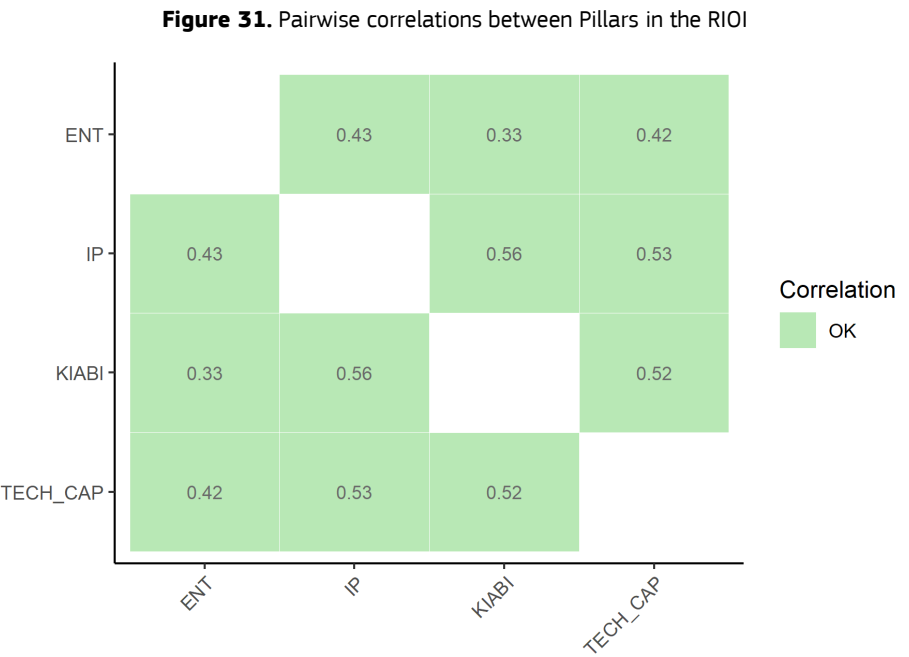
Figure 30: Pairwise Pearson correlations between indicators in the RIOI.



Source: Authors' calculation

Figure 31 shows correlations between the pillars of the RIOI. All correlations are between the thresholds suggested by the JRC/OECD (2008) guidelines and they are also quite close to each other, suggesting that there does not exist a particular pair of Pillars with very high correlation that could drive the RIOI results. The correlation between KIABI and ENT is close to the threshold of 0.3, while the correlations between ENT and

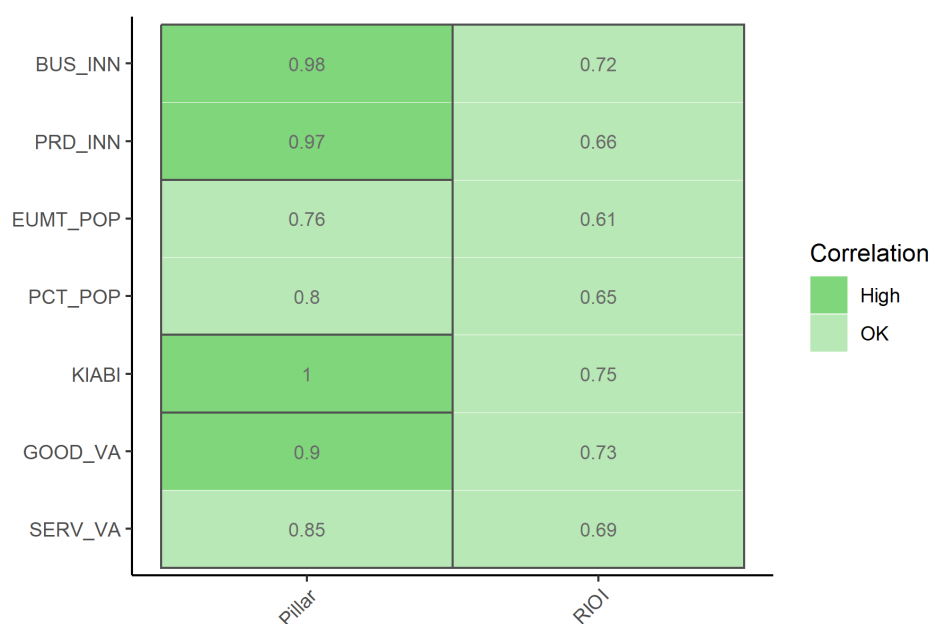
the other two indicators are also lower compared to those between pairs of IP, KIABI and TECH_CAP, suggesting a slightly lower rate of information transmission from ENT to the RIOI compared with other indicators. However, this is not an issue of concern as the ENT Pillar is adequately highly correlated with the RIOI (see the discussion below).



Source: Authors' calculation

Figure 32 shows the pairwise correlations between indicators with their respective pillars and the RIOI. From the left column, we see that all indicators are well correlated with their respective pillar and that correlations are rather balanced. An issue of concern is the high (above the suggested threshold of 0.92) correlations of PRD_INN and BUS_INN with the ENT Pillar, which follows from the very high correlation between the two indicators (see **Figure 30**). This may pose a risk of redundancy and a ‘double counting’ of the information contained in the two indicators. Despite this risk, accounting both for technological and non-technological innovation outputs is essential for the conceptual structure of the RIOI. However, this risk warrants further investigation and In future editions of the RIOI, JRC will reconsider the structure of the ENT pillar with the aim to improve its correlation structure. The right column shows indicator correlations with the RIOI. The strength of the correlation between indicator and aggregates most directly reflects the amount of information an indicator transmits to the index. These correlations all fall within the recommended guideline, providing evidence that the RIOI effectively summarises the information from its indicators and pillars. In addition, indicators’ correlation with the RIOI varies but not much: The KIABI indicators has the strongest correlation with the RIOI (0.75) while EUTM_POP has the lowest (0.61). When looking in the correlations of indicators of the same Pillar, we see that the two components of the ENT, TECH_CAP, and IP pillars have a rather balanced contribution to the RIOI scores.

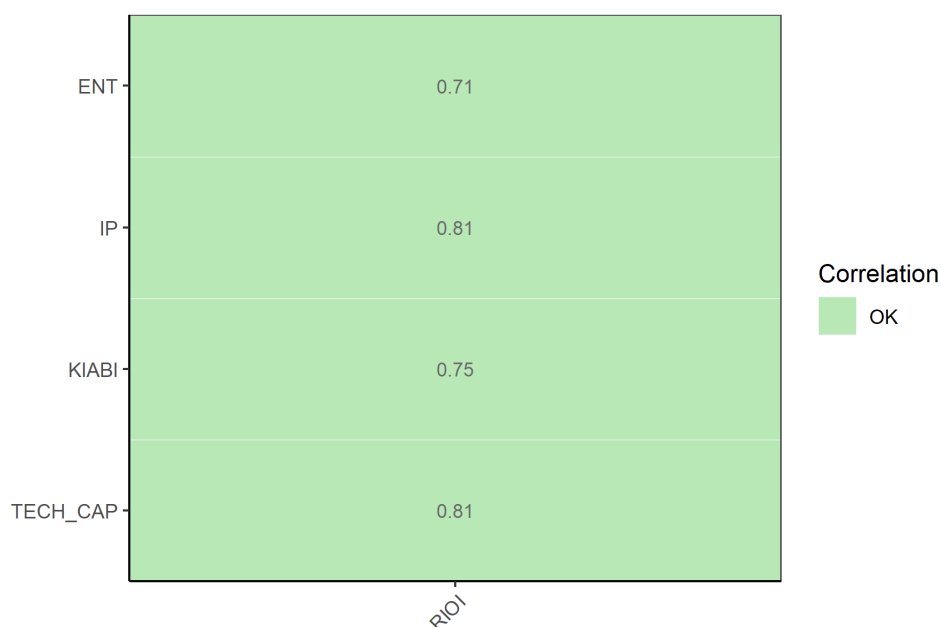
Figure 32. Pairwise correlations between indicators and their respective pillars and with the RIOI



Source: Authors' calculation

Figure 33 depicts the correlations between pillars and the RIOI, all of which are adequately high and within the JRC/OECD (2008) guidelines. In addition, the similarity of correlations (ranging from 0.71 to 0.81) indicates that each pillar contributes information to the RIOI to a rather similar degree. The KIABI and ENT pillars have a slightly lower correlation, as a result of their low correlation of ENT with KIABI and with the other two pillars. Nevertheless, all correlations are adequate and similar, providing evidence that the RIOI is a good summary measure of the information contained in the seven considered indicators.

Figure 33. Pairwise correlations between pillars and the IOI.



Source: Authors' calculation

Annex 6. Robustness of RIOI ranks

Uncertainty analysis

This section is similar to **Annex 2** and investigates the extent to which the modelling choices made during the development of the RIOI have an effect on the regions' RIOI ranking. The uncertainty analysis systematically changes four different modelling assumptions: the normalisation, imputation, the weights and aggregation. The alternatives considered in the uncertainty analysis of the RIOI are the same as those used for the national IOI and are shown in **Table 5**. Mean imputation is used as the alternative method to the RIS algorithm used in the nominal RIOI, the standardization of indicators with a mean of five and a standard deviation of 1.5 is used as the alternative normalisation method to min-max normalisation, the geometric mean is used as the alternative for aggregating the four RIOI pillars, while indicator and pillar weights are randomly varied by 20% from their original value.

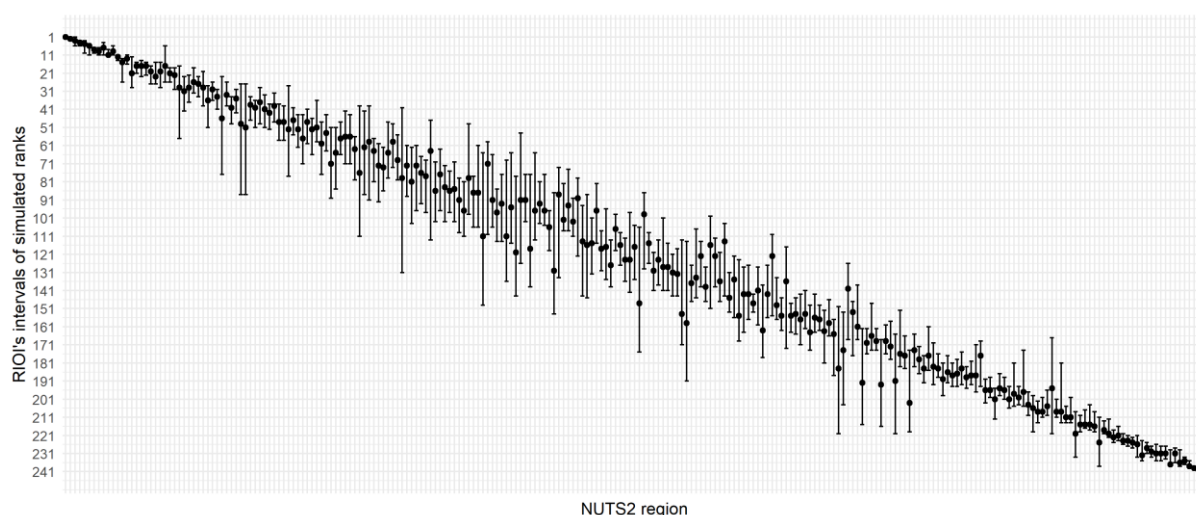
The combinations (original and alternative) of the normalisation, imputation and aggregation steps result in eight different versions of the RIOI. Four hundred alternatives are randomly drawn from these, and each is computed 100 times using a randomly selected set of weights from a distribution of weights with random perturbation of up to 20% of their original value. This results in 4000 versions of the RIOI. **Figure 34** plots for each NUTS-2 region the median rank, the 5th and the 95th percentile rank across the 4000 versions, while **Table 14** summarises the deviations between nominal and median ranks.

The ranks of the RIOI are generally robust against changes in methodological assumptions. In particular, for 60% of the NUTS-2 regions median rank up to five positions away from the nominal ranking (see **Table 14**). It should be noted that the absolute impact –in terms of changes in ranks– of changes in methodology is naturally expected to increase with sample size, as a given (absolute) change in a region's RIOI might cause larger changes in its ranking compared to, e.g. the national IOI where the number of the assessed units is considerably smaller. In this sense, with a sample of 241 NUTS-2 regions, rank deviations up to 5 positions are viewed as highlighting high stability and deviations up to 10 positions as highlighting satisfactory stability. From **Table 14** we see that for the vast majority of regions (83%), rank deviations from nominal rank are up to 10 positions. What is more, as can be seen from **Figure 34**, the rankings of the top and bottom-ranked regions are extremely stable: The average deviation of the median simulated rank from the nominal rank is 2.51 positions for the top-40 ranked regions and 1.98 positions for the bottom-40 ranked regions, suggesting that the RIOI can identify the top and the bottom 16% of regions in terms of innovation outputs with extreme precision.

There are 24 regions (10%) for which the median rank is more than 15 positions away from the nominal rank. These are all located around the middle of the RIOI distribution (their ranks range from 89th to 194th), where rank changes are in general more frequent even for smaller changes in absolute RIOI values due. In addition, for 67% (16/24) of these regions concern PT30, and French, Dutch, and Belgian regions, for which the PRD_INN and BUS_INN indicators are not available. Thus, the likely cause of the higher deviation of their median and nominal ranks is the imputation process, in which the default modelling choice (the RIS algorithm) takes into account the parent region's performance (which will unavoidably be the same for all NUTS-2 regions sharing the same parent), while the alternative mean imputation considers the own region's performance in the other five indicators of the RIOI. On the other hand, the remaining eight regions are Greek. These are ranked considerably better in terms of the two ENT indicators compared to the other five indicators of the RIOI, suggesting that their indicator profile is quite unbalanced. Thus, their rank deviations are likely caused by the aggregation process, where the alternative aggregation formula is the geometric aggregation which penalises unbalanced profiles, in combination with weights perturbation.

The confidence interval (CI) provides another indication of the uncertainty in the ranking due to modelling choices. The confidence interval reflects whether modelling alternatives result in a wide variety of rankings. In the uncertainty analysis, confidence intervals are computed as the 5th and 95th percentile of the observed distribution of ranks across simulations. The CI indicates highly stable rankings for the 15.8% of regions, which have a CI range of zero to ten. An additional 67% of regions have a CI range between 11 and 40 positions, indicating some uncertainty, but still allowing accurate inference. Lastly, the CI range is larger than 40 positions for the remaining 17% of regions, indicating a larger uncertainty in their ranking. The majority of these regions (29/42) include the French, Dutch, and Belgian regions, for which the PRD_INN and BUS_INN indicators are not available, and Greek regions which have quite unbalanced indicator profiles as discussed earlier.

Figure 34. Median rank in the Regional Innovation Output Indicator across simulations with alternative methods.



Source: Authors' calculation.

Table 14. Deviations of the medial simulated rank from nominal rank for NUTS2 regions.

Deviation from nominal rank	Share of NUTS2 regions
0 positions	12.03%
1 to 5 positions	59.75%
6 to 9 positions	11.20%
less than 10 positions	82.98%
10 to 15 positions	7.05%
>15 positions	9.97%

Source: Authors' calculation.

Sensitivity Analysis

The RIOI sensitivity analysis was implemented in a similar fashion with that of the national IOI. **Table 15** shows descriptive statistics of the sensitivity analysis, in particular the (absolute) deviations between the median rank and the nominal RIOI rank for each of the eight model specifications across 50 runs with random perturbations of the weights, averaged across the 241 regions.

In a similar fashion with the national IOI results, the random perturbation of the weight alone has little impact on rankings (the median rank in 98.3% of the regions deviates up to two positions from the nominal rank, while for four regions the rank deviates up to four positions). Changing one of the aggregation, imputation or normalisation functions in combination with the random weights impacts on the RIOI rankings, but not considerably. On average, the deviations between nominal and median ranks are up to 10 positions (see columns 2 to 4 of **Table 15**). For each of the three assumptions, at least 63% of the regions changes their rank for up to 10 positions. Among the three assumptions, aggregation appears to have the largest impact on rank deviations. Moreover, only 8.7% (21/241) regions change more than 30 rank positions ranks in comparison to the nominal ranks when using all three alternative functions (normalisation, imputation, aggregation) and random perturbations of the weights. In this case, the average of the absolute differences between nominal and median ranks is only 8.6 positions. Overall, the sensitivity analysis shows that the model specifications in the RIOI are robust to changes in the normalisation, imputation, weights and aggregation methods.

Table 15. Average deviation between RIOI nominal rank and median rank of eight model specifications across NUTS2 regions.

Only Weights	Alternative Specification						
	Weights +						
	Agg	Imp	Norm	Agg+Imp	Agg+norm	Norm+Imp	All
0.683	10.077	5.463	8.104	11.019	9.386	6.037	8.614

Source: Authors' calculation. *Note:* Agg refers to alternative aggregation formula, namely the geometric mean. Imp refers to the alternative imputation formula, i.e. mean imputation. Norm refers to the alternative normalisation procedure, namely standardization (Mean = 5 , S.D. = 1.5).

Table 16: Normalized and Imputed descriptive statistics of the RIOI 2024 and data for the year 2023

Indicator	Min	Max	Mean	Median	Std	Skew	Kurt	N. non-missing	% non-missing
GOOD_iVA	0	10	3.5	3.21	2.12	0.67	0.09	241	100%
SERV_iVA	0	10	3.57	3.22	1.81	0.76	0.55	241	100%
KIABI	0	10	5.45	5.54	1.78	-0.21	-0.23	241	100%
PCT_POP	0	10	1.74	0.91	2.37	1.95	3.67	241	100%
EUTM_POP	0	10	2.64	1.91	2.19	1.58	2.33	241	100%
PRD_INN	0	10	4.42	4.51	1.84	0.06	0.1	229	95%
BUS_INN	0	10	5.66	6.26	2.15	-0.47	-0.23	229	95%

Source: Authors' calculation.

Annex 7. Regional performance in the RIOI and components

Table 17: RIOI and pillar ranks, EU NUTS2 regions

NUTS code	Name	RIOI score	RIOI rank	IP rank	KIABI rank	TECH_CAP rank	ENT rank
SE11	Stockholm	190.2	1	1	2	8	23
DE21	Oberbayern	186.2	2	2	15	3	9
DE30	Berlin	161.9	3	16	11	6	30
CZ01	Praha	161.2	4	17	7	11	11
FR10	Ile-de-France	159.1	5	4	13	12	96
FI1B	Helsinki-Uusimaa	157.5	6	6	21	22	21
DE11	Stuttgart	154.5	7	10	27	13	29
DEA2	Köln	154.0	8	51	28	2	20
DE12	Karlsruhe	152.5	9	26	36	4	28
DE60	Hamburg	152.1	10	14	37	17	19
SE23	Västsverige	151.4	11	30	4	14	31
DE71	Darmstadt	150.4	12	28	34	5	34
DK01	Hovedstaden	148.5	13	8	3	53	49
IE06	Eastern and Midland	140.4	14	86	23	1	103
NL41	Noord-Brabant	139.9	15	5	55	66	-
BE31	Prov. Brabant Wallon	139.0	16	27	1	80	59
DE25	Mittelfranken	136.7	17	31	58	10	64
AT13	Wien	136.3	18	24	45	18	80
SE22	Sydsverige	135.1	19	41	9	40	36
SE12	Östra Mellansverige	134.6	20	61	5	20	75
DEA1	Düsseldorf	134.5	21	37	75	9	42
DE14	Tübingen	134.5	22	39	32	25	35
NL32	Noord-Holland	134.4	23	18	14	57	-
FRK2	Rhône-Alpes	132.8	24	11	74	24	106
DEB3	Rheinhausen-Pfalz	132.5	25	38	31	15	121
DE91	Braunschweig	130.9	26	65	19	39	27
CY00	Κύπρος (Kýpros)	126.7	27	13	143	77	12
BE21	Prov. Antwerpen	126.4	28	67	39	52	15
DE27	Schwaben	126.1	29	46	76	23	40
BE10	Région de Bruxelles-Capitale/Brussels Hoofdstedelijk Gewest	125.4	30	32	84	27	63
FI19	Länsi-Suomi	123.1	31	29	89	56	37
EL30	Αττική (Attiki)	122.5	32	128	61	49	8
ITC4	Lombardia	122.4	33	57	142	7	100
FRJ2	Midi-Pyrénées	121.9	34	22	40	59	126
BE24	Prov. Vlaams-Brabant	120.4	35	68	20	130	15
DEA4	Detmold	119.8	36	49	79	41	50
DE13	Freiburg	119.0	37	43	73	64	45
DEA5	Arnsberg	118.9	38	97	83	19	66
BE22	Prov. Limburg (BE)	118.3	39	55	64	96	15
DE26	Unterfranken	118.1	40	48	54	45	104
BE23	Prov. Oost-Vlaanderen	118.1	41	64	50	117	15
DE92	Hannover	117.8	42	56	41	28	133
DK05	Nordjylland	117.2	43	7	68	192	89
DEA3	Münster	116.4	44	101	66	26	53
DEF0	Schleswig-Holstein	116.0	45	82	94	33	39
DECO	Saarland	114.9	46	93	49	35	97
FRLO	Provence-Alpes-Côte d'Azur	114.8	47	45	53	62	122
DEB1	Koblenz	114.5	48	98	80	70	24
DK04	Midtjylland	114.3	49	36	65	90	85
AT34	Vorarlberg	114.1	50	3	161	205	71
DE23	Oberpfalz	114.1	51	73	57	48	87
AT31	Oberösterreich	113.7	52	23	131	78	71
SE21	Småland med öarna	113.3	53	103	30	69	78
LU00	Luxembourg	113.1	54	9	18	198	151
DE50	Bremen	113.0	55	74	70	46	83
DED5	Leipzig	112.7	56	130	69	21	92

NUTS code	Name	RIOI score	RIOI rank	IP rank	KIABI rank	TECH_CAP rank	ENT rank
FI1D	Pohjois- ja Itä-Suomi	111.8	57	66	108	60	44
DED2	Dresden	111.3	58	126	52	30	107
FRK1	Auvergne	110.8	59	12	111	150	106
SI04	Zahodna Slovenija	110.8	60	34	106	94	77
LT01	Sostinės regionas	110.8	61	15	77	204	76
DE24	Oberfranken	110.4	62	72	102	55	52
SE33	Övre Norrland	110.4	63	76	17	100	110
IE05	Southern	110.3	64	144	101	16	108
BE25	Prov. West-Vlaanderen	109.7	65	40	120	170	15
PL91	Warszawski stołeczny	109.2	66	42	26	83	160
PT17	Centro (PT)	109.0	67	125	81	34	86
HR05	Grad Zagreb	108.4	68	116	47	153	18
FRI1	Aquitaine	107.9	69	54	78	50	136
ITI4	Lazio	107.1	70	109	105	29	101
FI1C	Etelä-Suomi	106.4	71	104	117	47	65
NL31	Utrecht	106.4	72	91	10	101	-
DE93	Lüneburg	106.1	73	160	114	54	25
SK01	Bratislavský kraj	106.0	74	69	12	67	170
ITC1	Piemonte	104.9	75	84	145	63	38
DEG0	Thüringen	104.3	76	110	107	31	128
DE94	Weser-Ems	104.0	77	146	135	32	47
SE32	Mellersta Norrland	103.7	78	135	24	104	102
DE22	Niederbayern	103.7	79	83	100	68	98
NL33	Noord-Holland	103.5	80	81	22	102	-
FRC2	Franche-Comté	103.3	81	21	87	167	139
AT12	Niederösterreich	103.2	82	60	113	99	80
DE72	Gießen	103.2	83	117	63	109	51
DK03	Syddanmark	102.7	84	33	96	114	140
HU11	Budapest	102.3	85	87	29	72	164
ITH5	Emilia-Romagna	101.8	86	53	174	58	91
AT32	Salzburg	101.8	87	20	171	171	71
DE40	Brandenburg	101.5	88	136	92	51	94
ES30	Comunidad de Madrid	101.5	89	62	82	36	171
MT00	Malta	101.3	90	19	60	173	166
FRG0	Pays de la Loire	100.9	91	77	136	71	93
SE31	Norra Mellansverige	100.4	92	122	42	118	99
DED4	Chemnitz	99.5	93	179	112	38	95
CZ02	Střední Čechy	99.1	94	131	130	74	46
IE04	Northern and Western	98.9	95	139	118	73	62
FRH0	Bretagne	98.6	96	63	104	103	127
BE33	Prov. Liège	97.9	97	123	25	210	59
EE00	Eesti	97.8	98	25	149	189	90
DE73	Kassel	97.5	99	163	86	88	54
BE35	Prov. Namur	97.3	100	159	6	224	59
AT33	Tirol	97.2	101	50	155	142	71
FRD2	Haute-Normandie	96.9	102	44	91	112	159
NL22	Gelderland	96.7	103	100	35	115	-
FRF1	Alsace	96.3	104	70	99	116	131
CZ06	Jihovýchod	96.3	105	154	123	81	41
FRE1	Nord-Pas de Calais	96.2	106	105	85	82	144
BG41	Югозападен (Yugozapaden)	95.9	107	52	154	44	165
FRB0	Centre — Val de Loire	95.7	108	145	103	42	147
NL11	Groningen	95.5	109	134	16	126	-
FRJ1	Languedoc-Roussillon	95.2	110	108	98	93	126
EL52	Κεντρική Μακεδονία (Kentriki Makedonia)	95.0	111	175	210	97	3
ES51	Cataluña	94.8	112	58	147	43	168
ITH1	Provincia Autonoma di Bolzano/Bozen	94.6	113	35	190	110	112
BE32	Prov. Hainaut	94.5	114	166	38	178	59
NL21	Overijssel	94.4	115	78	48	139	-

NUTS code	Name	RIOI score	RIOI rank	IP rank	KIABI rank	TECH_CAP rank	ENT rank
EL54	Ἠπειρος (Ipeiros)	93.8	116	140	204	199	1
CZ05	Severovýchod	93.5	117	147	115	105	68
ITH3	Veneto	93.3	118	59	202	85	88
FRI3	Poitou-Charentes	92.7	119	79	119	107	136
AT22	Steiermark	92.3	120	71	129	133	118
SI03	Vzhodna Slovenija	91.1	121	47	150	166	116
CZ08	Moravskoslezsko	89.7	122	150	133	134	48
FRC1	Bourgogne	89.7	123	111	90	136	139
ES21	País Vasco	89.5	124	107	93	89	161
NL23	Flevoland	89.3	125	94	51	159	-
CZ07	Střední Morava	89.0	126	99	162	155	55
NL42	Limburg (NL)	88.5	127	89	67	141	-
FRE2	Picardie	88.4	128	75	138	123	144
FRI2	Limousin	88.2	129	151	71	127	136
ITH4	Friuli-Venezia Giulia	88.0	130	112	165	147	56
DK02	Sjælland	87.9	131	120	59	168	148
DEB2	Trier	86.8	132	114	134	181	84
DEE0	Sachsen-Anhalt	86.6	133	210	127	76	114
ITH2	Provincia Autonoma di Trento	86.1	134	121	152	135	109
CZ03	Jihozápad	86.0	135	182	144	131	67
BE34	Prov. Luxembourg (BE)	85.6	136	217	33	231	59
ITI3	Marche	85.6	137	96	196	187	33
FRF3	Lorraine	84.9	138	119	124	149	131
FRD1	Basse-Normandie	84.3	139	138	95	113	159
EL43	Κρήτη (Kriti)	84.3	140	170	235	129	4
EL63	Δυτική Ελλάδα (Dytiki Elláda)	84.1	141	178	224	208	2
AT21	Kärnten	83.3	142	113	151	157	118
ITC3	Liguria	83.2	143	92	156	184	111
FRF2	Champagne-Ardenne	83.0	144	102	122	194	131
NL12	Friesland (NL)	82.8	145	133	62	146	-
RO32	București-Ilfov	81.7	146	85	140	37	219
AT11	Burgenland	81.2	147	158	110	215	80
PT16	Centro (PT)	79.9	148	177	163	122	113
FRY4	La Réunion	79.3	149	234	72	125	156
EL61	Θεσσαλία (Thessalia)	78.8	150	201	221	206	7
ITI1	Toscana	78.6	151	88	191	140	145
NL34	Zeeland	78.5	152	129	116	128	-
PT11	Norte	78.2	153	142	195	75	149
ITI2	Umbria	78.1	154	124	197	203	43
FRY2	Martinique	78.1	155	235	88	121	156
ITF1	Abruzzo	77.9	156	165	178	182	74
ITF3	Campania	77.7	157	156	176	119	134
NL13	Drenthe	77.7	158	198	56	144	-
FRY3	Guyane	77.7	159	241	46	165	156
DE80	Mecklenburg-Vorpommern	77.6	160	171	125	79	172
SK03	Stredné Slovensko	77.2	161	213	126	65	173
HR03	Jadranska Hrvatska	76.3	162	195	193	92	124
EL51	Ανατολική Μακεδονία, Θράκη (Anatoliki Makedonia, Thraki)	75.5	163	219	214	174	22
EL42	Νότιο Αιγαίο (Notio Aigaio)	75.1	164	162	234	124	26
FI20	Åland	74.4	165	233	44	190	167
EL64	Στερεά Ελλάδα (Sterea Elláda)	73.8	166	212	240	162	6
ES24	Aragón	73.5	167	132	146	120	177
EL41	Βόρειο Αιγαίο (Voreio Aigaio)	73.4	168	236	203	137	73
FRY1	Guadeloupe	72.1	169	240	97	164	156
ES22	Comunidad Foral de Navarra	71.4	170	137	132	156	174
EL65	Πελοπόννησος (Peloponnisos)	71.4	171	200	236	219	5
ES52	Comunitat Valenciana	71.3	172	95	200	98	182
CZ04	Severozápad	71.0	173	184	166	201	119
HR06	Sjeverna Hrvatska	70.7	174	205	209	153	82

NUTS code	Name	RIOI score	RIOI rank	IP rank	KIABI rank	TECH_CAP rank	ENT rank
ES41	Castilla y León	69.5	175	155	177	91	183
PL21	Małopolskie	68.8	176	118	157	175	178
EL62	Ιόνια Νησιά (Ionia Nisia)	68.8	177	224	239	111	32
PL51	Dolnośląskie	68.1	178	148	121	145	208
HU12	Pest	67.4	179	152	137	160	186
ES64	Ciudad de Melilla	67.2	180	180	8	225	221
ITF4	Puglia	66.2	181	183	208	176	129
PL63	Pomorskie	65.3	182	141	128	197	190
ES61	Andalucía	65.0	183	168	183	87	207
EL53	Δυτική Μακεδονία (Dytiki Makedonia)	65.0	184	229	237	211	10
PT30	Região Autónoma da Madeira	64.9	185	127	173	216	162
HR02	Panonska Hrvatska	64.1	186	232	211	153	120
ES11	Galicia	64.0	187	153	188	108	196
ITF5	Basilicata	63.7	188	181	164	200	163
ITG1	Sicilia	63.6	189	208	175	188	153
PT18	Alentejo	63.5	190	185	181	213	146
ITF2	Molise	63.2	191	203	148	193	169
LV00	Latvija	62.6	192	90	182	207	200
ITF6	Calabria	62.6	193	225	185	221	115
HU31	Észak-Magyarország	62.6	194	226	139	163	185
HU21	Közép-Dunántúl	62.3	195	228	153	143	187
LT02	Vidurio ir vakarų Lietuvos regionas	61.7	196	149	207	234	123
ITG2	Sardegna	61.3	197	161	184	223	150
ITC2	Valle d'Aosta/Vallée d'Aoste	61.3	198	176	179	239	142
ES12	Principado de Asturias	59.6	199	157	189	148	198
HU23	Dél-Dunántúl	59.5	200	207	159	177	192
PL22	Śląskie	59.2	201	167	168	179	203
ES62	Región de Murcia	59.0	202	106	222	158	189
HU22	Nyugat-Dunántúl	58.7	203	220	180	151	188
ES13	Cantabria	57.7	204	211	169	180	191
SK02	Západné Slovensko	56.5	205	206	141	185	215
ES43	Extremadura	56.4	206	215	160	186	204
PT20	Região Autónoma dos Açores	56.1	207	221	167	241	152
HU33	Dél-Alföld	55.7	208	199	192	161	199
ES23	La Rioja	55.1	209	115	212	212	179
SK04	Východné Slovensko	55.0	210	216	158	202	206
FRM0	Corse	54.9	211	218	109	229	212
PL82	Podkarpackie	54.8	212	169	187	196	197
HU32	Észak-Alföld	54.6	213	237	170	169	209
PL42	Zachodniopomorskie	54.1	214	197	172	209	202
PT15	Algarve	53.1	215	196	215	237	141
RO42	Vest	52.4	216	222	186	61	225
ES42	Castilla-La Mancha	52.2	217	190	194	183	211
ES53	Illes Balears	51.7	218	80	218	218	216
BG42	Южен централен (Yuzhen tsentralen)	51.5	219	193	213	191	175
PL41	Wielkopolskie	50.8	220	143	227	172	205
PL71	Łódzkie	50.5	221	164	199	214	201
ES63	Ciudad de Ceuta	49.0	222	223	43	235	226
PL62	Warmińsko-mazurskie	46.4	223	204	198	233	194
ES70	Canarias	45.8	224	187	219	195	213
PL61	Kujawsko-pomorskie	45.1	225	186	205	217	214
PL81	Lubelskie	44.4	226	202	216	220	193
BG31	Северозападен (Severozapaden)	43.4	227	231	206	240	181
RO12	Centru	43.2	228	194	229	95	223
PL43	Lubuskie	42.4	229	172	201	227	220
BG34	Югоизточен (Yugoiztochen)	42.2	230	214	230	222	180
BG33	Североизточен (Severoiztochen)	42.1	231	188	225	236	184
BG32	Северен централен (Severen tsentralen)	42.0	232	191	231	230	176
PL84	Podlaskie	41.9	233	173	223	238	195

NUTS code	Name	RIOI score	RIOI rank	IP rank	KIABI rank	TECH_CAP rank	ENT rank
PL52	Opolskie	41.6	234	192	220	226	210
RO11	Nord-Vest	40.4	235	189	233	86	224
RO31	Sud-Muntenia	40.0	236	238	226	84	227
PL72	Świętokrzyskie	37.8	237	209	217	228	217
PL92	Mazowiecki regionalny	36.7	238	174	228	232	218
RO22	Sud-Est	30.7	239	230	232	138	228
RO41	Sud-Vest Oltenia	27.4	240	239	238	132	229
RO21	Nord-Est	26.1	241	227	241	106	222

Source: Authors' calculation. Note: Ranks for the ENT Pillar may involve imputed data from the corresponding NUTS-1 region. A dash in the ENT Pillar ranks refers to NUTS-2 regions for which the underlying regional CIS data at the NUTS-2 and the NUTS-1 level are not available. These are the 12 NUTS-2 regions of the Netherlands. See Annex 5 for more details on the imputation methodology used in the RIOI.

Table 18: Regional performance in the RIOI and individual indicators, summarised by country.

variable	RIOI			PCT_POP			EUTM_POP			KIA_REG		
country	Best-to-worst ratio	Regions above EU27	Regions below EU27	Best-to-worst ratio	Regions above EU27	Regions below EU27	Best-to-worst ratio	Regions above EU27	Regions below EU27	Best-to-worst ratio	Regions above EU27	Regions below EU27
AT	1.68	5	4	63.70	2	7	4.48	8	1	1.30	3	6
BE	1.62	7	4	6.88	5	6	7.55	5	6	1.47	10	1
BG	2.28	0	6	n	0	6	13.28	1	5	1.39	0	6
CY	-	1	0	-	0	1	-	1	0	-	0	1
CZ	2.27	1	7	8.21	1	7	12.93	1	7	1.46	1	7
DE	2.40	33	5	83.96	19	19	17.66	20	18	1.33	34	4
DK	1.69	4	1	6.15	4	1	4.24	3	2	1.29	5	0
EE	-	0	1	-	0	1	-	1	0	-	0	1
EL	1.88	1	12	2.80	0	13	6.00	0	13	2.09	1	12
ES	2.22	1	18	7.87	0	19	7.84	7	12	1.82	4	15
FI	2.12	4	1	3.71	3	2	15.77	1	4	1.22	4	1
FR	2.90	8	18	65.32	18	8	124.85	1	25	1.32	21	5
HR	1.69	1	3	1.02	0	4	7.60	1	3	1.55	1	3
HU	1.87	1	7	4.81	0	8	11.15	1	7	1.46	1	7
IE	1.42	2	1	1.87	0	3	3.25	1	2	1.22	2	1
IT	2.00	4	17	187.11	1	20	11.22	9	12	1.32	1	20
LT	1.80	1	1	1.00	0	2	4.06	1	1	1.38	1	1
LU	-	1	0	-	0	1	-	1	0	-	1	0
LV	-	0	1	-	0	1	-	0	1	-	0	1
MT	-	1	0	-	0	1	-	1	0	-	1	0
NL	1.80	4	8	67.83	1	11	7.25	8	4	1.27	11	1
PL	2.98	1	16	26.33	0	17	7.22	1	16	1.76	1	16
PT	2.05	1	6	1.00	0	7	4.69	2	5	1.46	1	6
RO	3.13	0	8	4.02	0	8	22.41	1	7	2.29	0	8
SE	1.89	8	0	6.94	5	3	5.78	3	5	1.24	8	0
SI	1.22	1	1	2.05	1	1	2.38	1	1	1.10	1	1
SK	1.93	1	3	n	0	4	6.25	1	3	1.37	1	3
EU27	7.29	92	149	1448.93	60	181	541.85	81	160	3.50	114	127

Source: Authors' calculation. Note: A dash refers to countries for which there is only one NUTS-2 region or countries for which regional NUTS-2 data are not available. Italic figures refer to ratios calculated over the minimum positive value because the real minimum was zero, and an "n" refers to countries for which all the value of an indicators is zero for all its regions.

Table 18: (contd).

variable		GOOD_IVA		SERV_I-VA			PRD_INN			BUS_INN		
country	Best-to-worst ratio	Regions above EU27	Regions below EU27	Best-to-worst ratio	Regions above EU27	Regions below EU27	Best-to-worst ratio	Regions above EU27	Regions below EU27	Best-to-worst ratio	Regions above EU27	Regions below EU27
AT	4.28	1	8	2.97	3	6	-	-	-	-	-	-
BE	6.96	1	10	3.38	4	7	-	-	-	-	-	-
BG	6.21	0	6	3.65	1	5	1.31	0	6	1.25	0	6
CY	-	0	1	-	1	0	-	1	0	-	1	0
CZ	1.95	0	8	3.64	3	5	1.62	7	1	1.37	8	0
DE	4.85	30	8	2.70	31	7	3.60	30	7	1.90	35	2
DK	1.66	2	3	2.53	1	4	1.27	4	1	1.52	3	2
EE	-	0	1	-	0	1	-	0	1	-	1	0
EL	41.35	0	13	1.92	9	4	1.58	13	0	1.63	13	0
ES	11.73	1	18	2.10	8	11	11.18	0	19	7.76	0	19
FI	5.49	0	5	1.85	5	0	3.19	4	1	1.43	4	1
FR	23.50	4	22	2.14	14	12	-	-	-	-	-	-
HR	1.10	0	4	1.52	1	3	1.65	4	0	1.32	3	1
HU	3.23	0	8	1.23	4	4	1.57	0	8	1.47	0	8
IE	1.56	3	0	2.52	2	1	1.09	3	0	1.15	3	0
IT	11.47	3	18	3.17	8	13	2.43	12	9	1.80	14	7
LT	1.24	0	2	2.41	0	2	1.24	2	0	1.10	2	0
LU	-	0	1	-	1	0	-	0	1	-	0	1
LV	-	0	1	-	0	1	-	0	1	-	0	1
MT	-	0	1	-	0	1	-	0	1	-	0	1
NL	3.07	1	11	1.65	5	7	-	-	-	-	-	-
PL	3.45	0	17	4.17	1	16	2.19	0	17	2.47	0	17
PT	43.32	0	7	3.22	3	4	1.46	1	5	1.32	2	4
RO	1.74	3	5	2.18	4	4	10.41	0	8	7.52	0	8
SE	3.16	4	4	1.66	8	0	1.66	7	1	1.19	8	0
SI	1.17	0	2	1.62	1	1	1.14	2	0	1.15	1	1
SK	1.58	0	4	3.46	2	2	1.66	0	4	1.51	0	4
EU27	139.55	53	188	8.11	120	121	42.53	91	91	57.07	99	83

Source: Authors' calculation. Note: A dash refers to countries for which there is only one NUTS-2 region or countries for which regional NUTS-2 data are not available. Italic figures refer to ratios calculated over the minimum positive value because the real minimum was zero, and an "n" refers to countries for which all the value of an indicators is zero for all its regions.

Annex 8. Definitions of IOI indicators

INDICATOR: PCT_POP		
	Numerator	Denominator
Definition	Number of PCT patent applications by country of residence of the inventor	Millions Population
Source	OECD MSTI (MSTI_PUB/P_PCT.) if available (PATS_IPC/PCT_A.INVENTORS..TOTAL.PRIORITY) OECD REGPAT otherwise	World Bank sp.pop.totl
Notes		
Most recent year used [Nr. Years lag vs. 2023]	2021 [2]	
Corresponding EIS indicator	NA	

INDICATOR: TRA_POP		
	Numerator	Denominator
Definition	Number of trademark classes in total trademark resident applications in direct and Madrid system by origin (equivalent count).	Millions of Population
Source	WIPO IP Statistics Data Center: https://www3.wipo.int/ipstats/index.htm?tab=trademark Indicator: "5 – Class count in total applications (direct and via the Madrid system)". Report type: "Total count by applicant's origin (equivalent count)". Type = "Resident".	World Bank sp.pop.totl
Notes	Missing time-series for EL and IE.	
Most recent year used [Nr. Years lag vs. 2023]	2022 [1]	
Corresponding EIS indicator	NA	

INDICATOR: KIABI		
	Numerator	Denominator
Definition	Employment in knowledge-intensive business industries	Total employment
Sources	Eurostat, htec_kia_emp2 (DOI: https://doi.org/10.2908/HTEC_KIA_EMP2); ILOSTAT, EMP_TEMP_SEX_EC2_NB_A; Japan Statistical Office, LFS; US BLS CPB; UK BRES; OECD, SSIS_BSC_ISIC4; KOSIS; Central Bureau of Statistics of Israel; New Zealand Statistical office	
Notes	US, JP, UK: data reporting discontinued on Eurostat website; figures were re-computed using national sources, following methodology described by Eurostat htec_esms. KR: Korean Office Statistical Information Service (KOSIS) from 2016-2021 IL: Central Bureau of Statistics of Israel, from 2016-2022 NZ: New Zealand Statistical office, from 2016-2020, figures re-computed following manual matching between ANZIC (Australian and New Zealand Standard Industrial Classification Revision 2.0) and NACE Rev. 2 No data available for CN.	
Most recent year used [Nr. year lag vs 2023]	2023 [0]	
Corresponding EIS indicator	4.1.1 Employment in knowledge-intensive activities as percentage of total employment	

INDICATOR: GOOD_VA		
	Numerator	Denominator
Definition	Domestic value added content of medium-high and high-tech manufacturing exports. Medium-high and high tech industries: ISIC Rev 4 Divisions 20,21,26,27,28,29,30.	Total manufacturing exports Manufacturing sectors: ISIC Rev. 4 Division 10-33.
Sources	OECD TIVA_2022_C1, EXGR_DVASH	
Notes	Data is missing for ME, MK, and RS.	
Most recent year used [Nr. year lag vs 2023]	2020 [3]	
Corresponding EIS indicator	NA	

INDICATOR: SERV_VA		
	Numerator	Denominator
Definition	Domestic value added content of knowledge-intensive service exports Knowledge intensive service sectors: ISIC Rev. 4 Divisions: 50,51, 58, 59, 60, 61, 62, 63, 64, 65, 66, 69, 70, 71, 72, 73,74,75, 84, 85, 86, 87, 88, 90, 91, 92, 93	Total service exports Service sectors: ISIC Rev. 4 Division 45-95.
Sources	OECD TIVA_2022_C1, EXGR_DVASH	
Notes	Sectors 78 and 80 are not considered as they are aggregated into group 77-82. Data is missing for ME, MK, and RS.	
Most recent year used [Nr. year lag vs 2023]	2020 [3]	
Corresponding EIS indicator	NA	

INDICATOR: ENT		
	Numerator	Denominator
Definition	Number of innovation-active enterprises Innovation-active firms include product and/or process innovative firms, as well as firms with only innovation activities	Total number of enterprises.
Sources	Eurostat Community Innovation Survey OECD Innovation Indicators database	
Notes	Figures are based on Eurostat CIS core industries (NACE Rev. 2 sections & divisions B-C-D-E-46-H-J-K-71-72-73). Data is missing for CA, IL, ME, ad MX.	
Most recent year used [Nr. year lag vs 2023]	2020 [3]	
Corresponding EIS indicator	NA	

Annex 9. Definitions of RIOI indicators

INDICATOR: PCT_POP		
	Numerator	Denominator
Definition	Number of PCT patent applications by region of residence of the inventor	Millions Population
Source	OECD REGPAT	World Bank sp.pop.totl
Notes	-	
Most recent year used [Nr. Years lag vs. 2023]	2021 [2]	
Corresponding RIS indicator	NA	

INDICATOR: EUTM_POP		
	Numerator	Denominator
Definition	Number of trademark applications applied for at EUIPO	Millions of Population
Source	DG RTD, European Union Intellectual Property Office (EUIPO).	World Bank sp.pop.totl
Notes	-	
Most recent year used [Nr. Years lag vs. 2023]	2022 [1]	
Corresponding RIS indicator	NA	

INDICATOR: KIA_REG		
	Numerator	Denominator
Definition	Employment in medium-high and high-tech manufacturing (C_HTC_MH) and knowledge-intensive services (KIS)	Total employment
Sources	Eurostat, htec_emp_reg2 (DOI: https://doi.org/10.2908/HTEC_EMP_REG2)	
Notes	C_HTC_MH sectors: NACE Rev.2 Divisions 20, 21, 26, 27-30 KIS sectors: NACE Rev.2 Divisions 50, 51, 58-66, 69-75, 78, 79, 80, 84-89, 90, 91-93	
Most recent year used [Nr. year lag vs 2023]	2023 [0]	
Corresponding RIS indicator	4.1.1 Employment in medium-high and high-tech manufacturing and knowledge-intensive services	

INDICATOR: GOOD_IVA		
	Numerator	Denominator
Definition	Domestic value added content of <i>international</i> medium-high and high-tech manufacturing exports. Medium-high and high tech industries: ISIC Rev 4 Divisions 20,21,26,27,28,29,30.	Total <i>international</i> manufacturing exports Manufacturing sectors: ISIC Rev. 4 Division 10-33.
Sources	JRC FIGARO-REG	
Notes	-	
Most recent year used [Nr. year lag vs 2023]	2017 [6]	
Corresponding RIS indicator	NA	

INDICATOR: SERV_IVA		
	Numerator	Denominator
Definition	Domestic value added content of <i>international</i> knowledge-intensive service exports Knowledge intensive service sectors: ISIC Rev. 4 Divisions: 50,51, 58, 59, 60, 61, 62, 63, 64, 65, 66, 69, 70, 71, 72, 73,74,75, 84, 85, 86, 87, 88, 90, 91, 92, 93	Total <i>international</i> service exports Service sectors: ISIC Rev. 4 Division 45-95.
Sources	JRC FIGARO-REG	
Notes	-	
Most recent year used [Nr. year lag vs 2023]	2017 [3]	
Corresponding RIS indicator	NA	

INDICATOR: PRD_INN		
	Numerator	Denominator
Definition	Number of SMEs introducing product innovations.	Total number of SMEs
Sources	DG RTD, Eurostat Community Innovation Survey	
Notes	Figures are based on Eurostat CIS core industries (NACE Rev. 2 sections & divisions B-C-D-E-46-H-J-K-71-72-73). Data is missing for 60 NUTS-2 regions [AT (all 9 regions), BE (10 regions, all regions except BE10), FR (all 27 regions), NL (all 12 regions), PT (1 region), DE (1 region)]	
Most recent year used [Nr. year lag vs 2023]	2020 [3]	
Corresponding RIS indicator	3.1.1 SMEs introducing product innovations as percentage of SMEs	

INDICATOR: BUS_INN		
	Numerator	Denominator
Definition	Number of SMEs introducing business process innovations.	Total number of SMEs.
Sources	DG RTD, Eurostat Community Innovation Survey	
Notes	Figures are based on Eurostat CIS core industries (NACE Rev. 2 sections & divisions B-C-D-E-46-H-J-K-71-72-73). Data is missing for 60 NUTS-2 regions [AT (all 9 regions), BE (10 regions, all regions except BE10), FR (all 27 regions), NL (all 12 regions), PT (1 region), DE (1 region)]	
Most recent year used [Nr. year lag vs 2023]	2020 [3]	
Corresponding RIS indicator	3.1.2 SMEs introducing business process innovations as percentage of SMEs	

Table 20. Orientation of the economy toward knowledge-intensive sectors

Country	Employment in the high and medium-high tech manufacturing sectors, as a share of total manufacturing employment				Employment in the knowledge-intensive service sector, as a share of total service employment				The share of medium- and high-tech products in total product exports (GOOD)				Knowledge-intensive services exports as share of total service (SERV) exports			
	2023	2022-23 Growth rate (%)	2021-23 Growth rate (%)	2013-23 Growth rate (%)	2023	2022-23 Growth rate (%)	2021-23 Growth rate (%)	2013-23 Growth rate (%)	2023	2022-23 Growth rate (%)	2021-23 Growth rate (%)	2013-23 Growth rate (%)	2023	2022-23 Growth rate (%)	2021-23 Growth rate (%)	2013-23 Growth rate (%)
AT	6.4	0.0	-1.5	10.3	40.6	2.0	1.0	8.0	56.2	2.2	0.0	-0.7	47.0	-0.6	-4.4	1.4
BE	4.2	0.0	-4.5	-10.6	50.1	-0.4	-2.5	7.3	50.2	5.7	-4.7	9.2	72.5	0.4	-1.1	3.2
BG	4.6	2.2	2.2	17.9	32.3	1.9	0.6	6.3	35.2	4.1	-3.8	32.7	53.2	0.5	-0.2	6.8
CY	0.9	-10.0	-18.2	-10.0	43.1	2.1	2.6	12.5	45.1	40.8	-16.7	4.4	84.5	-1.7	-3.5	10.1
CZ	10.7	-2.7	-5.3	1.9	36.3	2.0	1.1	11.3	68.4	4.6	4.1	9.4	50.4	-0.1	-1.9	3.3
DE	9.5	1.1	-5.9	-1.0	43.3	1.9	2.6	9.3	66.4	2.0	0.6	1.2	74.2	-1.0	-2.5	-0.5
DK	5.3	6.0	6.0	3.9	49.8	1.4	0.8	-0.2	49.0	-0.7	-3.4	12.5	78.4	-5.0	-4.7	0.0
EE	4.1	-2.4	-4.7	0.0	39.7	3.1	3.4	11.8	41.2	8.1	10.5	-0.8	66.9	0.9	0.5	9.8
EL	1.6	-5.9	0.0	33.3	37.9	-0.8	-3.1	3.3	23.5	8.5	-6.0	31.6	53.4	-3.3	-7.7	0.8
ES	4.0	0.0	0.0	2.6	38.1	0.0	0.3	6.1	45.4	5.5	3.5	-0.3	41.6	-0.5	-9.1	1.7
FI	5.0	8.7	6.4	-2.0	46.6	1.5	2.0	5.0	45.8	9.1	3.0	19.9	75.2	-2.1	-6.6	1.0
FR	3.9	2.6	2.6	-13.3	47.5	2.4	1.9	6.3	54.4	3.7	0.3	-4.5	65.0	-2.0	-4.4	0.8
HR	3.4	-5.6	-8.1	-5.6	37.2	4.8	6.6	13.4	38.4	18.5	7.2	2.0	23.2	0.2	-0.3	0.7
HU	8.9	-3.3	-4.3	4.7	36.3	-0.5	-1.9	0.6	70.8	10.0	6.3	8.4	52.7	-1.5	-0.9	2.4
IE	4.2	0.0	-8.7	-19.2	48.6	1.5	0.0	7.0	63.5	4.0	0.8	32.1	90.6	0.0	-0.1	4.9
IT	6.2	0.0	-3.1	5.1	35.6	2.0	0.6	4.1	51.4	4.4	3.0	2.2	51.2	-0.2	-5.9	-0.2
LT	2.7	-3.6	-3.6	50.0	38.5	1.0	3.8	16.3	36.6	5.0	-6.4	18.8	39.8	2.1	1.4	4.0
LU	0.6	20.0	-14.3	-33.3	58.1	-0.9	1.2	1.6	42.9	4.8	3.0	-12.2	85.1	0.2	-1.1	-1.9
LV	1.9	-9.5	-5.0	5.6	38.3	0.8	0.8	6.1	33.2	12.1	4.3	14.5	59.7	0.2	1.9	4.6
MT	2.8	12.0	0.0	-28.2	49.5	-1.2	-0.8	12.2	62.5	5.8	13.9	12.9	46.1	3.9	2.5	-9.3
NL	2.9	0.0	0.0	7.4	53.0	-0.2	1.7	13.5	50.1	8.8	-1.6	24.2	77.6	0.1	0.0	10.0
PL	5.3	1.9	1.9	6.0	35.4	4.7	5.7	13.5	49.7	4.3	1.9	2.1	49.7	1.1	0.3	4.4
PT	3.8	0.0	8.6	40.7	38.7	-1.8	-4.0	15.5	41.8	5.6	2.1	18.7	40.8	-0.3	-3.7	-1.3
RO	7.0	2.9	2.9	45.8	24.4	-5.1	-3.2	21.4	54.5	5.6	-1.2	9.4	53.4	0.8	0.2	3.9
SE	4.7	4.4	6.8	6.8	57.1	0.4	2.7	10.4	55.8	9.7	5.0	7.7	78.8	1.0	-2.1	-1.0
SI	8.6	-2.3	-2.3	3.6	37.2	-8.8	-11.0	8.1	65.8	9.2	8.0	20.5	39.2	-0.5	-1.4	1.9
SK	10.3	-8.8	-13.4	5.1	36.3	-1.4	0.0	10.7	69.9	6.1	3.1	9.9	45.3	1.5	-0.5	3.5
EU	6.0	1.7	-1.6	1.7	41.3	1.2	1.5	9.0	62.2	4.6	2.1	5.6	75.6	-1.5	-4.8	2.7

Source: Eurostat, accessed in December 2024.

Table 21. Entrepreneurship

Country	Birth rate, number of enterprise births on the total number of active enterprises				Employment share of enterprise births				Employment in fast-growing enterprises, as a percentage of total employment (DYN_rev)			
	2020	2019-20 Growth rate (%)	2018-20 Growth rate (%)	2011-20 Growth rate (%)	2020	2019-20 Growth rate (%)	2018-20 Growth rate (%)	2011-20 Growth rate (%)	2022	2021-22 Growth rate (%)	2020-22 Growth rate (%)	2012-22 Growth rate (%)
AT	5.4	-9.8	-13.2	-29.4	1.1	-16.8	-19.3	-46.3	7.8	8.6	11.5	-5.0
BE	6.9	-3.8	3.8	-2.3	2.3	-2.1	14.2	11	5.7	32.8	-1.5	-34.1
BG	9.1	-20.4	-20.2	-17.4	2.1	-21.1	-26	-27.8	13.7	2.0	-3.4	-25.0
CY	9.1	3.3	-12.7	100.4	2	6.4	-20.3	19.8	3.3	22.8	16.3	56.4
CZ	8.2	-6.5	-13.8	-22.7	1.9	-7.8	-10.4	-41.7	11.1	4.8	2.4	-1.5
DE	7.2	-21.6	-10.5	-17.3	1.2	-24.1	-1.6	-16.3	12.5	0.9	1.8	4.0
DK	11	0.5	0.2	-9.1	1.5	4.1	10.1	-12.1	15.8	-14.4	21.9	-3.1
EE	12.1	9.1	-0.7	1.9	3.5	21.5	9.4	21.5	15.4	7.8	21.4	3.3
EL	4.6	-3.8	-1.7	-10.7	2.7	-6.9	-2.2	-10.9	18.9	-14.8	-14.8	-26.2
ES	7.4	-20.7	-23.7	-7	2.5	-22.1	-26.7	-19.6	14.8	-5.1	-3.9	59.2
FI	8.8	1.7	10	-9.8	1.5	6.9	-26.3	108.1	14.5	2.6	-7.1	27.9
FR	11.3	-6.1	3.7	2.8	3.3	-4.9	9.3	32.3	4.4	2.3	-39.8	-61.7
HR	9.4	-26.1	-4.8	14.5	2.4	-26.3	-20.5	-25.8	11.6	7.0	-5.5	-3.4
HU	10.7	-13.2	-18.3	7.6	2.7	-41	-39.4	-26.8	16.4	15.0	15.0	-6.8
IE	6.7	0	25.6	7.2	1.7	0	50.9	97.7	22.1	6.3	6.3	15.9
IT	6.5	-11.9	-8.7	-2.4	1.9	-15.5	-15.5	-15.9	8.7	1.0	-29.1	-18.5
LT	18.1	-6.4	-4.6	-23.6	2.9	-8.3	-4.3	-47.5	14.7	31.6	-9.0	-6.3
LU	7.8	-14.4	-13.1	-19.4	1.3	-25.4	-17.1	-25.9	12.1	6.4	1.8	16.2
LV	11.3	-2.8	-7.4	-40.9	2.4	-5.9	-23.6	-53.9	15.8	18.8	29.4	-15.3
MT	14.1	1.9	-5.1	305.7	4.4	9	34.7	130.2	19.4	-14.3	-14.3	44.4
NL	10.4	-3	5.5	-6.5	1.4	-2.2	7.9	-17.1	16.2	15.8	-4.2	1.9
PL	10.4	-17.9	-21.4	-16.3	2.9	-17.8	-21.9	-28	14.0	-1.9	-1.9	1.3
PT	12.2	-22.6	-23.7	-2.8	3.8	-24.3	-24.6	-11.1	16.2	4.5	4.5	28.6
RO	10.8	-7.5	-0.9	-17.2	3.3	-5.8	4.2	-19.8	14.5	-39.9	177.8	137.6
SE	6.8	9.3	7.3	-17	1.8	-0.6	-2.2	-27	21.6	3.8	12.8	13.5
SI	9.1	-12.2	-14.2	-10.5	2.2	-11.4	-14.6	-11.1	8.6	0.5	-19.3	2.2
SK	10.4	-14.1	-17.2	-27.5	3.6	-16.7	-17.9	-24.1	11.8	22.5	15.5	-34.4
EU	8.9	-11.6	-8.7	-1.4	2.2	-14.7	-10.6	-10.6	12.0	8.7	-3.9	4.1

Source: Eurostat, accessed in December 2024.

Table 22. Innovation

Country	R&D investment (PPS, 2005 constant prices)				R&D expenditure, as a percentage of GDP				Employment in innovation-active firms, as a share of total employment			Share of turnover from new-to-market products		
	2023	2022-23 Growth rate (%)	2021-23 Growth rate (%)	2013-23 Growth rate (%)	2023	2022-23 Growth rate (%)	2021-23 Growth rate (%)	2013-23 Growth rate (%)	2020	2018-20 Growth rate	2011-20 Growth rate	2020	2018-20 Growth rate	2011-20 Growth rate
AT	9,679.0	2.6	5.4	25.7	3.3	3.5	0.9	10.4	80.7	-0.1	5.8	5.7	-9.5	38.5
BE	12,318.5	2.2	3.0	69.1	3.3	0.9	-2.6	43.1	86	1.7	12.7	5.5	34.1	36.5
BG	807.2	7.4	9.2	62.4	0.8	5.3	2.6	25.4	59.9	3.6	25.6	2.8	0	54
CY	181.2	-0.7	-3.4	114.9	0.7	-2.9	-12.8	41.7	76.5	-0.9	26.9	4.9	53.1	12.3
CZ	4,821.6	-3.1	-2.5	22.6	1.8	-3.2	-5.2	-2.1	76.3	13.5	9.8	6.1	-4.7	-7.5
DE	84,257.3	0.8	2.0	25.1	3.1	1.3	1.0	11.9	86.4	-0.1	0.8	3.4	-2.9	14.2
DK	5,866.5	6.7	13.3	25.9	3.0	4.2	9.1	1.0	68.2	-0.4	-6.2	4.7	51.6	-42.1
EE	469.8	1.3	1.9	34.5	1.8	4.5	5.1	8.2	77	-9.3	19.2	2.3	-68.3	-37
EL	3,073.5	3.4	12.7	106.3	1.5	0.7	4.2	81.7	82.5	9.4	17.7	9.7	-6.7	72.4
ES	17,751.8	9.0	16.6	42.4	1.5	5.7	6.4	17.3	58.2	2.1	-6.4	6.4	-7.2	19.7
FI	4,850.1	2.3	2.8	2.6	3.1	3.7	2.7	-5.8	81.8	1	8.9	5	-7.4	14.5
FR	41,655.8	-0.5	2.3	9.3	2.2	-1.4	-0.9	-1.8	82	3.9	6.2	3.6	-26.5	-42.7
HR	1,008.4	1.0	23.8	131.4	1.4	-2.1	12.1	75.9	72.3	6.3	11.1	3.9	21.9	8.1
HU	2,700.2	-1.2	-12.3	35.7	1.4	0.0	-14.7	0.7	53.5	2.7	-11.3	2.6	-36.6	-47.1
IE	3,331.6	10.9	17.6	49.3	2.2	0.0	102.8	40.9	76	42.3	-2.2	18.4	166.7	-66
IT	19,503.7	-3.2	-1.9	11.5	1.3	-4.4	-7.1	1.6	71.7	-6.5	-3.4	4.3	-31.7	-15.3
LT	688.4	-0.3	-2.2	51.3	1.1	0.0	-4.5	10.5	76.2	7.2	39.1	4.3	19.4	105.9
LU	408.1	-2.9	-0.4	3.8	1.0	-1.9	-1.0	-16.3	65.2	-2.2	-20.1	1.4	-41.7	-64.7
LV	280.7	3.7	11.2	64.4	0.8	2.5	7.8	29.7	51.6	-7	-1.5	3	-6.3	28.9
MT	111.9	13.7	14.1	59.6	0.6	5.2	1.7	-16.4	59.5	-5.1	-18.9	3.1	-13.9	18.8
NL	13,929.0	-4.2	-1.2	19.4	2.1	-4.6	-6.3	-2.8	70.6	7.8	3.5	4.9	2.1	-29.9
PL	13,221.7	8.5	16.3	155.7	1.6	8.3	9.9	77.3	60.9	20.8	11.5	3.6	63.6	35.6
PT	3,855.4	2.6	11.3	55.0	1.7	0.0	1.2	28.0	70.4	17.3	-2.1	4.3	-10.4	-33.2
RO	1,516.8	14.3	16.7	88.9	0.5	13.0	10.6	33.3	21.4	-13.7	-39.7	1	-54.5	10
SE	12,429.0	2.6	5.8	33.2	3.6	2.9	4.4	8.8	81.1	0.9	9.2	6.3	16.7	66.2
SI	1,180.9	3.8	4.6	10.5	2.1	1.4	-0.5	-17.8	75.7	9.2	5.4	3.6	38.5	-10.9
SK	1,292.5	8.1	17.7	60.9	1.0	6.1	15.6	26.8	61.7	13.4	5.8	11.9	58.7	-19.1
EU	258,558.9	0.2	2.8	27.2	2.3	0.0	0.4	8.2	73.2	3.4	2.5	4.6	-2.1	-3.2

Source: Eurostat, accessed in December 2024. Note: Data on the share of new-to-the market products refer to 2018 for Estonia.

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